Empowered lives.
Resilient nations.

United Nations Development Programme

Project Document template for projects financed by the various GEF Trust Funds

Project title: Enhancing sustainability in electric mobility for low carbon urban transport and an Extended Producer Responsibility (EPR) approach in batteries and vehicle components		
Country(ies): Peru	**Implementing Partner (GEF Executing Entity):** *Ministerio del Ambiente (MINAM, Ministry of Environment)*	**Execution Modality**: *Partial CO Support to NIM*
Contributing Outcome (UNDAF/CPD, RPD, GPD): *UNDAF Outcome 1: By 2021, people living in poverty and vulnerability improve access to decent livelihoods and productive employment by means of sustainable development that strengthens social and natural capital, integrating an adequate management of risk.* *UNDAF Outcome 3: By 2021, public management is more efficient, effective, transparent and equitable, thus increasing the confidence of the people in institutions.* *CPD Outcome 1: Inclusive and sustainable growth and development. (Output 1.1: National and subnational capacities strengthened to implement policies, plans or other instruments of sustainable and inclusive development).* *CPD Outcome 3: Institutions and transparency. (Output 3.1: National and subnational capacities in public management strengthened to increase efficiency and effectiveness and comply with international commitments and Output 3.3: National and subnational capacities strengthened for implementation of 2030 Agenda).*		
UNDP Social and Environmental Screening Category: *Moderate*	**UNDP Gender Marker:** *2*	
Quantum Award ID: *00115399.1*	**Quantum Project/Output ID:** *00115399*	
UNDP-GEF PIMS ID number: *6384*	**GEF Project ID number:** *10286*	
LPAC meeting date: *21 July 2021*		
Last possible date to submit to GEF: *13 December 2020*		
Latest possible CEO endorsement date: *13 June 2021*		
Project duration in months: *42*		
Planned start date: *15 April 2023*	**Planned end date:** *14 October 2026*	

Expected date of Mid-Term Review: *N/A*	**Expected date of Terminal evaluation:** *15 July 2026*
Brief project description: *The project aims at accelerating the shift towards electric mobility in the urban public transport systems of Peruvian cities in order to reduce GHG emissions and improve air quality. Based on an in-depth analysis of the existing legal and economic barriers, the project will propose reforms in the regulatory, technical, operational and economic framework to foster the use of electric vehicles in public and private passenger transportation, and will introduce various electric public buses and three-wheeler taxis (means of transport popular in certain urban areas in Peru) in two pilot cities, putting the mobility of vulnerable urban groups at the center of the project.* *The project will also provide a strategy for the uptake of the Extended Producer Responsibility (EPR) approach to end-of-life vehicle management, including batteries and other electric vehicle components, as well as appropriate finance schemes and business models. It will encourage private sector participation and investments, in the widespread deployment and commercialization of low carbon transport systems, to generate GHG emissions reduction schemes not limited to the implementation of public subsidies and improving air quality.*	
FINANCING PLAN	
GEF Trust Fund grant	USD 1,784,862
UNDP TRAC resources[1]	USD 0
Confirmed cash co-financing to be administered by UNDP	USD 0
(1) Total Budget administered by UNDP	**USD 1,784,862**
CO-FINANCIERS THAT WILL DELIVER PROJECT RESULTS INCLUDED IN THE PROJECT RESULTS FRAMEWORK (FUNDS NOT ADMINISTERED THROUGH UNDP ACCOUNTS)	
Ministerio del Ambiente	*USD 379,818*
Ministerio de Energía y Minas	*USD 2,000,000*
Municipalidad Provincial de Arequipa	*USD 2,670,623*
Engie Energía Perú S.A.	*USD 2,000,000*
QEY Tech Perú SAC	*USD 1,500,000*
Asociación de Emprendedores para el Desarrollo e impulso del vehículo Eléctrico en el Perú - AEDIVE	*USD 1,400,000*
Motores Diesel Andinos, S.A. - MODASA	*USD 700,000*
Agencia Francesa de Desarrollo - ADF	*USD 600,536*
SINOMAQ, S.A.	*USD 452,000*
Universidad de Ingeniería y Tecnología - UTEC	*USD 425,250*
Integra Arequipa SAC	*USD 388,000*
Oficina de Cooperación Suiza - COSUDE	*USD 349,097*
SENATI	*USD 94,311*
WUITO	*USD 88,500*
ABB	*USD 77,000*
CAVI Industrial SAC	*USD 62,000*
DEUMAN	*USD 41,900*
Fundación Transitemos	*USD 39,000*
ENEL	*USD 382,000*

[1] This is not a mandatory requirement.

CITE Energía	*USD 949,800*
UNDP	*USD 200,000*
Profonanpe	*USD 60,000*
(2) Total confirmed co-financing	**USD 14,859,835**
(3) Grand-Total Project Financing (1)+(2)	**USD 16,644,697**

SIGNATURES:

NOTE: IF THE PROJECT DOCUMENT IS IN FRENCH OR SPANISH, THE FINAL PROJECT DOCUMENT MUST BE CLEARED BY THE RTA BEFORE SIGNATURE.

Signature:	**Agreed by Government Development Coordination Authority**	**Date/Month/Year:**
Signature:	**Agreed by Implementing Partner**	**Date/Month/Year:**
Signature:	**Agreed by UNDP**	**Date/Month/Year:**

Key GEF Project Cycle Milestones:

Project document signature: benchmark date for completion no later than 6 months from GEF CEO endorsement.
First disbursement date: To be achieved by no later than 18 months from GEF CEO endorsement, provided that the RR shall make all efforts to achieve this by a benchmark date of 8 months from GEF CEO endorsement.
Inception workshop date: to be held by a benchmark date of 2 months from the date of first disbursement
Operational closure: Operational closure must happen at the end date calculated by the approved duration after the Project Document signature or at the revised operational closure date as approved in the project extension.
Financial closure: within 6 months of operational closure or after the date of cancellation

I. TABLE OF CONTENTS

I. Table of Contents....4
II. Development Challenge....6
III. Strategy....9
IV. Results and Partnerships....12
V. Project Results Framework....32
VI. Monitoring and Evaluation (M&E) Plan....36
VII. Governance and Management Arrangements....40
VIII. Financial Planning and Management....43
IX. Total Budget and Work Plan....48
X. Legal Context....53
XI. Risk Management....53
XII. Mandatory Annexes....57
Annex 1: GEF Budget Template....58
Annex 2: Project map and geospatial coordinates of the project area....60
Annex 3: Multi Year Work Plan....61
Annex 4: Monitoring Plan....66
Annex 5: UNDP Social and Environmental Screening Procedure (SESP)....71
Annex 6: UNDP Risk Register....84
Annex 7: Overview of Project Staff and Technical Consultancies....87
Annex 8: Stakeholder Engagement Plan....95
Annex 9: Gender Analysis and Gender Action Plan....105
Annex 10: Procurement Plan....117
Annex 11: GEF focal area specific annex (Estimates of Direct and Consequential GHG Emission Reductions and Beneficiaries)....121
Annex 12: Additional agreements....130
Annex 13: GEF Core indicators....131
Annex 14: GEF 7 Taxonomy....132
Annex 15: Funding scheme to public transport operators participating in e-mobility projects....138

List of Tables

Table 1: Knowledge Management Plan....38
Table 2: M&E Plan and Budget....39
Table 3: Confirmed Co-financing....46
Table 4: Total Project Budget....49
Table 5: Summary of funds....51
Table 6: Budget notes....52
Table 7: Data required by the E-mob Model....124

List of Figures

Figure 1: Source: UNEP, E-Mobility Program Framework Document....31
Figure 2: Project Organisation Structure....41

Acronyms and Abbreviations

3W	Three wheeler
ATU	Autoridad de Transporte Urbano para Lima y Callao (Lima and Callao Public Transport Authority)
CO_{2e}	CO2 equivalent
COFIDE	Corporación Financiera de Desarrollo, Development Bank of Peru
ELV	End-of-Life Vehicle
EV	Electric Vehicles
FACE	Funding Authorization and Certificate of Expenditures
GDP	Gross Domestic Product
GEF	Global Environment Facility
GEFTF	GEF Trust Fund
GHG	Greenhouse gas
HDV	Heavy Duty Vehicle
ICE	Internal Combustion Engine
ITB	Invitation to Bid
NDC	Nationally Determined Contributions
LDV	Light Duty Vehicle
M&E	Monitoring and Evaluation
MIMP	Ministerio de la Mujer y Poblaciones Vulnerables (Ministry of Women and Vulnerable Populations)
MINAM	Ministerio del Ambiente (Ministry of Environment)
MINEM	Ministerio de Energía y Minas (Ministry of Energy and Mining)
MRV	Monitoring, Reporting and Verification
MTC	Ministerio de Transportes y Comunicaciones (Ministry of Transport and Communications)
NPD	National Project Director
Osinergmin	Organismo Supervisor de la Inversión en Energía y Minería (Supervisory Agency for Investment in Energy and Mining of Peru)
PE	Project Executive
PM	Project Manager
PMC	Project Management Costs
PMU	Project Management Unit
PB	Project Board
RE	Renewable Energy
RFP	Request for Proposals
RFQ	Request for Quotations
SESA	Strategic Environmental and Social Assessment

SESP	Social and Environmental Screening Procedure
SIT	Sistema Integrado de Transporte, Arequipa
TA	Technical Advisor
UITP	Union Internationale des Transports Publics (International Public Transport Union)
UNDP	United Nations Development Programme
UNEP	United Nations Environment Programme
USD	USA Dollar
WG	Working group of the global e-mob programme

II. DEVELOPMENT CHALLENGE

1. As it is the case for most countries, Peru faces increasing GHG emissions from the inland transport sector (60% growth between 2005 and 2014[2]). Electrification of road transport (generally referred to as e-mobility) is one of the key game-changers, and developing countries are lagging behind in the adoption of the new technologies.

2. Urbanization trends in Peru are at the basis of the growth in transport emissions, so that policies tackling urban mobility provide a strong climate change mitigation potential: the percentage of Peruvians living in urban areas increased from 74.0% in 2010 to 76.4% in 2017[3]. The living conditions of this growing population are strongly influenced by the transport system, as it provides access to jobs and services; this is more even more relevant for women and other vulnerable groups, which cannot often afford the most convenient transport means. Air quality is a critical public health problem in cities and it can be improved as an associated benefit by GHG mitigation measures in urban transport. The mitigation potential in urban mobility is higher than in other transport subsectors, and so is the potential to attain favorable gender, social and environmental impacts.

3. The development challenge can be described as follows: Policies to reduce GHG emissions in Peruvian cities and to improve air quality cannot make full use of the potential provided by e-mobility. There are four immediate causes of this development challenge: (i) insufficient policies and regulations on e-mobility, necessary to remove current barriers to sustainable mobility and innovation and to reduce uncertainties about emerging technologies and political priorities (for a clear vision about the future urban mobility); (ii) insufficient knowledge of the potential of e-mobility (including myths, misperceptions, and biased decisions by stakeholders); (iii) a vehicle market monopolized by conventional technologies, so that accessing electric vehicles (EV) or chargers becomes a difficult, long and expensive task; (iv) end of life vehicle management insufficiently developed, which raises concerns among environmental authorities and the public about the ability of the country to adequately manage its potentially hazardous and valuable vehicle components, particularly its lithium batteries.

4. The lack of comprehensive policies and regulations derives from several root causes: a fragmented institutional framework with e-mobility partially addressed from different ministries without unified leadership; lack of formalized coordination among the governmental units with competences linked to e-mobility; lack of an e-mobility strategy providing a shared vision guiding the governmental sectoral actions, and reliance on outdated regulations, poorly suited to deal with e-mobility. Furthermore, e-mobility is linked to a variety of regulated sectors, from energy to transport or urban planning, each of them with

[2] Ministerio del Ambiente (2020). Inventario de emisiones de gases de efecto invernadero (INGEI 2014).

[3] Instituto Nacional de Estadística e Informática.

strong autonomy (technical, professional, entrepreneurial and regulatory cultures), which jeopardizes the development of a comprehensive approach to e-mobility as a game-changer.

5. The insufficient knowledge of the potential of e-mobility to tackle the challenges of urban mobility demand is due to several root causes: conservative urban policy trends in cities, favoring the expansion of high-emitting private cars within a *business as usual* approach; weak public transport systems without resources to invest in the potential of e-mobility solutions; most local stakeholders wrongly consider that there are high risks in transitioning to e-mobility solutions, due to their limited awareness of technical progress and success stories in other countries.

6. The difficulties of e-mobility solutions to reach the Peruvian market (particularly in what refers to public transport services) are due to a variety of root causes: high costs and limited information on EV options available; absence of some key market actors in the country (manufacturers, dealers, repairs…), business models poorly tailored (if at all) to the Peruvian context; financial schemes unable to meet the needs of public transport operators to transition towards e-mobility, public transport concession schemes discouraging the introduction of innovative technologies, such as EVs, and informal operators unfairly competing with formal concessions.

7. The insufficient development of end-of-life vehicle (ELV) management is due to insufficient awareness among regulators and stakeholders about the requirements for end-of-life management of EVs; also to the fact that the existing solid waste disposal system does not directly consider EV's batteries; furthermore, to the slow development of Extended Producer Responsibility (EPR) mechanisms in the country[4](particularly in the automotive sector); and also to the gap in knowledge and awareness of the environmental risks associated to ELVs.

8. There are environmental, social and economic consequences of lagging behind in the deployment of e-mobility: (i) the growing negative environmental impact of urban mobility; (ii) insufficient quality of public transport services; (iii) the economic inefficiency of the urban transport sector. The environmental impacts of urban mobility are primarily reflected in terms of increasing GHG emissions from road transport in Peru: in 2012, road transportation accounted for 15.26 million tons CO2e, or 8.9% of total GHG emissions (up from 8.2% of emissions in 2005 and 5.3% of emissions in 2000)[5]. Low air quality in Peruvian cities is also largely due to transport emissions: The World Health Organization (WHO) estimates that annually, 4,239 deaths in Peru are correlated to air pollution, with ischemic heart disease as the most common cause. Furthermore, the cost of respiratory diseases caused by air pollution is estimated to be around 1,935 million USD per year. The estimated average of $PM_{2.5}$ concentration in Lima is among the highest in the region (28.0 $\mu g/m^3$ in 2018, compared to the WHO recommended limit of 10 $\mu g/m^3$)[6]. Lastly, electric vehicles (EV) entering the country may become a serious environmental hazard in the absence of adequate end-of-life management, due to the polluting potential of some of the components of the battery pack, such as cobalt, nickel and manganese.

9. E-mobility could contribute to improve the quality of urban public transport services in terms of noise reduction, comfort and technological reliability. E-mobility could bring up the use of cutting-edge technologies, that could serve as a catalyst to attract young talent and innovation to a sector in Peru that, in spite of governmental and local actions to improve regular public transport services, has been unable to increase its share of urban trips. For example, in Metropolitan Lima, regular public transport (metro and buses) provided 56% of motorized trips in 2018[7]; although high, it has substantially decreased from a 70% share in 2007[8]. The share of private cars in motorized trips has increased from 10% in 2004 to 19% in 2009, consistent with the relatively low motorization rate in the country, 114 vehicles/1000 inhabitants.

[4] There is basic legislation on EPR, but only one scheme has been implemented thus far, addressing waste from electrical and electronic equipment (*DECRETO LEGISLATIVO N°1278, que aprueba la Ley de gestión integral de residuos sólidos*).

[5] Ministerio del Ambiente (2016). Inventario Nacional de Gases de Efecto Invernadero (INGEI 2012): Fuentes móviles.

[6] IQ Air (2019). World Air Quality Report 2018.

[7] Fundación Transitemos (2018). Informe de Observancia situación del transporte urbano en Lima y Callao.

[8] JICA (2007). Estudio de Factibilidad de Transporte Urbano para el Área Metropolitana de Lima y Callao.

Nevertheless, the main reason for the reduction of the share of public transport is the increase of trips made by taxis and informal services, reaching 25% of all motorized trips, probably due to the poor quality of public transport. The availability of updated mobility data is also limited, although in Lima and other cities the authorities are working in developing plans based on reliable information.

10. Lacking access to e-mobility technologies, public transport in Peruvian cities cannot benefit from the reduced operating costs provided by e-mobility. In the absence of adequate enabling conditions, particularly in what refers to financial mechanisms for electrification and more specifically in public transport systems (a core issue to be addressed by the project's outputs), investors remain cautious about investing in the sector. The government finds it difficult to implement the intended regulatory reforms and informal transport thrives as an affordable alternative for vulnerable groups. Public transport suffered from excessive deregulation in the previous decades, resulting in a general loss of quality, poor working conditions and financial difficulties for operators. Regulatory reforms are being implemented at a slow pace; for example, in Metropolitan Lima, the number of small PT vehicles (minibuses with up to 33 seats and microbuses with up to 16 seats) is still high (4,827 and 7,492 units respectively), compared to only 1,537 full-size buses[9], in spite of the central and municipal governments' efforts to expand the latter through the creation of bus corridors with higher level of service[10] (*corredores complementarios*), currently served by 669 buses[11].

11. Addressing the project challenges described above is consistent with national priorities as stated in the following documents:

- General national priorities are identified in the long-term planning document: Strategic Plan for National Development (*Plan Estratégico de Desarrollo Nacional*, PEDN), published in 2010 and with a 2050 horizon approved by Decree N. 095-2022-PMC . It refers to urban mobility in relation with the improvement of air quality in cities (100% compliance with PM10 limits), and the reduction of road accidents.
- The National Environmental Policy (*Política Nacional del Ambiente, 2030*- approved by Decree N. 023-2021 MINAM) includes a general call for the use of appropriate technologies for climate change mitigation, the promotion of clean technologies and the promotion of energy efficiency.
- Climate change mitigation priorities are identified in the National Climate Change Strategy (*Estrategia Nacional ante el Cambio Climático*, *ENCC*, *2015*) and in the Nationally Determined Contributions (NDC), submitted by Peru to the United Nations Framework Convention on Climate Change (UNFCCC) and approved in December 2020[12]. The Peruvian State compromises that its net GHG emissions do not exceed 208.8 MtCO2eq in the year 2030, being able to reach a maximum level of 179.0 MtCO2eq[13] . In relation to electric vehicles, its foreseen the participation of 5% of the total fleet by 2030 (mitigation measure, "Promotion of electric vehicles (EV) at the national level"), likewise reference is made to sustainable mobility through the mitigation measure "a National Sustainable Urban Transport Program, by the Ministry of Transport and Communications (MTC).
- The National Energy Policy (Política Energética Nacional 2010-2040) approved by Supreme Decree N° 064-2010-EM includes the development of e-mobility as one of the required actions to improve energy efficiency. This approach is also followed in the National Energy Plan (Plan Energético Nacional, 2014-2025), the Sectoral Strategic Multiannual Plan (Plan Estratégico Sectorial Multianual (PESEM) 2016-2021), and the Efficient Energy

[9] In accordance with the WB's Urban Bus Toolkit (https://ppiaf.org/sites/ppiaf.org/files/documents/toolkits/UrbanBusToolkit/assets/1/1c/1c7.html), the ratio of buses per 1,000 population would be expected to be between 0.5 and 1.2. In the case of the metropolitan area of Lima-Callao, this ratio was 1.5 (2018 data), confirming the need to reduce the fleet by replacing smaller units by bigger ones.

[10] https://www.uitp.org/buses-high-level-service-final-report.

[11] Fundación Transitemos (2018). Informe de Observancia situación del transporte urbano en Lima y Callao.

[12] Perú (2020). Contribuciones determinadas a nivel nacional del Perú. Reporte de actualización período 2021-2030. Accesible en: https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Peru%20First/Reporte%20de%20Actualizacio%CC%81n%20de%20las%20NDC%20del%20Peru%CC%81.pd

[13] Contribuciones Determinadas a Nivel Nacional del Perú. Reporte de Actualización Periodo 2021-2030.

Use Master Plan (Plan Referencial del Uso Eficiente de la Energía 2009-2018): the government is developing new legislation to promote electric vehicles ([14] in accordance with all these plans).

- The National Plan for Competitiveness and Productivity (Plan Nacional de Competitividad y Productividad), approved by Supreme Decree DS N° 237-2019-EF, promotes the electrification of the transport sector as a means to diversify the electricity matrix.
- The Sectoral Strategic Multiannual Plan (Plan Estratégico Multianual Sectorial (PESEM 2018-2020) PESEM 2018-2023), approved by Ministerial Resolution N° 1029-2018-MTC-01 and updated by Ministerial Resolution N° 284-2019 MTC/01 y N° 1305-2019 MTC/01, of the Ministry of Transport and Communications (MTC) includes references to government's investments in the Metro system in Lima and Callao, in the Public Transport Master Plan for the metropolitan area as well as public transport systems in other cities and traffic management systems. PESEM includes guidelines that are closely linked to the project contents, such as "increasing the quality of the transport system" and developing "environmentally sustainable transport systems", as well as a commitment for the promotion of clean energy in all transport modes.
- With the support of British and German cooperation, MTC has also developed the TRANSPerú-NAMA project on sustainable urban transport. Key actions included in this NAMA have subsequently been implemented, such as the legislation passed in 2019 approving the National Policy on Urban Transport[15], a National Program on Sustainable Urban Transport (Promovilidad) and an urban transport authority for the metropolitan region of Lima and Callao[16]. Similar urban transport authorities are operating or being implemented in other cities, including Arequipa, the second largest metropolitan area in Peru (Sistema Integrado de Transporte). As stated in the National Policy on Urban Transport, all these actions, aim at "providing safe, reliable, high-quality, institutionally coordinated and socially, economically and environmentally sustainable urban transport systems", including the key areas addressed by this project (public transport, implementation of technological innovations, and institutional coordination).

12. The promotion of e-mobility is linked in these documents to other national priorities, such as the favourable framework provided by the very low carbon intensity of Peruvian's electricity grid[17], the industrial and economic development opportunities linked to the significant Peruvian lithium reserves, and the potential of EV manufacturing by the national industry, at least in the segments of buses and 3-wheelers. To these, it is worth adding the interest of various global players (such as BYD, ENEL and ENGIE) to introduce e-mobility in Peru, with some short pilots conducted in the last years[18].

III. STRATEGY

13. The project strategy addresses the four causes identified above in full alignment with the approach of the UNEP proposal to GEF for a Global Programme to Support Countries with the Shift to Electric Mobility, and will contribute to its component on "regional support and investment platform". To be effective, the project objective of reducing GHG emissions by stimulating the deployment of electric vehicles has to build upon five action blocks: (i). institutions and regulations, in order to provide the overall policy vision at the national level and the necessary institutional and legal instruments and implementation capacities; (ii).

[14]

[15] Decreto Supremo N° 012-2019-MTC, que aprueba la política nacional de transporte urbano.

[16] Programa Nacional de Transporte Urbano Sostenible (Promovilidad) established by Decreto Supremo N°027-2019-MTC. Autoridad de Transporte Urbano para Lima y Callao (ATU) established in December 2018 by Law N°30900.

[17] 412 gCO_2/kWh in 2016, as calculated by MINEM.

[18] Such as the e-bus operating in Arequipa in the context of the Mining Convention (*Convención Minera, septiembre 2019*), the e-bus in San isidro, Lima (October-December 2018) and the on-going operation of one e-bus in Javier Prado corridor in Lima (2020).

demonstrations of EV implementation, in order to remove short-term barriers, such as the perception of a high technological risk due to the lack of knowledge about the performance of EVs on the field; (iii). market conditions, facilitating the new technology to reach the envisaged public transport systems; (iv). anticipated deployment of end-of-life vehicle management to avoid the environmental hazards raised by EVs; and (v). project monitoring, evaluation and knowledge management.

14. Firstly, e-mobility requires to be institutionalized, i.e., for the government to establish a national strategy for the expansion of EVs in the country and the adequate institutional framework to implement it. The project will build upon existing energy and transport plans and policies, and the pilots that have taken place in the past. It will build with the government a comprehensive long-term strategy and develop the capacities for its implementation by the relevant stakeholders, while building up wide support for this transition. The strategy provides the framework for the regulatory reforms and other actions to be implemented within and beyond the project.

15. In the second place, it is necessary to provide evidence of the feasibility and sustainability of e-mobility options in the country, putting the new technologies at the service of a consistent sustainable transport vision, rather than the other way round. Consistent but advancing the current transport policy in Peru, the project will focus on the deployment of EVs in urban public transport systems, working with two pilot cities with bold sustainable mobility ambitions and a good performance record in the improvement of urban mobility: Arequipa and Lima. These cities are making consistent efforts to improve public transport and reduce private car use, and the project will provide insight on how EVs can further strengthen such sound policy practices[19].

16. In the third place, to be financially sustainable, the availability of EVs in Peru has to be guaranteed by a solid regulatory framework that -among other aspects- includes regular importation or manufacturing and maintenance support. EV manufacturers are concentrating their market strategies in countries with a sufficiently vehicle market and attractive incentives for potential EV users. These conditions will primarily target the urban public transport sector, but they will also benefit the deployment of EVs in other transport subsectors and other vehicle categories such as taxis, vans and private EVs.

17. Finally, to be environmentally sustainable the introduction of e-mobility needs to provide adequate measures to manage EVs at their end-of-life, preventing them from becoming an environmental hazard for Peru. At a time where these measures are still at their infancy globally, it is necessary to align Peru with the regulations and market conditions already successfully established elsewhere. This implies fostering the Extended Producer Responsibility for ensuring re-use of the batteries and its adequate disposal.

Alignment with GEF focal area strategy

18. The Global Programme, that supports countries with the Shift to Electric Mobility, and all its national child projects are aligned with Objective 1 of the Climate Change Focal Area, which is to “Promote innovation and technology transfer for sustainable energy break-throughs”, and more specifically with CCM1-2, to “Promote innovation and technology transfer for sustainable energy breakthroughs for electric drive technologies and electric mobility”.

19. Whereas the global programme will provide tools, policy best practices, training materials and strategies to foster large-scale market introduction of electric mobility, the child project in Peru will put in place enabling conditions for this transformation. In particular, through their working groups, the global programme is expected to provide updated information on progress in EV technologies, that the child project can consider during the design of its demonstrations (component 2), and scaling-up actions (component 3); likewise, it can facilitate access to international progress in the management of EV and components at their end of life

[19] Arequipa is implementing its Integrated Public Transport System (*Sistema Integrado de Transporte*, *SIT*) with the support of different International Financial Institutions. A metropolitan Public Transport Authority (*Autoridad de Transporte Urbano*) has been established for Lima and Callao, and a Master Plan for the public transport system (Plan Maestro de Lima 2050) is under preparation.

(as they affect global manufacturing chains rather than local ones), so that the project (component 4) can focus on the integration of Peru in such value chains. The section on barriers provides more information on the enabling conditions this project will need to strengthen.

20. By creating a favorable institutional and regulatory framework for EVs, the project in Peru will contribute to point 112 of the GEF-7 Programming Directions to accelerate "the speed and scale of sustainable energy investment in developing countries"; by establishing adequate business models for a stronger EV market, the project will contribute to point 113, developing "innovative business models that go beyond business as usual".

21. In addition, the demonstrations on EV operations are expected to accelerate the transition to low-emission urban mobility. They directly address the root cause of weak public transport systems, which are unable to afford the introduction of EV solutions and therefore cannot provide the higher quality mobility services associated to attract car users. They also address the root cause of the high risks perceived by public transport managers of making use of innovative electric technologies. Besides addressing the short-term barriers linked to these root causes, the project provides long-term sustainability for the adoption of EVs, providing adequate market conditions and management of potential environmental hazards generated by them.

Incremental cost reasoning

22. Electric mobility is making quick progress in those countries and regions offering some key enabling conditions: dynamic vehicle and battery manufacturing industries, strong research and innovation ecosystems in these fields, financial and political support from the public sector, users' financial capacity and interest in innovation and public and private commitment with GHG emission mitigation targets. These enabling conditions are supported by consistent strategies, providing a long-term vision, and inspiring the legislative and institutional reforms to encourage the development and implementation of innovations.

23. After more than two decades of efforts, technological innovation is successfully delivering decreasing EV prices, delivering higher performances in terms of range and lifespan, and new models, increasingly adapted to particular niche markets such as urban delivery or public transport.[20] New legislation is being deployed on vehicle approval, roadworthiness and end-of-life management, as well as on the use of EVs for the provision of public services, and innovative financing schemes are being proposed.

24. In the absence of the enabling conditions mentioned above, developing countries risk to lag behind in the transition towards low-carbon mobility. EVs are being designed to meet the needs of those developed markets with higher actual or potential demand, while the other markets seem prone to become the dumping sites for old fashion technologies, as it has been the case in the past (and even today, as outdated technologies such as EURO III diesel buses remain being the standard for public transport concessions in some cities). EV manufacturers are unlikely to push forward on their own, as they carefully choose the most promising markets for development, to limit costs and risks.

25. GEF-7 is in a unique position to support a step change, by contributing to supporting governments in developing countries to establish the required legislation. Channeling the EV introduction towards the public transport sector first make producers and consumers become more confident about the robustness of the local EV market, to, benefits a larger population, and attract global players (from EV manufacturers or battery managers to financial institutions and global regulators) interested in the needs and priorities EVs can meet in developing countries.

26. The potential size of the Peruvian EV market, the sustained effort of national and local governments to improve urban public transport services and the rapid fleet growth in the last 5 years (particularly of the 10 and 12 m long bus segment, due to the implementation of new concession contracts requiring larger buses) provides an excellent environment for GEF-7 action. The benefits of current public transport enhancement

[20] https://www.transportenvironment.org/sites/te/files/publications/2019_07_TE_electric_cars_report_final.pdf

actions can be expanded through the integration of considerations still not included, such as fleet electrification, women empowerment (as users and workers) in this sector, and widening the options for concession contracts, in order to speed up quality service improvements and the adoption of environmentally-friendly technological developments. In terms of GHG emissions, the project may accelerate the transition towards EVs in public transport by 5 to 10 years, with a potential for spill over effects in other types of vehicles. In the absence of action on EV promotion, the bus fleet growth over the next years will largely rely on conventional technologies which will remain on the roads for decades, slowing down the introduction of electric technologies.

27. The GEF support provides the resources and political leadership necessary to undertake a comprehensive approach, addressing the necessary coordination among institutions and political capital for regulatory changes enabling a transition towards electric mobility. In the absence of GEF support, it is unlikely that local stakeholders interested could move beyond current small demonstrations unable to be expanded due to an unsupportive regulatory framework and the political and human capital resources needed to reform it.

28. In addition, the project is supported by the global e-mob programme, and this programmatic approach seeks to bundle demand in the region and thus reduce the incremental costs of electric technologies in a cost-effective way:

    - Generic tools are expected to be produced at the global level, disseminated though regional support and investment platforms and adapted to the needs in Peru at the country level; in this way, return on investment for development of tools and methodologies is maximized.
    - Bundling demand for e-vehicles for demonstration in a certain region can lead to lower vehicle prices, reducing investment risk.
    - Through adequate training of vehicle operators and exchange between numerous projects, the industry is less likely to face misuse of technology.

29. Last but not least, Peru requires GEF support to accelerate the preparation of adequate national strategic planning on transport electrification and to engage public transport authorities and private and public stakeholders in the transport community in the transition towards e-mobility through capacity building and practical evidence of the operation of these new technologies in Peruvian cities.

IV. RESULTS AND PARTNERSHIPS

Expected Results

30. The expected result of the project is the provision of higher quality public transport (buses and 3-wheeler services) with the use of electric vehicles, providing substantial emission reductions of GHG and local air pollutants. The introduction of EVs in public transport will subsequently spread to other vehicle categories, providing additional GHG emission savings.

31. This result is expected to be a consequence of the achievement of a supporting legal and institutional framework, a wider support to the use of EVs in public transport and to the development of a strong EV market in Peru.

32. The project provides these conditions through four expected outcomes and a M&E outcome:

COMPONENT 1: Institutionalization of low-carbon electric mobility

33. **Outcome 1**: Institutional, regulatory and fiscal frameworks established for the promotion of e-mobility with the support of all key stakeholders, including potential EV providers and users.

34. The risks identified under the Social and Environmental Screening Procedure (SESP) point to potential conflicts with social and environmental safeguards. Related to policy and market introduction of eMobility, specific mention is given to: (1) a human rights approach to ensure inclusiveness and equitable access to

the benefits of the Project, both short- and long-term; (2) incorporation of gender-oriented elements in policies and action plans, as to avoid any discriminatory effects; moreover, and (3) Potential environmental hazards associated with waste generated during manufacture of electric vehicle batteries as well as handling batteries that have reached their end-of-life. UNDP and the IP will ensure that safeguards are monitored by applying a Strategic Social and Environmental Assessment (SESA) throughout Component 1. The SESA will include FPIC as determined appropriate and necessary for SES compliance in the course of that assessment.

35. This outcome requires the consolidation of an internal space of coordination, the formulation of a widely-supported strategy supporting the deployment of EVs in Peru, the identification and implementation of required changes in the existing regulatory framework in order to make it compatible with the new strategy, and the involvement of a growing number of stakeholders in support of such policy. To achieve these, this outcome is based on four outputs that will provide the institutional structure, policies, regulations and capacities. It also includes the liaise with the Global E-mob Programme.

36. ***Output 1.1***: An intersectoral and multilevel e-mobility coordination space established, including all relevant governmental stakeholders. The need for such coordination space has also been indicated within the legislative proposal for the promotion of electric transport[21] published by the Ministry of Energy and Mines[22]. This space will primarily include national and subnational governmental institutions, and may also extend participation to the private sector, academia and civil society (which are expected to coordinate more closely under the e-mobility network established through Output 3.1). Thus, this coordination space will enable convergence among the various ministries and public institutions involved in e-mobility, replacing the current non-binding discussion platforms established by each ministry related to e-mobility, which are convened by different public authorities with rigid sector divisions and which focus on partial issues: vehicle technical approval, tariffs, charging standards, fiscal incentives, among others. The coordination space will facilitate the approval of strategies, policies and regulations, and will also serve as an entry point for partnerships with non-governmental stakeholders. Two stages are envisaged: initially, the key stakeholders will be convened with support from the project to establish and agree upon a mandate and working plan for the coordination space. Once finalized, the joint working plan will be delivered to the various governmental institutions involved in the official ratification of the space, establishing the adequate structure within the government.

37. Activity 1.1.1: Convene and carry out periodic meetings with all key stakeholders identified, while the interim coordination space is being structured, as well as when it has been established as a new official e-mobility coordination space or within an operating official government initiative.

38. Activity 1.1.2: Provide technical assistance to participating governmental stakeholders for the preparation and delivery of a draft document, which will include the mandate, structure, governance and working plan for the continuation of the coordination space. This document will include procedures for the screening and involvement of interested parties and the subsequent strategic assessment of policies, plans and regulations during its deliberation within the coordination space. This activity also includes the coordination of the liaise with the Global E-mob Programme, in particular in what refers to the participation in the various working groups and LAC regional platform activities, and the travel expenses required[23].

[21] Resolución Ministerial N° 250-2019 –DM publicando el proyecto del Decreto Supremo (DS) que declara de interés nacional y necesidad pública la promoción de vehículos eléctricos e híbridos, y facilita las condiciones para el desarrollo de la infraestructura para su abastecimiento de energía.

[22] Acciones Nacionales Apropiadas de Mitigación (NAMA) en sector de generación de energía y su uso final en el Perú.

[23] The travel budget (USD 50,000) includes 12 international trips to participate (one participant each) at 3 meetings of each of the four Working Groups @ USD 1,900, 18 regional trips to participate (2 participants each) at the kick-off meeting, 3 regional group meetings, 2 capacitation workshops, 2 market place events and 1 replication event, @USD 1,400 and USD 2,000 for travel contingencies. Due to COVID-19 restrictions, some activities could be held on-line.

39. *Output 1.2*: National e-mobility strategy submitted to the government and launched, including economic impacts (competitiveness, industrial development), geographical deployment (e.g. charging network) and environmental and social dimensions (gender-responsive).

40. The comprehensive national e-mobility strategy addresses three key areas: (i) the expansion of EVs within the vehicle fleet in Peru; (ii) the deployment of the charging network; and (iii) the electrification of public transport. Economic impacts, geographical deployment and environmental and social dimensions that are key for the development of these three intervention areas will be considered and integrated into the strategy. To this end, the national e-mobility strategy will provide guidelines on technological and industrial development, legal and regulatory framework reforms needed, and a strategy to raise awareness and build capacities in support of e-mobility implementation. A gender analysis and action plan will be included in the strategy to ensure EV deployment in Peru is gender-responsive and contributes to gender-equality in the energy and transport sectors.

41. This output will include the following activities:

42. Activity 1.2.1: Study of electrification options for public transport by road, with a focus on urban buses and 3-wheelers. This study will provide a screening of electric bus and 3-wheelers manufacturers and models, and their availability and fitness for current and future public transport conditions in Peru[24]. This study will benefit from the contributions of the Working Group (WG) on HDV and e-buses, the WG on 2&3 wheelers and the regional (LAC) marketplace events of the Global E-mob Programme.

43. Activity 1.2.2: Study of electrification options for light vehicles, with a focus on public and private fleets. This deliverable provides a screening of light EV manufacturers and models, and their availability and fitness for current and future mobility conditions in Peru[8]. This study will benefit from the contributions of the WG on LDV and the regional (LAC) marketplace events of the Global E-mob Programme.

44. Activity 1.2.3: Study of the charging network necessary to expand electric routes in the cities of Lima and Arequipa. This activity will identify the most suitable locations and characteristics of public charging points to provide opportunity charging to e-buses and 3-wheelers. This activity contributes to the implementation of the regulation of charging points developed by the NAMA-energy project. In order to properly manage any environmental, social or economic impacts, within the study a Strategic Environmental and Social Assessment (SESA) approach will be applied, to ensure compliance with the SES. The study will avoid any places in which any environmental, social or economic impacts may arise, and for doing so it will (1) identify a sufficient number of alternative locations for each charging point; (2) undertake an assessment of potential traffic generation created by the envisaged infrastructure; (3) identify the eventual needs for changes in the electricity distribution grid; (4) mitigation measures, in case concerns are raised from any of the relevant local stakeholders. This study will benefit from the contributions of the WG on batteries and charging infrastructure and the regional (LAC) marketplace events of the Global E-mob Programme.

45. Activity 1.2.4: Capacity needs assessment and elaboration of a capacity-building and communications plan. This deliverable builds upon the materials provided by the NAMA-Energy project and its activities to promote substantial electrification of the fleet in the 2030 horizon[25].

46. Activity 1.2.5: Development of a gender analysis and action plan to address gender gaps in the energy and transport sectors related to e-mobility deployment in Peru, including the analysis of gender-based violence and harassment, in order to include appropriate gender-responsive measures in the e-mobility strategy.

[24] This activity builds upon the proposal on regulatory changes and enabling conditions delivered by the NAMA-Energy project.

[25] In particular, the Diagnostic on Clean Transport (July 2017), the Estimate of the energy efficiency of the bus fleet in Lima (2019), and the Pilot to operate an electric bus in Lima (Javier Prado corridor).

47. Activity 1.2.6. Prepare an e-mobility strategy through participatory and multistakeholder consultations, including its strategic impact assessment for identifying interested parties (and avoid the exclusion of marginalized actors), the potential environmental and social impact of the strategy, and a monitoring and evaluation mechanism for its implementation. The preparation of the strategy will adopt a Strategic Environmental and Social Assessment (SESA) approach, to ensure compliance with the SES. The strategy will also provide charging infrastructure location criteria, to avoid any places in which any environmental, social or economic impacts may arise, including guidelines to (1) identify a sufficient number of alternative locations for each charging point; (2) undertake an assessment of potential traffic generation created by the envisaged infrastructure; (3) identify the eventual needs for changes in the electricity distribution grid; (4) define mitigation measures, in case concerns are raised from any of the relevant local stakeholders.

48. Activity 1.2.7 Submission of final e-mobility strategy to the government for adoption. The project provides the necessary technical support to get the strategy endorsed by the e-mobility coordination space and submitted to the government.

49. *Output 1.3*: Revised regulatory and fiscal framework for deployment of electric vehicles delivered for governmental approval.

50. The envisaged framework is expected to combine the deployment of EVs in the various categories (with a focus on public transport) with the phasing out of high-carbon vehicles (through vehicle scrapping with incentives for acquisitions for EV, VAT exemptions/reductions, incentives for EV in public bids for routes, etc), and the protection of the Peruvian market against the import of outdated technologies and outdated second-hand vehicles. This output will include the following activities:

51. Activity 1.3.1: Submission to the government of national regulations reforms on vehicle approval, including EVs, building upon current government initiatives on vehicle labeling and energy efficiency; this document will include a strategic assessment of the potential social and environmental impacts of this regulation.

52. Activity 1.3.2. Design of national regulations and standards to install public and private charging infrastructure, including installation guidelines addressing professional installers and consumers. These will be based on the regulations and standards developed by the electric transport standardization technical committee and other projects; this document will include a strategic environmental and social assessment (SESA) of the potential social and environmental impacts of this regulation, as well as requirements on professional training to prevent health risks associated to battery banks and EV repairs.

53. Activity 1.3.3. Reform proposals for vehicle taxes, to incentivize EV and discourage the purchase of high-carbon vehicles. Reforms have been identified within the NAMA-Energy project, dismissing the introduction of tax incentives; the project will provide support for their implementation, and for the consideration of additional incentives; this document will include a strategic assessment of the potential social and environmental impacts of this regulation.

54. *Output 1.4*: Strengthened capacities and awareness of public and private decision-makers, technical professionals and consumers regarding e-mobility policy design and implementation, highlighting the social and environmental benefits of EV deployment.

55. As the lack of updated information jeopardizes deliberation and slows-down the deployment of e-mobility options, the project will disseminate information strategically tailored to the specific needs of three target groups: the general public, private and public decision-makers and transport and energy professionals. The capacity building and awareness activities included under this output will be based on the capacity needs assessment and related capacity-building and communications strategy developed through the Activity 1.2.4.

56. This output includes the following activities:

57. Activity 1.4.1: Implementation of the communications strategy, specifically addressing educational centers and the public to foster the use of EVs. Whereas current initiatives are focusing on the communication of regulations and benefits, the project will expand these actions to also support professionals in the educational and social media sectors.

58. Activity 1.4.2: Implementation of training activities on e-mobility policies, targeting public and private decision-makers and other professionals, which will include the design and use of capacity-building materials. They will include training in health risks associated to battery banks and EV repairs. The contents of the training activities will integrate the materials and methodologies developed by the Global E-mob Programme and by the capacitation sessions on vehicles and charging infrastructure of the regional platform.

59. Activity 1.4.3: Design of technical training modules for PT technical staff (drivers and operators) and transport and electricity professionals on EV and charging infrastructure maintenance and assembly. Awareness towards gender-based violence and harassment and measures to address these risks will also be addressed. The contents of the training activities will integrate the materials and methodologies developed by the Global E-mob Programme and by the capacitation sessions on vehicles and charging infrastructure of the regional platform.

COMPONENT 2: Short term barrier removal through low-carbon e-mobility demonstrations

60. **Outcome 2: EV demonstrations in urban public transport systems provide evidence of technical, financial and environmental sustainability to scale-up e-mobility**.

61. A few short-term pilots have been completed thus far in Peru, giving bus and 3-wheelers operators the opportunity to make use of one EV for a few weeks[26], providing them with direct contact with the technology. Yet, they have not engaged operators in managing the EV as an asset and rethinking their operational plans based on the strengths of this new technology. The project builds upon this evidence and upon consistent national and local plans to promote sustainable mobility options in cities, and focuses on bus services, and a few short pilots demonstrating the technical, financial and environmental sustainability of electric buses and 3-wheelers for subsequent up-scaling.

62. The project's challenge is to demonstrate that electric buses and 3-wheelers can be consistent with current efforts to improve public transport quality in Peruvian cities, that they can provide additional advantages compared to ICE vehicles, and that their current higher upfront costs can be compensated through adequate financial and non-financial schemes without requiring additional public resources, at least in the mid-term, as battery costs continue to decrease. Furthermore, it will be necessary to make financially-weak bus and 3-wheelers operators confident in EVs reliability and operational costs, and to identify reforms in the urban transport systems that could expand the use of EVs (such as appropriate routes and operational conditions, fares and public subsidies, schedules and circulation permits…). Furthermore, the demonstrations will provide evidence to incentivize public transport authorities in the two pilot cities to include EVs as a part of their public transport network planning activities, and to review concession and authorization rules accordingly to make the most of the opportunities associated to e-mobility. In addition to testing electric vehicles, data will be collected for the evaluation of the pilots to generate evidence and recommendations on best practices for replicability. Operators participating in the demonstration are requested to prepare and implement a gender policy, to be monitored during the project lifespan.

63. In order to facilitate the procurement and operation of the EV in the demonstrations, the Executing Agency (EA) will establish a competitive procedure and sign with the selected contractors contract of services in

[26] Although jointly covering different seasons, so that performance in summer and winter can be compared.

which the contractor is requested to operate an e-bus on a regular urban line (or 3W taxis) for a minimum period of time and a minimum of km travelled. The EA is regularly paying the contractor with an amount according to the verified km travelled. The details of such contracts are described in Annex 15.

64. The financial compensation to the PT operators is described in detail in Annex 15 of the ProDoc, in which a justification of the budget initially estimated for the demonstration is also provided. The selection of operators is made through a competitive Request for Proposals (RFP) for the provision of the EV demonstration services envisaged in activities 2.2.1 and 2.3.1, in which offerors will include the compensation they are requested per km travelled during the demonstrations, and the selection of the most advantageous proposal for the project. Market competition will therefore provide a mechanism for checks and balances. There is a cap in the total compensation, so that only the km established in the ProDoc as necessary for a successful demonstration will be entitled to be subsidized. The preparation of the RFP requirements and the subsequent review and assessment of the proposals received will include a third-party expert (the contractor selected for activity 2.1.2: Elaboration of a design proposal for the pilots in Lima and Arequipa), to verify that the subsidy does not exceed the incremental costs of the EVs compared to conventional vehicles of similar performance.

65. Payments to the selected PT operators will be calculated based the actual km driven by the EVs during the demonstration, for which they will be required to provide detailed justification (with a GPS system accessible to the PM) of the actual EV mileage driven.

66. ***Output 2.1***: E-mobility pilots in public urban transport systems designed in Lima and Arequipa, integrating gender-responsive measures.

67. The project will collect information on successful implementation of electric buses and 3-wheelers in other countries, in order to identify best practices that facilitate the selection of the vehicles' and charging points' manufacturers and models, the identification of a suitable service route and service conditions, and the selection of the operator. Once the best practices guidelines for pilot design are developed they will be applied to the design of the pilots in Lima and Arequipa.

68. Activity 2.1.1: Development of best practice guidelines and design recommendations for e-mobility public transportation pilots, which shall include guidelines on gender-responsive design based on participatory analyses. These guidelines will build upon previous experiences in the cities from the private sector, the NAMA-Energy project and others[27], and will be consistent with the strategies of the public transport authorities in the pilot cities (Lima and Arequipa). They will include recommendations for public transport reforms, if needed, to allow the operation of the electric vehicles, and will support the design of comparable and scalable pilots nationwide. The guidelines will include recommendations for safe ridership under COVID-19 pandemic conditions. For the preparation of the guidelines, it is expected to get continued support from the Assistance Service provided by the LAC Regional Platform of the Global E-mob Programme in what refers to the identification and integration in the guidelines of best international and regional practice.

69. Activity 2.1.2: Elaboration of a design proposal for the pilots in Lima and Arequipa, following the guidelines for e-mobility pilot design. The design will include the assessment and identification of best routes for the implementation of electric buses and three-wheelers for the pilot, considering potential for scale-up, technical and financial feasibility, and social and environmental sustainability principles as described in the safeguards screening, in particular the affordability of fares and awareness towards gender-based bias in bus design and operations. To address these risks, the project will undertake a livelihood analysis of the potential impact electrification may have on the living and working conditions of 3-wheeler owners and

[27] Such as the results on the small demonstration carried out with one e-bus in Lima, operated for some months on the "Javier Prado" and "Panamericano" corridors. 30 electric cars are also envisaged to be used for taxi services in Lima, within the NAMA-Energy project.

drivers, in conformity with UNDP SES standard 5 requirements[28]. The pilot design proposal will be developed in close coordination with the city government and the local public transport authority.

70. Activity 2.1.3 Design and implementation of multi-media campaigns against gender-based harassment, discrimination and violence, including against other vulnerable groups (children, elderly, persons with disabilities, LGBTQIA people).

71. ***Output 2.2*:** EV pilot in Lima implemented, monitored and evaluated: two electric buses and ten 3-wheelers (indicatively) in public transport lines.

72. Within this output, the selection of the relevant pilot partners (EV and charger provider, public transport operator) will be undertaken in Lima as the first pilot city. Subject to the results of activity 2.1.2, it is expected that an open call for proposals will be made for interested vehicle manufacturers and public transport operators, in order to choose the most appropriate vehicles and partners to conduct the pilot.

73. Activity 2.2.1: Selection of EV operators for e-buses in Lima in compliance with selection criteria established based on best practice guidelines, including gender considerations for gender-responsive design and selection of drivers.

74. Activity 2.2.2: Integration and operation of at least 2 e-buses and 10 electric 3-wheelers in operator's fleet in Lima along one selected route (through incremental cost support for vehicle purchase or long-term leasing). The 3-wheelers will be operated as *mototaxi* services.

75. Activity 2.2.3: Identification and systematization of results, best practices and lessons learned for the route where the pilot was implemented, including the assessment of social and environmental impacts and recommendations regarding how to escalate to other routes. This will be integrated and fed into MINAM's knowledge management processes and systems. This activity will benefit of assistance support from the regional (LAC) platform and present and discuss results at the LAC regional platfom group and at the WG on HDV and e-buses of the Global E-mob Programme.

76. Monitoring and evaluation of this demonstration is undertaken within the M&E Component.

77. ***Output 2.3***: EV pilot in Arequipa implemented, monitored and evaluated: two electric buses and ten 3-wheelers (indicatively) in public transport lines.

78. Within this output, the selection of the relevant pilot partners (EV and charger provider, public transport operator) will be undertaken in the second pilot city. Subject to the results of the activity 2.1.2, it is expected that an open call for proposals will be made for interested vehicle manufacturers and public transport operators, in order to choose the most appropriate vehicles and partners to conduct the pilot.

79. Activity 2.3.1: Selection of EV operators for e-buses in Arequipa in compliance with selection criteria established based on best practice guidelines, including gender considerations for a gender-responsive design and selection of drivers. Operators participating in the demonstration are requested to prepare and implement a gender policy, to be monitored during the project lifespan.

80. Activity 2.3.2: Integration and operation of at least 2 e-buses and 10 electric 3-wheelers in operator's fleet in Arequipa along one selected route (through vehicle incremental cost support for vehicle purchase or long-term leasing). The 3-wheelers will be operated as 3W *taxi* services.

[28] UNDP Social and Environmental Standard 5, N° 2: Would the Project possibly result in economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?

81. Activity 2.3.3: Identification and systematization of results, best practices and lessons learned for the route where the pilot was implemented, including the assessment of social and environmental impacts and recommendations regarding how to escalate to other routes. This will be integrated and fed into MINAM's knowledge management processes and systems. This activity will benefit of assistance support from the regional (LAC) platform and present and discuss results at the LAC regional platfom group and at the WG on 2&3 wheelers of the Global E-mob Programme.

82. Monitoring and evaluation of this demonstration is undertaken within the M&E Component.

83. ***Output 2.4*:** Technical training delivered (focused on EV driving and maintenance)

84. In line with the objectives of output 1.4 with regards to the strengthening of capacities of technical professionals, (in particular activity 1.4.2), the project will carry out technical training activities on EV driving and maintenance specifically targeted at those who will be directly involved with the operation of the pilots, including drivers, maintenance technicians, transport inspectors and/or others. Additional participants not directly involved in the operation of the pilot may be included to strengthen capacities for scalability and replicability. Female participants will be sought and encouraged to promote gender equality in the access to jobs generated by the operation of EVs in public transport. Participants involved in the operation of e-buses will also receive training in gender-responsive measures.

85. Activity 2.4.1: Implementation of technical training activities on EV driving and maintenance, that also address gender issues and gender-responsive measures in public transportation. They will include training in health risks associated to battery banks and EV repairs. Awareness towards gender-based violence and harassment and measures to address these risks will also be addressed. Training activities are targeting professionals involved in the operation and maintenance of the pilots, although they can include additional participants. Participation of women will be prioritized.

86. Although focusing on facilitating the implementation of the pilots, these training activities on EV driving will be open to future trainers from institutions interested in providing such educational services in the future, such as the academia and technical institutes participating in the project, which will be authorized to make use of the activity's training materials.

87. **Output 2.5:** Evidence and best practices on EV pilot implementation communicated to key stakeholders and incorporated in e-mobility guidelines for replicability.

88. Communications materials will be developed and activities will be carried out to disseminate the pilot results, so that national and local decision-makers and stakeholders, as well as the general public, can have detailed knowledge of the contents and conclusions of the demonstrations. It is expected that this activity will address possible misperceptions, biases and/or myths regarding the viability of using EVs in public transport, so as to build the necessary support from the various stakeholders – including public transport users – for replicability and scaling-up. The communications materials and activities will be based on the systematized best practices and lessons learned of the pilots, which fall under activities 2.2.3 and 2.3.3. Lastly, to strengthen the potential for replicability of the pilots, these best practices and lessons learned will be further integrated into the e-mobility guidelines developed under activity 2.1.1.

89. This output will include the following activities:

90. Activity.2.5.1: Dissemination of project pilot results through communication materials and initiatives (ie. articles, infographics, videos, social media, etc.).

91. Activity 2.5.2: Incorporation of best practices and lessons learned in e-mobility best practice guidelines. The preparation of the guidelines will benefit of assistance support from the regional (LAC) platform and present and discuss results at the LAC regional platform group, at the WG on HDV and e-buses and at the WG on 2&3 wheelers of the Global E-mob Programme.

COMPONENT 3: Preparing for scale-up and replication of low-carbon electric mobility.

92. Outcome 3: Conditions established to support the inclusion of EVs in the vehicle market and to accelerate their deployment

93. This outcome bridges the gap between institutionalization and demonstration on the one side and the market on the other side. It will support the consolidation of a market for EVs, primarily in the vehicle categories included in the pilots (buses and 3-wheelers) and subsequently in other categories (mainly light duty vehicles for passengers (cars) and freight (vans)). The NAMA-Energy project and the Interamerican Development Bank-COFIDE initiative are expected to provide alternative business and financing models for fleet electrification in Peru, covering inter alia vehicle loans and leasing (see Annex 15 for a description of the financing of the demonstrations, which will provide guidance for the development of such schemes); this project will focus on updating them based on Component #2 results and supporting the implementation of such models, taking into consideration relevant changes during the project life (e.g. a significant reduction in EV prices due to technological progress).

94. ***Output 3.1*:** Assessment of e-mobility market potential in Peru carried out for the introduction of electric vehicles and electrification of road public transport systems.

95. Based on the analysis of current demand and supply trends in the vehicle market, forecasts of EV demand for the different vehicle categories will be developed, as well as and an in-depth analysis of the demand potential for urban public transport fleets. This will consider the results from the NAMA-Energy project demonstrations, paying particular attention to how the interaction with national and local governments and other key stakeholders was developed during the demonstrations. In this way, it is expected that these activities will expand the market analysis to be provided by the Ministry of Energy and Mines in order to make it consistent with the Global Programme requests to facilitate comparison among participating countries.

96. Activity 3.1.1: Study of scenarios on the potential of the electric vehicle market in Peru. This deliverable will cover all vehicle categories, building upon the experience of previous initiatives such as the NAMA-Energy project activities (e.g. taxis) and other actions (private fleets, delivery vans).

97. Activity 3.1.2: Study of the electrification potential of road public transport systems in Lima and Arequipa. It will cover all public transport road services (including the last-mile link), in line with the global programme's guidelines, and with the emerging best-practice following the COVID-19 pandemia.

98. ***Output 3.2*:** E-mobility network established in Peru, including manufacturers, importers, technology and service providers, as well as public transport operators and other final users.

99. A variety of private stakeholders are already participating in the private network AEDIVE-Perú (*Asociación de Emprendedores para el Desarrollo e Impulso del Vehículo Eléctrico- Perú)*, as well as in a dedicated Committee within the AAP (*Asociación Automotriz de Perú*). However, these spaces do not include the academia, NGOs and civil society organizations. Therefore, it is expected that a space where all non-state actors can engage and integrate their different approaches to support the advancement of an e-mobility agenda The project will act as a contact and networking point, facilitating regular interactions among the e-mobility coordination space (output 1.1), AEDIVE, local public transport authorities and public transport operators, and the engagement of civil society stakeholders.

100. Activity 3.2.1: E-mobility network engagement with the project, providing regular contacts among e-mobility stakeholders, the government and civil society.

101. Activity 3.2.2: Elaboration of the network action plan (information platform and website, quarterly workshops, quarterly position papers) to accelerate the deployment of e-mobility. The action plan defines the network activities to be implemented until project conclusion and beyond.

102. ***Output 3.3*****:** Detailed financial and business models for EV commercialization and operation designed.

103. The main objective of this output is to overcome the financial barriers involved in the purchase of electric vehicles by creating and strengthening strategies to finance the initial capital cost requirements, enabling the participation and investment of the private sector. Whereas fiscal incentives are identified within component 1, this output will analyze the existing vehicle trading chain, will take stock of the business models developed in various initiatives such as the NAMA-Energy project and the studies made by COFIDE (Peruvian Public Development Bank) with the support of IADB, and will foster their implementation.

104. Although building upon the pilots conducted in Component 2, and focusing on urban public transport services, this output will also provide business models for other categories, including cars in public fleets, taxis, 3-wheelers and private fleets and vans for urban delivery.

105. Building upon the results of the NAMA project, the design of business models and financial schemes includes considerations on regulations (e.g. for the application of accelerated depreciation to electrified vehicles in corporate fleets), fiscal incentives (e.g. temporary exemption from VAT on the sale of electrified vehicles and their spare parts; exemption from Ad-Valorem for the importation of electrified vehicles), public contracts (e.g. duration of transportation concessions aligned with the lifespan of the vehicles), or scrap bonds for the purchase of electrified vehicles.

106. The design of financial schemes will consider different leasing and vehicle loans options, based on international experience. In particular, it will build upon previous studies made by COFIDE with IADB on the feasibility of a program of low-interest private loans, also benefiting from lessons learned with the COFIGAS facility and the Vehicle Gas Bond of the Social Energy Inclusion Fund.

107. The financial schemes will be developed primarily with COFIDE, although these activities will be open to the participation of the private financial sectors, and will be tailored to the particular needs (terms of lending, concessionality level) of different economic sectors, with a focus on public transport operators and large corporate fleets.

108. Additionally to vehicle loans, leasing schemes could provide an attractive alternative to vehicle procurement, provided some changes are made in the regulation of concession contracts. Linked to the expansion of EVs, leasing schemes concerning vehicles and batteries, and alternative payment options including energy supply have been proposed and, in some cases, implemented by some fleet managers around the world. The advantages of leasing options are strongly linked to their taxing regime and to the development and competition within the local financial sector. It will also consider the design of financial strategies for those who are not part of a fleet; all these conditions will be included in the development of innovative business models for EVs:

109. Activity 3.3.1: Development of alternative business models for the electrification of urban public transport (buses and 3wheelers). Building on and collaborating COFIDE's experience in financing bus fleet renewal (COFIGAS and the future COFIELECTRICO facility). The preparation of the business models will benefit from the exchanges and materials developed by the marketplace of the LAC regional platform.

110. Activity 3.3.2: Development of alternative business models for the electrification of public and private corporate fleets (with focus on 3-wheeler and taxi fleets). The preparation of the business models will benefit from the exchanges and materials developed by the marketplace of the LAC regional platform.

111. Activity 3.3.3: Submission to the government of alternative concession schemes to facilitate financing of EV in public transport. Addressed to public transport authorities, in order to incentivize within their concession

agreements the use of e-buses by operators. These may include, among other specifications, minimum fleet ownership conditions, differentiated public service compensations for different bus technologies and performance.

112. ***Output 3.4*:** EV procurement guidelines for urban public transport operators and authorities developed.

113. The consideration of EVs as a feasible option in fleets providing public services requires careful consideration of their differences with ICE vehicles while defining contract specifications. This includes a comprehensive consideration of the vehicle lifespan, total energy consumption and emissions, maintenance obligations, etc. The guidelines will take into consideration the existing national regulatory framework and its likely changes in the future (integrating the changes proposed by the project in Component 1 and those under consideration by the national government), showing their impact on the above-mentioned parameters and the comparative advantage EVs offer to urban public transport operators and authorities under the most likely regulatory scenarios. In 2020, the NAMA-Energy project is expected to provide guidelines to support public authorities in making this transition; the project will provide support for the implementation of these procurement recommendations to interested PT authorities and operators in Peruvian cities beyond Arequipa and Lima, as a way to scale-up the project results. Furthermore, the project will incorporate in these guidelines' recommendations for the adoption of gender-responsive measures in EV procurement, recruitment and training strategies of urban public transport operators.

114. Activity 3.4.1: Provision of tailored support and development of guidelines for operators of public transport services (buses and 3-wheelers) for the procurement of EVs for their fleets. The project will provide customized technical support for up to 4 operators interested in including EV in their fleets. This support will build on previous experiences and work the details of financing EVs. It will bring the strategy from concept to implementation.

115. Activity 3.4.2: Provision of tailored support to interested public authorities for promoting EVs in their own fleets and in those associated to public service concessions. The project will provide customized technical support for up to 4 public transport authorities in Peruvian cities interested in including EV in their systems. It will bring the strategy from concept to implementation.

COMPONENT 4: Long-term environmental sustainability of low-carbon electric mobility

116. Outcome 4: **Conditions created for a circular economy perspective in the electrification of mobility in Peru with EPR (Extended Producers Responsibility)**.

117. There is limited documented information on current end-of-life vehicle ELV management practices in Peru (no information on the rate of vehicles that are properly disposed or on the vehicle withdrawal rate); as ELV management generates valuable products, it usually is organized on a B2B basis, however without the intervention of producers; this can generate environmental hazards due to inadequate (or lack of treatment) of particular components (e.g. tires, lead batteries or lubricants). As identified at the Social and Environmental Screening, the future management for EVs at their end-of-life raises therefore two challenges in Peru: on the one hand, the need to strengthen current waste management chains for ELV, or at least for some components (where an EPR approach could eventually play a role); on the other hand, the need to introduce an adequate management approach for the EV battery. The first electric bicycles and motorbikes imported in Peru a few years ago are already reaching its lifespan, and the project will consider the management of their batteries as a first case study.

118. Although still at its infancy, EV battery management is likely to be organized around the use of batteries for energy storage (second life), rather than recycling. It is unclear whether the original EV (or battery) producer should be held responsible for the second-life use of its batteries, or whether this can be left to market forces, due to the relatively high value of used batteries. There is however a case to implement an EPR approach, in order to extend the lifespan of batteries as much as possible, as the production of a Lithium

battery can emit between 39 kg CO2e/kWh and 196 kg CO2e/kWh: finding a new use for EV batteries when they are no longer fit for their original purpose is a smart way to reduce their carbon footprint, before recycling. Finally, battery recycling would need to include some kind of credit system so that manufacturers using recycled materials could claim the associated emissions reduction.

119. In order to address all these issues comprehensively, while focusing on the promotion of second-life battery use, this component will provide the following outputs:

120. ***Output 4.1*:** Comprehensive diagnosis on national end-of-life vehicle management completed.

121. Currently, there are no reliable data available on ELV management, which jeopardizes any attempts to introduce an EPR approach to any vehicle components. The following deliverables will serve to provide a national ELV diagnosis:

122. Activity 4.1.1. Study on the description and analysis of local management conditions of ELVs and their components. This should consider the lifecycle of the components as well as value chains involved throughout their lifespan, and provide a comprehensive stakeholder mapping, identifying their interests and potential roles within ELV management. It should also provide off- and on-site research of the waste generation regarding vehicle components at the end-of-life, and the identification of ELVs components management and disposal.

123. Activity 4.1.2. Estimate of projected ELVs components generation, considering internal combustion and electric vehicles.

124. ***Output 4.2:*** Proposal on end-of-life vehicle management regulation, based on the EPR approach, delivered to the government for approval.

125. Framework regulation for EPR implementation has been issued in Peru in the last couple of years, but it needs to be complemented with specific regulation and norms for each particular waste stream to which EPR is intended to be applied. This output will provide the following deliverables:

126. Activity 4.2.1: Research and systematization for international and regional regulation on EPR-based ELV management, with a focus on EV components and battery. This activity will exchange information with the WG on batteries and charging infrastructure of the Global E-mob Programme.

127. Activity 4.2.2: Comprehensive review of existing and planned international regulations on second-life battery use.

128. Activity 4.2.3: Formulation of proposals for regulation in line with national priorities and in close coordination with the identified relevant private stakeholders.

129. ***Output 4.3*:** Cost-efficiency analysis and business models designed for implementation of an EPR approach to ELV management.

130. The assessment of business models for EPR implementation will ease the decision to invest for the private sector; the following deliverables will be provided by the project:

131. Activity 4.3.1: Research and systematization of successful financial and business models on ELV components and second-life battery use, based on the EPR concept. This activity will exchange information with the WG on batteries and charging infrastructure of the Global E-mob Programme and with the marketplace of the LAC regional platform.

132. Activity 4.3.2: Development of commercially viable EPR business models for ELV components and EV batteries, including cost benefit analysis estimating investment needs and financing schemes.

133. ***Output 4.4***: Roadmap for EPR approach to end-of-life vehicle management submitted to the government for approval.

134. The Road Map for implementation of EPR in ELV management will consider ELVs components, with a focus on EVs and their batteries, including the promotion of second-life use. The project will provide this road map, with short, medium and long term visions for country action in the field, through the following deliverables:

135. Activity 4.4.1: Development of a road map for EPR with ELVs components, built through a participatory process and in close coordination with the Ministry of Environment.

136. Activity 4.4.2: Advocacy and promotional activities to secure timely approval and enforcement of the roadmap as a planning instrument.

137. ***Output 4.5***: Awareness-raising and capacity-building activities completed, targeting government, consumers and private sector stakeholders on end-of-life vehicle management**.**

138. The following project activities are vital to ensure the engagement of private and public stakeholders as implementers, as well as the civil society as consumers and generators of ELVs components:

139. Activity 4.5.1: Design and delivery of a training programme for policymakers, to be implemented by local academia (trained beforehand if necessary), and to be later available for wider dissemination. It will include training in health risks associated to manipulation of batteries and other EV components at their end of life. This activity will exchange materials and methodologies with the capacitation activities of the LAC regional platform.

140. Activity 4.5.2: Provision of technical and managerial support to producers and importers. With a focus on the local industry (particularly 2-3 wheeled vehicle manufacturers) in order to strengthen their roles within the new ELV management framework.

141. Activity 4.5.3: Support to the government, industry and research sector in the design of a national campaign to raise awareness amongst consumers regarding the benefits of EPR for ELVs components, and the importance of second life battery use.

COMPONENT 5: Project monitoring and evaluation

142. Outcome 5: The project monitoring and evaluation plan is implemented

143. Under this component, all the project's monitoring and evaluation activities are undertaken.

144. Results from monitoring and evaluation activities will be regularly shared with the E-mobility Global Programme in order to facilitate global monitoring, and to support knowledge management activities at the global and regional levels.

145. ***Output 5.1:*** The project monitoring and evaluation plan and knowledge-management strategy are designed and implemented, providing regular reporting to the global programme

146. The project monitoring and evaluation plan is outlined in Annex 3, and the monitoring and evaluation budget is presented in Section VI. This plan will be revised and updated during the inception workshop and will ensure the regular reporting on the project's indicators established in the PRF. The GEF/UNDP NAMA Project has structured a robust MRV system for road transport emissions, including EVs, that will provide the adequate framework for the MRV plans in Arequipa and Lima.

147. The project knowledge management (KM) strategy is outlined in the knowledge management section of this document, including a liaise with the Global Programme. This strategy will be revised and updated during the inception workshop and will ensure the regular exploitation and dissemination of the materials developed under the other project components, including training and dissemination materials, policy and technical recommendations and technical tools.

148. Activity 5.1.1: Updating and implementation of project monitoring and evaluation plan.

149. Activity 5.1.2: Development and implementation of monitoring, reporting and verification (MRV) plan for the demonstration in Lima, including passenger satisfaction surveys to assess service quality of e-buses under a gender approach.

150. Activity 5.1.3: Development and implementation of monitoring, reporting and verification (MRV) plan for the demonstration in Arequipa, including passenger satisfaction surveys to assess service quality of e-buses under a gender approach.

151. ***Output 5.2:*** Best practices and lessons learnt reports developed and shared with the global programme

152. The Terminal Evaluation will be undertaken after the completion of all main project activities, as described in Section VI.

153. Activity 5.2.1: Terminal evaluation

154. Activity 5.2.2: Final best practices and lessons learnt report prepared and shared with the Global Programme its Regional Platform.

Partnerships:

155. The project intends to build upon the activities already initiated in Peru to promote e-mobility. Partnerships with these projects and stakeholders will allow the project to focus on implementation, bridging the gap between the on-going initiatives and the ambition to create the conditions for full-fledged deployment of EV in the Peruvian market, particularly in public transport. The following stakeholders and projects are currently active in the field of e-mobility in Peru, and this project has already established links with most of them during its design stage. Stakeholders participating in these projects will be invited to the inception workshop and to the e-mobility network to be set up in component 3; more details are provided in Annex 8 (Stakeholder Engagement Plan):

- GEF/UNDP NAMA-Energy project. The project "Peru: Nationally Appropriate Mitigation Actions in the Energy Generation and End-Use Sectors" is expected to conclude in April 2021 and will provide relevant recommendations on regulatory changes and acceleration of EV deployment in Peru. Building upon these deliverables, the project will take the demonstrations to a new stage (EVs owned as an asset of the operators participating in Component #2) and will take forward those recommendations to government's approval and to implementation by relevant stakeholders. Furthermore, this project will provide the adequate platform for the MRV system of the demonstrations in Arequipa and Lima. The MRV Platform developed by the NAMA project provides MRV protocols for its 12 mitigation measures, included those on electric mobility, in one centralized, integrated and automated online system[29]. This platform helps to ensure a transparent, traceable, understandable, and automated MRV process.
- COFIDE is currently working with BID in the preparation of a USD 20 million financial facility to support the electrification of bus fleets. If this facility is established, it would significantly expand the number of electric buses operating in the country during the project's lifespan, offering an opportunity to establish a shared monitoring

[29] http://sismrv.minem.gob.pe. See also:
https://undpgefpims.org/attachments/4679/213487/1737554/1760682/Annex_12_Diagnostic_Study_NAMA_Electric_Transport.pdf
https://undpgefpims.org/attachments/4679/213487/1737554/1760682/Annex_22a_Plataforma_MRV_NAMAs.pdf

and evaluation system and to multiply the impact of the demonstration activities of the project. COFIDE will also contribute with its past experience in financing bus fleet renewal (COFIGAS).

- In addition, the GEF/UNEP global e-mobility programme will support the project through participation in the global working groups and networking with the other projects in the Latin America and Caribbean region (Antigua and Barbuda, Chile, Costa Rica, Grenada, Jamaica, Saint Lucia) through a regional platform. This is expected to maximize the impact of the incremental financing provided by GEFTF, through the optimization of capacity building and knowledge management activities, bundling of EV procurement, provision of generic tools for business and financial modelling, etc.

Risks:

156. The project analysis has identified 13 risks for the achievement of the intended results, as described in Annex 6. This includes environmental, social, financial, operational, political, regulatory and strategic risks. Of them, the political risk is critical, as the project intends to achieve changes in the institutional and regulatory framework that will require political will and action in a highly polarized context. Among the main risks, it is also worth pointing out the social risk associated to the achievement of the project's ambitions in facilitating access to e-mobility jobs to women and to mobilize the financial resources required by operators to get access to EVs.

157. The mitigation measures for the risks mentioned above (and others) build upon the already substantial experience of the Peruvian government in climate change mitigation policies in the transport area (including the on-going NAMAS for the energy sector), the national sectoral plans on transport and energy and local plans on sustainable urban mobility. These policies are being successful in building and widening social and political consensus on sustainable mobility, with a similar approach to this project.

158. Furthermore, the project is including activities addressing these risks: for example, the project will prioritize women in professional training activities, and will provide financial incentives to operators interested in using EVs during the demonstrations, in cooperation with COFIDE and other financial institutions.

159. It is important to highlight that the project will have to face a high level of informality in the transport sector, both in buses and 3W taxis. This can lead to disloyal competition, difficulties with getting permits (both in legal and social terms) and other complex social issues.

160. The project risk associated to the COVID-19 pandemic has been included. The eventual implementation of lockdown and social distancing measures is expected to have a low impact on the project, as its activities can be undertaken under mobility-restricted conditions. In fact, the deterioration of economic and social conditions due to the pandemic has highlighted the critical importance of urban mobility and increased the attention of governments and key stakeholders to safe mobility. Urban mobility demand has decreased, and non-collective transport modes (walking, cycling and private transport) have been perceived as safer by many users[30]. Public transport authorities and operators have reacted by establishing safety protocols, and the critical importance for cities of safe urban public transport systems has become even more relevant, and pre-COVID PT ridership levels have recovered in many cities around the way, although not in Peru yet. The project provides additional arguments for the use of PT, as EVs reduce air pollution (also inside the vehicle) and provide a positive contribution for a safe ride, and the demonstrations are expected to include innovative protocols for a safe trip by the EVs participating in the project, but its ability to reach the expected number of beneficiaries will largely rely on the success of public transport authorities and operators to regain the trust of their customers.

161. Climate risk screening.

- Climate hazards. In accordance with the Third National Communication to UNFCCC (2016), Peru is a country with ecosystems particularly vulnerable to climate change: coastal areas, arid and semiarid areas, areas prone to drought or flooding, fragile montanious ecosystems, disaster-exposed areas and urban areas with high air pollution levels. The key climate hazards include an increase in average temperature and precipitations, and an

[30] https://www.limacomovamos.org/noticias/informe-de-encuesta-los-efectos-del-covid19-en-la-movilidad-de-lima-y-callao/

increase in extreme weather events. Based on the 2015 National Strategy on Climate Change (ENCC 2015) .Peru has developed climate scenarios and is currently completing its National Adaptation Plan. The ENCC 2015 is being prioritized with focus on 2050 and also the National Adaptation Plan was approved in June 07 2021, bu Ministerial Resolution N°096-2021-MINAM.

- Vulnerability and exposure. The project's vulnerability to climate change is related to the disruption in the operation of EVs due to their inability to be recharged. This vulnerability may come from (a) disruption in electricity supply or (b) lack of access or disruption of the charging infrastructure. The Third National Communication acknowledges the rise in the economic costs of climate-related electricity disruption events in Peru , and the potential decresase in hydro-generated electricity due to changes in climate (between 3.3% and 5% of GDP); the national energy policy is reducing this vulnerability through the expansion of renewables and distributed generation, as well as with improvements in the integrated distribution grid ; the future availability of charging infrastructure relies on the adequate deployment of the network, for which the project will contribute through the preparation of the e-mobility strategy. The exposure of the project to changes in climate is low, due to the fact that the electricity network in Peru is already highly integrated, the availability of charging infrastructure is already growing, and the primary charging points for the fleets targeted by the project (3W taxis and PT buses) will be located in depots, with lower exposure to the effecs of climate events.
- The probability of occurrence of climate events can be rated as moderate, considering the vulnerability of Peru, and the impact of such events on the project can be rated as low, considering the likelihood of major electricity disruptions in Lima and Arequipa, the expected location of the charging stations, the availability of alternative charging ports in both cities, and the quick response capacity, through the installation of alternative charging points. This risk is accordingly rated as low.
- The adaptation measures considering in the project include the preparation of the national e-mobility strategy and the study of the charging networks in Lima and Arequipa, in which the resilience of the system will be integrated; during the preparation of the pilots, contingency plans will be prepared to make it possible to recharge the EVs in case of disruption or failure of the project's charging points and to provide quick recovery of these points.

Stakeholder engagement plan and South-South cooperation:

162. The stakeholder engagement plan is attached as annex 8. The plan includes all relevant stakeholders: national and local government services, public transport authorities, electricity utilities, transport operators, EV providers, academia and civil society, as well as vulnerable groups.

163. During the project design stage, the process of identifying the private sector actors that are co-financiers of the project consisted of two stages. First, an initial mapping was carried out by UNDP in conjunction with the Ministry of the Environment. This was undertaken through a selection of key stakeholders of the GEF5 project under implementation "Appropriate National Mitigation Actions (NAMA) in the power generation sector and its final use in Peru", mainly the Electric Transportation NAMA which already had identified the private and public sector actors working with E-mobility in Peru. Second, workshops were held to socialize the project with representatives of manufacturers and marketers of public transport vehicles (buses, 3 wheelers) as well as transport infrastructure enterprises. This mapping stage considered the following main elements for the selection and analysis of stakeholders:

- Identification of actors that could benefit and be disadvantaged with the development of project's activities.
- Level of knowledge / information that the actors have on e-mobility.
- Type of relationships that preponderate between actors (collaboration, conflict, etc.), and their level of interest
- The likely benefits for each actor?

164. Finally, the second phase consisted of verifying the capacities of the actors identified through the workshops mentioned above as well as in bilateral meetings and interviews that were carried out with

them. The possibility is left open that later they may be responsible parties to implement project activities in which case they will be involved and selected in accordance with UNDP policies and rules

165. The engagement plan focuses on initial consultations with stakeholders since the first months and during implementation, information disclosure through public reports, website and publications, as required, and periodic reporting. Engagement activities aim at increasing the involvement of key partners (national and local government, public transport stakeholders, electricity sector) during all project activities, and particularly during the demonstrations. They also aim at providing a collaborative space for the preparation of regulatory proposals. Finally, the engagement plan will also aim at identifying vulnerable public transport users affected by the services provided by EVs, and at integrating their mobility needs during the demonstrations, and in the scaling-up and replication plans after project conclusion.

166. The engagement tools are based on different meetings: regular (quarterly) meetings for project implementation and *ad hoc* meetings with particular stakeholders, as project activities are deployed. The project includes training activities and awareness-raising workshops that will also facilitate the engagement of stakeholders.

167. Resources for stakeholder engagement activities are included within design and awareness-raising activities, as well as within the project management budget.

168. South-South and Triangular Cooperation (SSTrC) opportunities and technology transfer from peer countries will be further explored during project implementation. To present opportunities for replication in other countries, the project will codify good practices and facilitate dissemination through global ongoing South-South and global platforms, such as Africa Solutions Platform, the UN South-South Galaxy knowledge sharing platform and PANORAMA[31]. The project is included in the UNEP's global e-mob programme, which provides an excellent platform for cooperation with the other participating countries and particularly with those from the Latin American and Caribbean region: Antigua and Barbuda, Chile, Costa Rica, Jamaica and Saint Lucia. The project will facilitate the participation of local stakeholders and the Peruvian government in the working groups and other activities organized within the global programme. The project will also network with other transport-related initiatives and opportunities for regional and global cooperation, in which Peru is already participating, such as the Heavy Duty Vehicles Initiative of the Climate and Clean Air Coalition (CCAC), and MOVE-LATAM[32] (*Movilidad Eléctrica en América Latina*), a capacity-building platform supported by UNEP. It will also benefit from CALAC+[33], an on-going international cooperation initiative in which MINAM is participating.

169. The MOVE platform will be considered a priority space for sharing lessons and experiences in the region together with the regional Platform. In addition, to bring the voice of local e-mobility stakeholders to global and regional platforms, the project will explore opportunities for meaningful participation in specific events where UNDP could support engagement with the global development discourse on e-mobility. The project will furthermore provide opportunities for regional cooperation with countries that are implementing initiatives on e-mobility in geopolitical, social and environmental contexts relevant to the proposed project in Peru.

[31] https://panorama.solutions/en

[32] MOVE Electric Mobility in Latin America and the Caribbean is a community of practice launched by UN Environment to accelerate deployment of electric mobility in the region. MOVE provides capacity building, knowledge creation, resource mobilization and technical assistance to countries and cities in the region. It has following objectives:

- Capacity building: MOVE has a network of +600 public sector technical officers and provides monthly webinars on diverse topic related to electric mobility.
- Knowledge creation: MOVE has conducted technical assessments related to air quality as well as an annual status report on the deployment of electric drive technology in the region
- Technical assistance and resource mobilization: MOVE assists countries to access to climate finance and is currently implementing technical assistance projects in more than 10 countries in the region, fostering the establishment of an enabling environment for the uptake of high-use urban electric fleets and related charging infrastructure.

[33] https://programacalac.com

Gender equality and Women's Empowerment:

170. The gender analysis and action plan were integrated within the project design stage, and are presented in Annex 9. The project follows the guidance of the government's climate change policy, as established in the Gender and Climate Change Action Plan (*Plan de Acción en Género y Cambio Climático*[34]), to integrate the gender perspective in the design and implementation of GHG emission reduction projects and programmes in the energy and transport sectors. It aims at facilitating the access of women to the know-how and job opportunities associated to EV technologies, with a focus on urban public transport services. Additionally, it aims at properly identifying differences between women and men in their urban mobility requirements (e.g. in terms of vehicle design, quality of services, security…) and integrate the lessons learned within the demonstrations, and subsequently within the guidelines for procurement of EV of urban transport.

171. The project activities in this area include EV training activities, in which women will be given priority for enrollment, and where gender-equality challenges will be considered in the selection of topics and trainers. They also include the design and monitoring of demonstrations, in which gender-sensitive information (EV operation and use, relevance of the assessment criteria for women and men) will be embedded.

172. On top of the internal governmental assessment procedures, the project will provide an assessment on how each proposed regulation improves women's access to services, create opportunities to women and promote equitable treatment. The project will also assess how the proposed regulation will increase the participation of women in the transport sector, and how it will engage women directly in decision-making.

Innovativeness, Sustainability and Potential for Scaling Up:

173. The technological **innovativeness** of the project is very high, as the deployment of EVs is still low in most countries, and the characteristics of the EVs available in the market have been developed attending to contexts in other countries, not necessarily similar to the mobility conditions in Peruvian cities. Although a few EVs have been recently tested in Peru, the project will be innovative in engaging in permanent use, integrating the new EVs within the existing fleets of some operators. As the project is partnering with vehicle manufacturers, there is also the potential to innovate in vehicle design aspects that can better tailor the service provided to the preferences of Peruvian users.

174. Another relevant innovation is the EPR approach to the management of EVs and its components at their end of life. Peru has still limited experience in the practical implementation of the EPR concept, and EVs offer an excellent ecosystem for implementation, considering the reduced number of stakeholders initially involved. Furthermore, action at the international level is expanding in order to establish global recycling chains, at least for batteries, and regional legislation is being drafted (e.g. in the European Union). The project will facilitate access to the innovations associated to these recycling chains and their adaptation to national conditions.

175. The innovative potential of the project also includes financial schemes to facilitate the operators' access to EV, in spite of their higher upfront costs. The project will explore and propose incentive packages for early adopters, as well as alternative schemes to vehicle purchase, with the engagement of project partners from the financial and electricity sectors.

176. **Replication and scaling up** is included in the project design at three levels: i. within the pilot cities, through the support to public transport operators participating in the demonstrations to expand their EV fleets based on their first-hand experience gained from the project; ii. at the broader level of cities in Peru, through the dissemination of demonstration results and the preparation of EV procurement guidelines addressed to public transport authorities and operators; and iii. at the country level, through the adoption

[34] Ministerio del Ambiente, Ministerio de la Mujer y Poblaciones Vulnerables (2016). Plan de Acción en Género y Cambio Climático. Disponible en http://www.minam.gob.pe/cambioclimatico/wp-content/uploads/sites/11/2015/12/PLAN-G%C3%A9nero-y-CC-16-de-JunioMINAM+MIMP.pdf.

of regulatory recommendations creating a levelled playing field for e-mobility, the project will facilitate the deployment of other EV categories, such as cars or cargo vans. It is expected that thanks to these activities the project will be able to leverage on the pilots and have an impact beyond the vehicles acquired by the project:

- Besides the 4 electric buses in the project demonstration, operational in 2022, it is expected that additional electric buses will be added to the fleet as a result of project activities, so that 5% of the new bus sales are electric by 2025 (reaching that year a total fleet of 233 e-buses). Since then, the percentage of electric buses in new sales would grow reaching 10% of total bus sales in 2032, 15% in 2036 and 20% in 2040.
- Besides the 20 electric 3W in the project demonstration, operational in 2022, it is expected that additional electric 3W will be added to the fleet as a result of project activities, so that 2.5% of the new 3W sales are electric by 2025 (reaching that year a total fleet of 484 3W. Since then, the percentage of electric vehicles in 3W new sales would grow 0.5 percentage points per year, reaching 5% in 2030, and at a quicker path since then, reaching 9.5% in 2036 and 13% in 2040.

177. Moreover, the project will facilitate the efforts to accelerate the shift towards EVs globally, through the cooperation channels envisaged by the e-mob programme it participates in.

178. **Sustainability.** The project sustainability and exit strategy will mainly rely on the government and some of the project partners, based on their empowerment through the project activities and deliverables, and the institutional and informal networks established. Through its deliverables, the project is expected to support the implementation of a sound regulatory framework for the deployment of EVs, to implement business models and financing schemes (including incentives as required) for potential users, and to create a robust network of stakeholders interested in e-mobility. Furthermore, awareness-raising and training activities will provide the necessary know-how within the national and local governments, and in the transport sector at large. This friendly environment is expected to facilitate action from key stakeholders:

- Financial institutions will have access to the necessary factual evidence on EV performance to design adequate financial schemes, without unfair conditions compared to ICE vehicles.
- EV manufacturers will find a consistent legal framework in Peru as a growing market, in a time that bus fleets in cities are quickly growing and authorized 3-wheeler operators need to rely on more efficient technologies to reduce their operating costs and improve quality of service (noise, pollutants…).
- Managers of public transport and other fleets with find electric alternatives to their vehicle needs with reliable information and wide local expertise.
- The e-mobility network sustained by the project is a key instrument for the project's exit and sustainability strategy, as it has brought together all market actors behind a shared plan of action. This network (and the government) will be able to monitor future progress in the electrification of road transport mobility in Peru based on the project's results and the continuation of the operation of the MRV platform used by the project and initially provided by the NAMA energy project.

179. The project aligns its sustainability and exit strategy with the EV market transition design provided by the e-mob programme (figure below). It is a first step towards mainstreaming of EVs making them competitive in all market segments, while embedding the effort within the national and local governments priorities to improve and expand the public transport system.

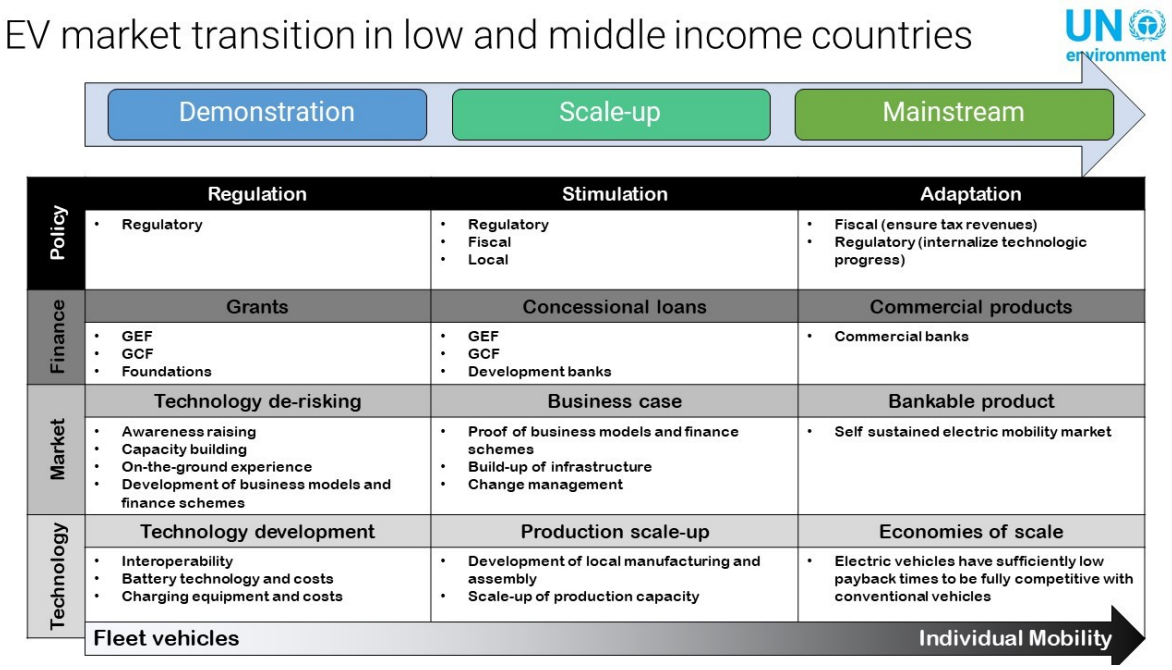


Figure 1: Source: UNEP, E-Mobility Program Framework Document

V. PROJECT RESULTS FRAMEWORK

This project will contribute to the following Sustainable Development Goal (s): 7 (Affordable and Clean Energy), 9 (Industry, Innovation and Infrastructure), 11 (Sustainable Cities and Communities) and 13 (Climate Action).				
This project will contribute to the following country outcome (UNDAF/CPD, RPD, GPD): CPD Outcome 1 (Inclusive and sustainable growth and development) and Outcome 3 (Institutions and transparency)				
	Objective and Outcome Indicators	**Baseline**	**Mid-term Target**	**End of Project Target**
Project Objective: *Reduce GHG emissions from urban transport by creating the conditions to accelerate market transformation of electric mobility in Peru*	**Mandatory** Indicator 1: # direct project beneficiaries disaggregated by gender (individual people)	0	1,300 women and 1000 men (1.17 million trips/year)	8,000 women and 6,000 men (7 million trips during the project)
	Mandatory Indicator 2: # indirect project beneficiaries disaggregated by gender (individual people)	0	26,500 women and 26,500 men beneficiaries	26,500 women and 26,500 men beneficiaries
	Mandatory GEF Core Indicators 6.2: Expected metric tons of CO_2e avoided (at the time of measurement)	0	35,692 t CO_{2e}	178,460 t (direct) end of life of project's vehicles 236,657 t (indirect) end of life of project's vehicles
Component 1	**Institutionalization of low-carbon electric mobility**			
Outcome 1 *Institutional, regulatory and fiscal frameworks established for the promotion of e-mobility, with the support of all key stakeholders, including potential EV providers and users*	Indicator 1.1: A national coordination space to promote the uptake of low-carbon e-mobility is established and operational	No coordination space	Coordination space established	4 meetings/year
	Indicator 1.2: National e-mobility strategy launched by the government	No e-mobility strategy	Draft e-mobility strategy delivered to government, including gender analysis and action plan	E-mobility strategy launched
	Indicator 1.3: Regulatory and fiscal schemes to incentivize the uptake of e-mobility (technical approval of vehicles, public transport operations, electricity fares and charging stations) approved by the government	Reduced excise on some taxes	At least one draft regulatory scheme (on PT vehicles, car fleets or 3-wheelers) delivered by the project. At least one draft fiscal scheme delivered by the project	At least one regulatory or one fiscal scheme tabled for approval by the government
	Indicator 1.4: Number of public and private decision-makers trained on legal and fiscal issues of e-mobility	0 trainees	5 women, 15 men trained	25 women, 25 men trained
Outputs to achieve Outcome 1	Output 1.1: An intersectoral and multilevel e-mobility coordination space established, including all relevant governmental stakeholders Output 1.2: National e-mobility strategy submitted to the government and launched, including economic impacts (competitiveness, industrial development), geographical deployment (e.g. charging network) and environmental and social dimensions (gender-responsive). Output 1.3: Revised regulatory and fiscal framework for deployment of electric vehicles delivered for governmental approval. Output 1.4: Strengthened capacities and awareness of public and private decision-makers, technical professionals and consumers regarding e-mobility policy design and implementation, highlighting the social and environmental benefits of EV deployment).			
Component 2	**Short term barrier removal through low-carbon e-mobility demonstrations**			
Project Outcome 2 *EV demonstrations in urban public transport systems provide*	Indicator 2.1: Number of kilometers served by the project's 4 electric buses and 20 3-wheelers per city (Lima and Arequipa)	0	Buses: 108,000 in each city by end of year 2 3W: 140,000 km in each city by	Buses: 216,000 in each city by end of year 3 3W: 280,000 km in each city by end

evidence of technical, financial and environmental sustainability to scale-up e-mobility			end of year 2	of year 3
	Indicator 2.2: Number of women and men receiving professional training	0	15 women and 15 men	30 women and 30 men
	Indicator 2.3: Number of EV purchases approved by PT operators in Lima and Arequipa	No purchase orders	At least 1 purchase order submitted to one operator's Board in one city	At least 1 purchase order approved in one city
Outputs to achieve Outcome 2	***Output 2.1***: E-mobility pilots in public urban transport systems designed in Lima and Arequipa, integrating gender-responsive measures ***Output 2.2***: EV pilot in Lima implemented, monitored and evaluated: two electric buses and ten 3-wheelers (indicatively) in public transport lines. ***Output 2.3***: EV pilot in Arequipa implemented, monitored and evaluated: two electric buses and ten 3-wheelers (indicatively) in public transport lines. ***Output 2.4***: Technical training delivered (including EV driving and maintenance) **Output 2.5:** Evidence and best practices on EV pilot implementation communicated to key stakeholders and incorporated in e-mobility guidelines for replicability.			
Component 3	**Preparing for scale-up and replication of low-carbon electric mobility**			
Outcome 3 *Conditions established to support the inclusion of EVs in the vehicle market and to accelerate their deployment*	Indicator 3.1: Procurement guidelines on EV for PT	No guidelines	Draft guidelines delivered, including guidance on women's PT quality priorities	Guidelines endorsed by PT authorities and government
	Indicator 3.2: At least one financial mechanism to facilitate the scaling-up of e-mobility in public transport fleets	No financial mechanism available	Financial mechanism proposed	Financial mechanism established
	Indicator 3.3: Number of new EV registered in Peru in (public transport) fleets	1	5 buses and 20 3-wheelers	21 buses and 100 3-wheelers
Outputs to achieve Outcome 3	Output 3.1: Assessment of e-mobility market potential in Peru carried out for the introduction of electric vehicles and electrification of road public transport systems. Output 3.2: E-mobility network established in Peru, including manufacturers, importers, technology and service providers, as well as public transport operators and other final users Output 3.3: Detailed financial and business models for EV commercialization and operation designed Output 3.4: EV procurement guidelines for urban public transport operators and authorities developed.			
Component 4	**Long-term environmental sustainability of low-carbon electric mobility**			
Outcome 4 *Conditions created for a circular economy perspective in the electrification of mobility in Peru with EPR*	Indicator 4.1: EPR-based roadmap for collection, reuse and disposal of used EV batteries	No roadmap in place	Diagnostic on ELV management completed	Roadmap endorsed by government
	Indicator 4.2: Number of women and men professionally trained on end-of-life EV management	Baseline 4.3: 0	Mid-point target 4.3: 6 women, 4 men	25 women, 15 men
Outputs to achieve Outcome 4	Output 4.1: Comprehensive diagnosis on national end-of-life vehicle management completed Output 4.2: Proposal on end-of-life vehicle management regulation, based on the EPR approach, delivered to the government for approval Output 4.3: Cost-efficiency analysis and business models designed for implementation of an EPR approach to ELV management Output 4.4: Roadmap for EPR approach to end-of-life vehicle management submitted to the government for approval Output 4.5: Awareness-raising and capacity-building activities completed, targeting government, consumers and private sector stakeholders on end-of-life vehicle management			
Project component 5	**Project monitoring and evaluation**			
Outcome 5 *The project monitoring and*	Indicator 5.1: Number of reports on best practices and lessons learned on the Peru project shared by the	Baseline 5.1: 0	Mid-point target 5.1: 0	4 reports (corresponding to activities 2.2.3, 2.3.3, 2.5.2 and 4.3.1)

evaluation plan is implemented	national coordination space with the global programme			
Outputs to achieve Outcome 5	Output 5.1: The project monitoring and evaluation plan and knowledge-management strategy are designed and implemented, providing regular reporting to the global programme Output 5.2: Best practices and lessons learnt reports developed and shared with the global programme			

PRF target notes:

Mandatory indicator 1: See Annex 11. Mid-term target based on the expected number of passenger trips provided by the project's electric buses and 3W in the first year of project demonstration. Final target based on a replication factor of 4 achieved through component 3 activities.

Mandatory indicator 2: See Annex 11. Final target is based on the assumption that the routes served by the vehicles in the project demonstrations could be convenient to at least 5% of the residents in the area of influence of such services. No replication factor is considered, as the additional EV resulting from component 3 activities are likely to be placed in the same areas. Therefore, the mid-term and the final targets are the same.

Mandatory indicator 3. See Annex 11. The average fuel consumption is estimated at 52.0 liters diesel/100 km for buses (the average value obtained from public operators in Lima) and 6 liters gasoline/100 km for 3-wheelers (CO_2 GEF Manual). The average annual mileage is estimated at 54,000 km for buses (average value for the targeted bus lines in Lima) and 14,000 km for 3-wheelers. A fuel efficiency improvement of 1% per year is assumed by the model. Only emissions saved from activities until the end of the project are considered. Mid-term targets assumed all demonstration EVs start operations at the beginning of year 2. Final targets assume operations continue until the end of the project and include a replication factor of 4 since year 3.

Indicator 1.1. As the coordination space brings together mainly governmental bodies, regular meetings provide effective monitoring of its activity. It is assumed that the coordination space is established by mid-term, with at least quarterly meetings. This indicator corresponds to one of the indicators necessary to report to the global programme.

Indicator 1.2. This indicator corresponds to one of the indicators necessary to report to the global programme.

Indicator. 1.3. This indicator corresponds to one of the indicators necessary to report to the global programme.

Indicator 2.1. This indicator corresponds to one of the indicators necessary to report to the global programme. It assumes 54,000 km per bus and year and 14,000 km per 3W and year (see Annex 11).

Indicator 2.2. The number of women and men receiving professional training is linked to the number of electric buses and 3W expected to be operational by mid-term (demonstrations) and by the end of the project.

Indicator 3.1. This indicator corresponds to one of the indicators necessary to report to the global programme.

Indicator 3.2. This indicator corresponds to one of the indicators necessary to report to the global programme. It refers to the design and availability of financial instruments offered by financial institutions for the procurement of EVs.

Indicator 3.3. At the time of project design, there was already one e-bus providing occasional services. My mid-term, the project is expected to make 4 e-buses and 20 electric 3W operational for the demonstrations. By the end of the project, it is expected that component 3 activities will foster the procurement of 16 additional buses and 80 additional electric 3W.

Indicator 4.2. This indicator corresponds to one of the indicators necessary to report to the global programme

Indicator 5.1. This indicator corresponds to one of the indicators necessary to report to the global programme.

VI. MONITORING AND EVALUATION (M&E) PLAN

180. The project results, corresponding indicators and mid-term and end-of-project targets in the project results framework will be monitored annually and evaluated periodically during project implementation. If baseline data for some of the results indicators is not yet available, it will be collected during the first year of project implementation. The Monitoring Plan included in Annex 4 details the roles, responsibilities, and frequency of monitoring project results.

181. Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP and UNDP Evaluation Policy. The UNDP Country Office is responsible for ensuring full compliance with all UNDP project monitoring, quality assurance, risk management, and evaluation requirements.

182. Additional mandatory GEF-specific M&E requirements will be undertaken in accordance with the GEF Monitoring Policy and the GEF Evaluation Policy and other relevant GEF policies. The costed M&E plan included below, and the Monitoring plan in Annex 4, will guide the GEF-specific M&E activities to be undertaken by this project.

183. In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed during the Project Inception Workshop and will be detailed in the Inception Report.

184. Additional GEF monitoring and reporting requirements:

Inception Workshop and Report:

185. A project inception workshop will be held within 60 days of project CEO endorsement, with the aim to:

a. Familiarize key stakeholders with the detailed project strategy and discuss any changes that may have taken place in the overall context since the project idea was initially conceptualized that may influence its strategy and implementation.
b. Discuss the roles and responsibilities of the project team, including reporting lines, stakeholder engagement strategies and conflict resolution mechanisms.
c. Review the results framework and monitoring plan.
d. Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP and other stakeholders in project-level M&E.
e. Update and review responsibilities for monitoring project strategies, including the risk log; SESP report, Social and Environmental Management Framework and other safeguard requirements; project grievance mechanisms; gender strategy; knowledge management strategy, and other relevant management strategies.
f. Review financial reporting procedures and budget monitoring and other mandatory requirements and agree on the arrangements for the annual audit.
g. Plan and schedule Project Board meetings and finalize the first-year annual work plan.
h. Formally launch the Project.

GEF Project Implementation Report (PIR):

186. The annual GEF PIR covering the reporting period July (previous year) to June (current year) will be completed for each year of project implementation. Any environmental and social risks and related management plans will be monitored regularly, and progress will be reported in the PIR. The PIR submitted to the GEF will be shared with the Project Board. The quality rating of the previous year's PIR will be used to inform the preparation of the subsequent PIR.

Knowledge management:

187. Knowledge management includes all the methods, processes, and systems in the project to support the storage, analysis, and access to the knowledge created during the project. Knowledge management activities are integrated within the project logical framework to collect and share experiences, lessons and results in a structured and user-friendly manner. It also facilitates the relations between the Peruvian child project and the global e-mob programme.

188. The project team will ensure extraction and dissemination of lessons learned and good practices to enable adaptive management and upscaling or replication at local and global scales. Results will be disseminated to targeted audiences through relevant information sharing platforms and networks. The project will contribute to scientific, policy-based and/or any other networks as appropriate (e.g. by providing content, and/or enabling participation of stakeholders/beneficiaries). The project will apply an open data policy, providing full access to the project's databases and reports to those interested, through the project's website whenever possible.

189. The knowledge management approach, integrated into the project and outlined in this section, will contribute to its overall impact, making special emphasis on lessons learned and good practices. The knowledge management activities will ensure that the knowledge generated is aligned and integrated into MINAM's, and UNDP's knowledge management systems. The capacities of some project stakeholders (particularly AEDIVE and Profonanpe) will also contribute to the knowledge management and dissemination effort.

190. In the case of MINAM, its contribution will be critical to facilitate replicability and up-scaling of the overall project activities. This is in accordance with the RM 349-2019-MINAM (Plan de Modernización Institucional del Ministerio del Ambiente 2020 – 2022) and MINAM's Institutional Strategic Plan, which dictates the implementation of its Knowledge management system. Every knowledge product generated by the project will be incorporated into MINAM's knowledge management system.

191. The project will actively participate in the global e-mob programme's activities at the regional and global levels, channelled through its component 1. Peru will actively participate and contribute to the knowledge exchange in the regional knowledge and investment (marketplace) platform and in all the global working groups, as well as by providing insights and knowledge (monitored through indicator 5.1). A budget is reserved within component 1 for travel necessary for participation in these international activities.

192. The project's knowledge management approach builds upon previous UNDP projects in the GHG mitigation sector, and particularly from the recently completed project to support the implementation of NAMA in the energy sector. The approach considers four steps in the knowledge management of project's activities:

- Identification and collection of project's reports, results, lessons learnt and other experience. The focal point at the PMU for this task is the Project Technical Assistant, which participates in all the project activities with a potential to generate such materials and will actively engage with project participants to get them involved in this effort.
- Analysis and formatting of the collected material, to be integrated in the knowledge management system. The Project Technical Assistant will prepare the relevant templates at the beginning of the project and complete them for each relevant output on a quarterly basis.
- Accessibility and dissemination of knowledge materials. This is mainly provided through the project website and the project's e-mobility network (output 3.2). Some stakeholders with networking experience and capacity to reach out to a wider audience (e.g. AEDIVE, Profonanpe) are expected to actively contribute to the dissemination of the project's knowledge materials.
- Experiences in adaptive management during the implementation of the project are also targeted within project management. This includes the materials produced and lessons learned during the various participatory, deliberating and training activities envisaged in the project, as well as in along the internal administrative procedures. The Project Manager, with the support of the project administration team provided by Profonanpe, will act as focal point for this.

193. A summary of the knowledge management activities, timeline and budget is presented in the table below..

Activity	Timeline	Budget	Deliverables
Project Technical Assistant (PTA) collection of knowledge materials from outputs 1.1, 1.3, 1.4 and liaise with the global programme KM system	Quaterly, from M6 to M34	8,300	Content management system (templates for storing information and tracking tools for changes).
Project Technical Assistant (PTA) collection of knowledge materials from outputs 2.1, 2.2, 2.3, 2.4, 2.5	Quarterly, from M7 to M28	9,200	Databases. Analytical tools. Video or virtual conferencing. Content management system (templates for storing information and tracking tools for changes). Storytelling.
Project Technical Assistant (PTA) collection of knowledge materials from outputs 3.1, 3.2, 3.3, 3.4	Quarterly, from M18 to M29	9,000	Knowledge platform. Video or virtual conferencing.
Project Technical Assistant (PTA) collection of knowledge materials from outputs 4.1, 4.2, 4.3, 4.4	Quarterly, from M19 to M33	7,500	Content management system (templates for storing information and tracking tools for changes).
Project Technical Assistant (PTA) monitoring of KM (outputs 5.1, 5.2)	Quarterly, from M6 to M36	2,000	Database. Knowledge platform.
Profonanpe (project administration team)	Quarterly, from M6 to M42	10,500	Content management system (templates for storing information and tracking tools for changes).
Management of publications and website (outputs 1.4, 2.5, 3.2)	Ongoing, from M6 until project closure	20,400	Knowledge platform. Content management system (templates for storing information and tracking tools for changes).
	Total	**66,900**	

Table 1: Knowledge Management Plan

GEF Core Indicators:

194. The GEF Core indicators included as Annex 13 will be used to monitor global environmental benefits and will be updated for reporting to the GEF prior to TE. Note that the project team is responsible for updating the indicator status. The updated monitoring data should be shared with TE consultants prior to required evaluation missions, so these can be used for subsequent groundtruthing. The methodologies to be used in data collection have been defined by the GEF and are available on the GEF website.

Terminal Evaluation (TE):

195. An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The terms of reference, the evaluation process and the final TE report will follow the standard templates and guidance for GEF-financed projects available on the UNDP Evaluation Resource Center.

196. The evaluation will be 'independent, impartial and rigorous'. The evaluators that will be hired to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project being evaluated.

197. The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the BPPS/GEF Directorate.

198. The final TE report and TE TOR will be publicly available in English and posted on the UNDP ERC by *8 March 2025*. A management response to the TE recommendations will be posted to the ERC within six weeks of the TE report's completion.

Final Report:

199. The project's terminal GEF PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

200. The project's terminal GEF PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

201. Agreement on intellectual property rights and use of logo on the project's deliverables and disclosure of information**:** To accord proper acknowledgement to the GEF for providing grant funding, the GEF logo will appear together with the UNDP, Government of Peru and MINAM logos on all promotional materials, other written materials like publications developed by the project, and project hardware. Any citation on publications regarding projects funded by the GEF will also accord proper acknowledgement to the GEF. Information will be disclosed in accordance with relevant policies notably the UNDP Disclosure Policy[35] and the GEF policy on public involvement[36].

Monitoring and Evaluation Plan and Budget: This M&E plan and budget provides a breakdown of costs for M&E activities to be led by the Project Management Unit during project implementation. These costs are included in Component 5 of the Results Framework and TBWP. For ease of reporting M&E costs, please include all costs reported in the M&E plan under the one technical component. The oversight and participation of the UNDP Country Office/Regional technical advisors/HQ Units are not included as these are covered by the GEF Fee.		
GEF M&E requirements	**Indicative costs (US$)**	**Time frame**
Inception Workshop	*3,500*	Within 60 days of CEO endorsement of this project.
Inception Report	*None*	Within 90 days of CEO endorsement of this project.
Project monitoring platform development &compilation of KM products	*20,000*	Annually and at mid-point and closure.
GEF Project Implementation Report (PIR)	*None*	Annually typically between June-August
Supervision missions	*None*	Annually
Monitoring of social and environmental safeguards management measures, stakeholder engagement activities and gender action plan	*30,800*	Annually
Independent Terminal Evaluation (TE)	*30,000*	*31/03/2024*
TOTAL indicative COST	*84,300*	

Table 2: M&E Plan and Budget

[35] See http://www.undp.org/content/undp/en/home/operations/transparency/information_disclosurepolicy/

[36] See https://www.thegef.org/gef/policies_guidelines

VII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

Roles and responsibilities of the project's governance mechanism:

Implementing Partner:

202. The Implementing Partner for this project is Ministry of the Enviroment (MINAM).

203. The Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in this signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document.

204. The Implementing Partner is responsible for executing this project. Specific tasks include:

- Project planning, coordination, management, monitoring, evaluation and reporting. This includes providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
- Risk management as outlined in this Project Document;
- Procurement of goods and services, including human resources;
- Financial management, including overseeing financial expenditures against project budgets;
- Approving and signing the multiyear work plan;
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

Responsible Parties:

205. The Peruvian Trust Fund for National Parks and Protected Areas (PROFONANPE), a private sector environment fund not for profit, has been selected to provide execution support. PROFONANPE will develop support activities in the execution of the project in components 1, 2, 3, 3, 4 and 5, providing operational assistance including, but not limited, to administrative and procurement functions for the entire project (Detailed activities in the ToR of project administrator). The Project Management Unit will be located within the Ministry of Environment.

206. PROFONANPE is the most important environmental management fund at the national level, and as such it will participate in the technical committees of the project in order to identify possibilities to accelerate processes or mobilize additional co-financing resources - particularly from the private sector - that could enhance project interventions.

Project stakeholders and target groups:

207. All national stakeholders will be represented and actively participate in the implementation and supervision of the project activities, and will be entitled to provide guidance to the project through their participation

at the Advisory Committee. Key project stakeholders will be engaged in the project decision making processes through their participation as full members or observers in the Project Board (PB). Project consultants will be required to identify and involve the target groups and stakeholders relevant to their activity throughout their technical consultancy services.

UNDP:

208. UNDP is accountable to the GEF for the implementation of this project. This includes oversight of project execution to ensure that the project is being carried out in accordance with agreed standards and provisions. UNDP is responsible for delivering GEF project cycle management services comprising project approval and start-up, project supervision and oversight, and project completion and evaluation. UNDP is also responsible for the Project Assurance role of the Project Board/Steering Committee.

209. A firewall will be maintained between the delivery of project oversight and quality assurance performed by UNDP and charged to the GEF Fee and any support to project execution performed by UNDP (as requested by and agreed to by both the Implementing Partner and GEF) and may be charged to the GEF project management costs (only if approved by GEF). The segregation of functions and firewall provisions for UNDP in this case is described in the next section.

Project organisation structure:

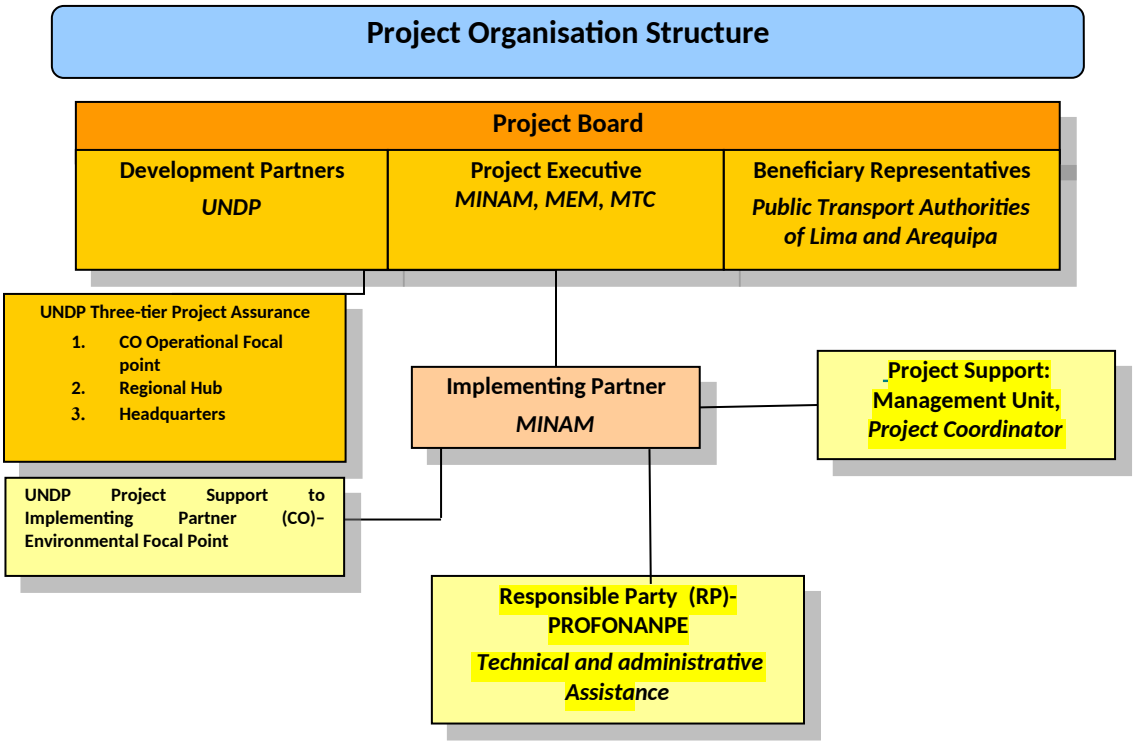


Figure 2: Project Organisation Structure

210. UNDP project support: The Implementing Partner and GEF OFP have requested UNDP to provide support services for the full duration of the project, and the GEF has agreed for UNDP to provide such execution support services provided however that such costs will not be charged to the project budget but will be fully covered by non-GEF resources. The execution support services – whether financed from the project budget or other sources - have been set out in detail and agreed between UNDP Country Office and the Implementing Partner in a Letter of Agreement (LOA). This LOA is attached to this Project Document.

To ensure the strict independence required by the GEF and in accordance with the UNDP Internal Control Framework, these execution services will be delivered independent from the GEF-specific oversight and quality assurance services.

Section 3: Segregation of duties and firewalls vis-à-vis UNDP representation on the project board:

211. As noted in the Minimum Fiduciary Standards for GEF Partner Agencies, in cases where a GEF Partner Agency (i.e. UNDP) carries out both implementation oversight and execution of a project, the GEF Partner Agency (i.e. UNDP) must separate its project implementation oversight and execution duties, and describe in the relevant project document a: 1) Satisfactory institutional arrangement for the separation of implementation oversight and executing functions in different departments of the GEF Partner Agency; and 2) Clear lines of responsibility, reporting and accountability within the GEF Partner Agency between the project implementation oversight andexecution functions.

212. The Project Board (also called Project Steering Committee) is responsible for taking corrective action as needed to ensure the project achieves the desired results. In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition.

213. In case consensus cannot be reached within the Board, the UNDP Resident Representative (or their designate) will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed.

214. The Project Board will consist of representatives of the Ministry of Environment, the Ministry of Energy and Mines, the Ministry of Transport and Commmunications, Local transport authorities from Lima and Arequipa, and UNDP.

215. Specific responsibilities of the Project Board include:

- Provide overall guidance and direction to the project, ensuring it remains within any specified constraints;
- Address project issues as raised by the project manager;
- Provide guidance on new project risks, and agree on possible mitigation and management actions to address specific risks;
- Agree on project manager's tolerances as required, within the parameters set by UNDP-GEF, and provide direction and advice for exceptional situations when the project manager's tolerances are exceeded;
- Advise on major and minor amendments to the project within the parameters set by UNDP-GEF;
- Ensure coordination between various donor and government-funded projects and programmes;
- Ensure coordination with various government agencies and their participation in project activities;
- Track and monitor co-financing for this project;
- Review the project progress, assess performance, and appraise the Annual Work Plan for the following year;
- Appraise the annual project implementation report, including the quality assessment rating report;
- Ensure commitment of human resources to support project implementation, arbitrating any issues within the project;
- Review combined delivery reports prior to certification by the implementing partner;
- Provide direction and recommendations to ensure that the agreed deliverables are produced satisfactorily according to plans;
- Address project-level grievances;
- Approve the project Inception Report, Mid-term Review and Terminal Evaluation reports and corresponding management responses;
- Review the final project report package during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

- Ensure highest levels of transparency and take all measures to avoid any real or perceived conflicts of interest.

216. The composition of the Project Board must include the following roles:

a. Project Executive: Is an individual who represents ownership of the project and chairs the Project Board. The Executive is normally the national counterpart for nationally implemented projects. The Project Executive is *the Director General de Calidad Ambiental (MINAM).*

b. Beneficiary Representative(s): Individuals or groups representing the interests of those who will ultimately benefit from the project. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. The Beneficiary representatives are the Public Transport Authorities of Arequipa and Lima.

c. Development Partner(s): Individuals or groups representing the interests of the parties concerned that provide funding and/or technical expertise to the project. The Development Partner is the UNDP Resident Representative.

d. Project Assurance: UNDP performs the quality assurance and supports the Project Board and Project Management Unit by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed, and conflict of interest issues are monitored and addressed. The Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. UNDP provides a three – tier oversight services involving the UNDP Country Offices and UNDP at regional and headquarters levels. Project assurance is totally independent of project execution.

217. **Advisory committee**. If considered necessary by the PB, the project will also be advised by stakeholders and representatives from the civil society organizations related to electro-mobility in Peru, a previous screening of suitable institutions for integrating the advisory committee has been made when requesting the co-financing support for this project. This Advisory committee should not be confused with the e-mobility network the project will support under component 3, although it can be expected that the most active members of the network will participate in the Advisory committee. While the former is intended to play a role in the governance of the project, the latter is a flexible network to facilitate the interaction among any stakeholders in Peru interested in the promotion of e-mobility. PROFONANPE will take part of the technical discussions in the advisory committee.

218. **Coordination space.** It will primarily include national and subnational governmental institutions, and may also extend participation to the private sector, academia and civil society (which are expected to coordinate more closely under the e-mobility network established through Output 3.1). This coordination space will constitute a meeting point for the various ministries and public institutions involved in e-mobility, replacing the current non-binding discussion platforms established by each ministry related to e-mobility, which are convened by different public authorities with rigid sector divisions and which focus on partial issues: vehicle technical approval, tariffs, charging standards, fiscal incentives, among others. The coordination space will facilitate the approval of strategies, policies and regulations, and will also serve as an entry point for partnerships with non-governmental stakeholders. Two stages are envisaged: initially, the key stakeholders will be convened with support from the project to establish and agree upon a mandate and working plan for the coordination space. Once finalized, the joint working plan will be delivered to the various governmental institutions involved in the official ratification of the space, establishing the adequate structure within the government.

219. **Project extensions:** The UNDP Resident Representative and the UNDP-GEF Executive Coordinator must approve all project extension requests. Note that all extensions incur costs and the GEF project budget

cannot be increased. A single extension may be granted on an exceptional basis and only if the following conditions are met: one extension only for a project for a maximum of six months; the project management costs during the extension period must remain within the originally approved amount, and any increase in PMC costs will be covered by non-GEF resources; the UNDP Country Office oversight costs in excess of the CO's Agency fee specified in the DOA during the extension period must be covered by non-GEF resources.

VIII. FINANCIAL PLANNING AND MANAGEMENT

220. The total cost of the project is *USD 16,644,697*. This is financed through a GEF grant of *USD 1,784,862* administered by UNDP and additional support of *USD 14,859,835*. UNDP, as the GEF Implementing Agency, is responsible for the oversight of the GEF resources and the cash co-financing transferred to UNDP bank account only.

Confirmed Co-financing:

221. The actual realization of project co-financing will be monitored during the terminal evaluation process and will be reported to the GEF. Note that all project activities included in the project results framework that will be delivered by co-financing partners (even if the funds do not pass through UNDP accounts) must comply with UNDP's social and environmental standards. Besides the co-financing letters summarized in the table below, the project has received supporting letters from Lima and Callao Public Transport Authority (*Autoridad de Transporte Urbano para Lima y Callao*, *ATU)* and the Ministry of Transport and Communications (*Ministerio de Transportes y Comunicaciones*, *MTC)*. Co-financing will be used for the following project activities/outputs:

Co-financing source	Co-financing type	Co-financing amount	Planned Co-financing Activities/Outputs	Risks	Risk Mitigation Measures
Ministerio del Ambiente	In-Kind	379,818	Activities to support project monitoring and management	Emergence of other political priorities	Keep alignment of project with governments' energy and transport priorities
Ministerio de Energía y Minas	In-Kind	2,000,000	Technical and operational support to on electric vehicles, charging stations and associated regulations. Consultations through the Technical Committee on Transport Standards. Support to dissemination and capacity-building events. Drafting of EV regulations. Support to project management	Emergence of other political priorities	Keep alignment of project with governments' energy and transport priorities
Municipalidad de Arequipa	In-Kind	2,670,623	New PT infrastructure (SIT)	Infrastructure investment delayed	Identification of alternative PT investments
Engie Energía Perú S.A.	In-Kind	2,000,000	Technical staff contribution and provision of urban and 3-W vehicles to be associated to the pilots	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility
QEY Tech Perú SAC	Equity Investment	1,500,000	Corporate activities and investments to promote and deploy e-mobility in Peru	Slow deployment of the company's e-mobility	Project creates favorable conditions to e-

Co-financing source	Co-financing type	Co-financing amount	Planned Co-financing Activities/Outputs	Risks	Risk Mitigation Measures
				portfolio	mobility
Asociación de Emprendedores para el Desarrollo e impulso del vehículo Eléctrico en el Perú - AEDIVE	In-Kind	1,400,000	Activities of the companies and professionals associated to this network to promote e-mobility through regulations, capacity building and business models	Slow deployment of e-mobility	Project creates favorable conditions to e-mobility
Motores Diesel Andinos, S.A. - MODASA	Equity Investment	700,000	R&D on electric buses, aiming at manufacturing EVs in Peru. Business models linked to the e-bus segment. Professional training and capacity building on e-bus manufacturing, operation and maintenance	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility
Agencia Francesa de Desarrollo - ADF	Grant	600,536[37]	Design of the Sustainable Urban Mobility Plan in Arequipa (with financial support from EUROCLIMA)	Slow progress in the design of the SUMP	Project team can be involved in SUMP and approach relevant local stakeholders
SINOMAQ, S.A.	Equity Investment	452,000	Public charging infrastructure to be installed in within the company's premises in Lima. New workshop for e-buses. Training and capacity building on operation and maintenance of EVs. Participation of company staff in promotion of e-mobility in Peru, networking and capacity-building activities. E-bus pilot project.	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility
Universidad de Ingeniería y Tecnología - UTEC	In-Kind	425,250	Research studies on air quality, e-mobility and other related topics in Peruvian cities	Uncertain sources for funding	Project's promotion of e-mobility to raise potential funders' interest
Integra Arequipa SAC	In-Kind	388,000	Participation in the coordination space to develop e-mob regulations and business plans. Contribution to the design and implementation of the project pilots. Participation in capacity building activities on e-mobility	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility
Oficina de Cooperación Suiza - COSUDE	Grant	349,097[38]	Promotion of regulatory initiatives, e-mobility strategy. Analysis of e-mobility impacts (including health). Dissemination targeting consumers, decision-makers and professionals. Business models and public procurement for clean urban mobility. Networking activities	None	None

[37] EUR 500,000 (Exchange rate 1 USD = 0.83259 EUR, 20 April 2021, https://www1.oanda.com/lang/es/currency/converter/)

[38] CHF 320,000 (Exchange rate 1 USD = 0.91665 CHF, 20 April 2021, https://www1.oanda.com/lang/es/currency/converter/)

Co-financing source	Co-financing type	Co-financing amount	Planned Co-financing Activities/Outputs	Risks	Risk Mitigation Measures
SENATI	In-Kind	94,311	Implementation of training and capacity building programmes on e-mobility at their premises in Lima and Arequipa	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility
WUITO	In-Kind	88,500	Training on manufacturing and maintenance of electric 3W; business model and financial analysis to undertake the manufacturing of 3W taxis in Peru	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility
ABB	In-Kind	77,000	Networking and technical support, as well as demonstrations of fast-charging infrastructure	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility
CAVI Industrial SAC	Equity Investment	62,000	Two pilots foreseen to manufacture and operate 3W taxis in Lima	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility
DEUMAN	In-Kind	41,900	Technical consultancy on e-mobility, with a focus on 3W (pilot in progress, results to be analyzed with the project team), and development of business models	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility
Fundación Transitemos	In-Kind	39,000	Public participation (workshops…); regular monitoring and assessment of urban mobility in Lima and other cities. Publication on e-mobility	Emergence of other mobility priorities requiring the scarce resources available	Project raises and keeps the government's and public's interest in e-moblity
UNDP	In-Kind	200,000	Technical support to the preparation of key deliverables and reports; stakeholders' engagement	None	None
ENEL X Peru SAC	In-Kind	382,000	Chargers to be installed in the country since 2021 , capacity building activities and networking	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility
Profonanpe	In-Kind	60,000	Specialized technical assistance for project management activities	None	None
CITE energía	In-Kind	949,800	Awareness-raising, technological developments, technical performance assessment, business model developments and end-of-life management	Slow deployment of the company's e-mobility portfolio	Project creates favorable conditions to e-mobility

Table 3: Confirmed Co-financing

222. As explained in the responsible parties section**, t**he Implementing Partner and GEF OFP in Peru have requested UNDP to provide support services without any cost to IP for the full duration of the project. The GEF has agreed to this request. The GEF execution **support letter** (signed by the GEF OFP) detailing these support services is included as an annex. To ensure the strict independence required by the GEF and in accordance with the UNDP Internal Control Framework, these execution services will be delivered independent from the GEF-specific oversight and quality assurance services (i.e. not done by same person to avoid conflict of interest).

Budget Revision and Tolerance:

223. As per UNDP requirements outlined in the UNDP POPP, the project board will agree on a budget tolerance level for each plan under the overall annual work plan allowing the project manager to expend up to the tolerance level beyond the approved project budget amount for the year without requiring a revision from the Project Board.

224. Should the following deviations occur, the Project Manager/CTA and UNDP Country Office will seek the approval of the BPPS/GEF team to ensure accurate reporting to the GEF:

    a) Budget re-allocations among components in the project budget with amounts involving 10% of the total project grant or more;
    b) Introduction of new budget items that exceed 5% of original GEF allocation.

225. Any over expenditure incurred beyond the available GEF grant amount will be absorbed by non-GEF resources (e.g. UNDP TRAC or cash co-financing).

Audit:

226. The project will be audited as per UNDP Financial Regulations and Rules and applicable audit policies. Audit cycle and process must be discussed during the Inception workshop. If the Implementing Partner is an UN Agency, the project will be audited according to that Agencies applicable audit policies.

Project Closure:

227. Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP. All costs incurred to close the project must be included in the project closure budget and reported as final project commitments presented to the Project Board during the final project review. The only costs a project may incur following the final project review are those included in the project closure budget.

Operational completion:

228. The project will be operationally completed when the last UNDP-financed inputs have been provided and the related activities have been completed. This includes the final clearance of the Terminal Evaluation Report (that will be available in English) and the corresponding management response, and the end-of-project review Project Board meeting. **Operational closure must happen at the end date calculated by the approved duration after the Project Document signature or at the revised operational closure date as approved in the project extension. Any expected activity after the operational date requires project extension approval**. The Implementing Partner through a Project Board decision will notify the UNDP Country Office when operational closure has been completed. At this time, the relevant parties will have already agreed and confirmed in writing on the arrangements for the disposal of any equipment that is still the property of UNDP.

Transfer or disposal of assets:

229. In consultation with the Implementing Partner and other parties of the project, UNDP is responsible for deciding on the transfer or other disposal of assets. Transfer or disposal of assets is recommended to be reviewed and endorsed by the project board following UNDP rules and regulations. Assets may be transferred to the government for project activities managed by a national institution at any time during

the life of a project. In all cases of transfer, a transfer document must be prepared and kept on file[39]. The transfer should be done before Project Management Unit complete their assignments.

Financial completion (closure):

230. The project will be financially closed when the following conditions have been met: a) the project is operationally completed or has been cancelled; b) the Implementing Partner has reported all financial transactions to UNDP; c) UNDP has closed the accounts for the project; d) UNDP and the Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).

231. The project will be financially completed **within 6 months of operational closure or after the date of cancellation**. Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to the BPPS/GEF Unit for confirmation before the project will be financially closed in Quantum by the UNDP Country Office.

Refund to GEF:

232. Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the BPPS/GEF Directorate in New York. No action is required by the UNDP Country Office on the actual refund from UNDP project to the GEF Trustee.

[39] See https://popp.undp.org/_layouts/15/WopiFrame.aspx?sourcedoc=/UNDP_POPP_DOCUMENT_LIBRARY/Public/PPM_Project%20Management_Closing.docx&action=default.

IX. TOTAL BUDGET AND WORK PLAN

Total Budget and Work Plan			
Quantum Award ID:	00115399.1	Quantum Output Project ID:	00115399
Quantum Proposal or Award Title:	Movilidad Electrica PRODOC		
Quantum Business Unit	PER10		
Quantum Primary Output Project Title	Movilidad Electrica PRODOC		
UNDP-GEF PIMS No.	6384		
Implementing Partner	*Ministry of Environment (MINAM)*		

Quantum Activity (GEF Component)	**Quantum Implementing Agent (Resp. Party[40] , IP, or UNDP)**	**Quantum Fund ID**	**Donor Name**	**Quantum Budgetary Account Code**	**Quantum Budget Account Description**	**Amount Year 2021 (USD)**	**Amount Year 2022 (USD)**	**Amount Year 2023 (USD)**	**Amount Year 2024 (USD)**	**Total (USD)**	**See Budget Note:**
COMPONENT 1 *(as per Table B of CEO endorsement request)*	**MINAM**	**62000**	**GEF Trustee**	71800	Contractual Svcs-indiv ImpPtnr	23,600	21,000	14,000	4,000	62,600	*1*
				72100	Contractual services - Companies	120,000	80,000	0	0	200,000	*2*
				71600	Travel	14,000	14,000	14,000	8,000	50,000	*3*
				75700	Training, workshop, conference	66,500	10,500	10,500	0	87,500	*4*
				74200	Audio Visual and Printing Costs	7,000	0	0	0	7,000	*5*
					sub-total GEF	**231,100**	**125,500**	**38,500**	**12,000**	**407,100**	
					Total Outcome 1	**231,100**	**125,500**	**38,500**	**12,000**	**407,100**	
COMPONENT 2: *(as per Table B of CEO endorsement request)*	**MINAM**	**62000**	**GEF Trustee**	71800	Contractual Svcs-indiv ImpPtnr	9,400	7,000	20,000	2,000	38,400	*6*
				72100	Contractual services - Companies	45,902	683,000	0	0	728,902	*7*
				71600	Travel	2,000	2,000	2,000	0	6,000	*8*
				75700	Training, wokshop, conference	6,750	6,750	6,750	0	20,250	*9*
				74200	Audio Visual and Printing Costs	750	750	750	0	2,250	*10*
					sub-total GEF	**64,802**	**699,500**	**29,500**	**2,000**	**795,802**	

[40]*Only the responsible parties to be created as Quantum Implementing Agent as part of the COAs should be entered here. Sub-level responsible parties reporting directly to NIM Implementing Partners should not entered here. For example, if under NIM, UNOPS signs LOA with the IP to manage component 2, and a department of Ministry X will manage component 3, this means that UNOPS will be listed as the responsible party under component 2. The rest of the components will list the IP as the responsible party.*
Only cash co-financing actually transferring to UNDP bank accounts and to be budgeted and used under this project should be entered in the main TBWP table and in Quantum. Additional support should only be shown in the summary table.

Quantum Activity (GEF Component)	Quantum Implementing Agent (Resp. Party , IP, or UNDP)	Quantum Fund ID	Donor Name	Quantum Budgetary Account Code	Quantum Budget Account Description	Amount Year 2021 (USD)	Amount Year 2022 (USD)	Amount Year 2023 (USD)	Amount Year 2024 (USD)	Total (USD)	See Budget Note:
					Total Outcome 2	**64,802**	**699,500**	**29,500**	**2,000**	**795,802**	
COMPONENT 3: *(as per Table B of CEO endorsement request)*	**MINAM**	**62000**	**GEF Trustee**	71800	Contractual Svcs-indiv ImpPtnr	11,000	15,000	14,000	10,000	50,000	*11*
				72100	Contractual services - Companies	5,000	80,000	65,000	0	150,000	*12*
					sub-total GEF	**16,000**	**95,000**	**79,000**	**10,000**	**200,000**	
					Total Outcome 3	**16,000**	**95,000**	**79,000**	**10,000**	**200,000**	
COMPONENT 4: *(as per Table B of CEO endorsement request)*	**MINAM**	**62000**	**GEF Trustee**	71800	Contractual Svcs-indiv ImpPtnr	4,000	11,000	12,000	8,400	35,400	*13*
				72100	Contractual services - Companies	10,000	25,000	55,000	0	90,000	*14*
				75700	Training, workshop, conference	0	4,500	4,500	0	9,000	*15*
				74200	Audio Visual and Printing Costs	0	500	500	0	1,000	*16*
					sub-total GEF	**14,000**	**41,000**	**72,000**	**8,400**	**135,400**	
					Total Outcome 4	**14,000**	**41,000**	**72,000**	**8,400**	**135,400**	
COMPONENT 5: KM and M&E[41] *(as per Table B of CEO endorsement request)*	**MINAM**	**62000**	**GEF Trustee**	71800	Contractual Svcs-indiv ImpPtnr	6,000	13,600	4,800	6,400	30,800	*17*
				72100	Contractual services - Companies	3,500	20,000	0	0	23,500	*18*
				71200	International consultants	0	0	0	30,000	30,000	*19*
					sub-total GEF	**9,500**	**33,600**	**4,800**	**36,400**	**84,300**	
					Total Outcome 5	**9,500**	**33,600**	**4,800**	**36,400**	**84,300**	
PROJECT MANAGEMENT UNIT[42] (This is not to appear as an Outcome in the Results Framework)	**MINAM**	**62000**	**GEF Trustee**	71800	Contractual Svcs-indiv ImpPtnr	6,000	4,400	7,200	5,200	22,800	*20*
				72100	Contractual services - Companies	30,000	30,000	30,000	15,000	105,000	*21*
				72500	Supplies	2,000	2,000	2,000	1,250	7,250	*22*
				72200	Equipment and furniture	4,176	1,392	1,392	0	6,960	*23*
				72800	Information technology equipment	3,000	1,000	1,000	0	5,000	*24*
				74100	Professional services	2,000	2,000	2,000	2,000	8,000	*25*
				74500	Miscellaneous expenses	2,000	2,000	2,000	1,250	7,250	*26*
					sub-total	**49,176**	**42,792**	**45,592**	**24,700**	**162,260**	
					Total Management	**49,176**	**42,792**	**45,592**	**24,700**	**162,260**	
					PROJECT TOTAL	**384,578**	**1,037,392**	**269,392**	**93,500**	**1,784,862**	

[41] *This must include the M&E budget included in section IV Monitoring and Evaluation Plan.*

[42] *Should not exceed 5% of total project budget for FSPs and 10% for MSPs. PMU costs will be used for the following activities: Audit; Full time or part time project manager (and or coordinator); Full time or part time project administrative/finance assistant; Travel cost of the PMU project staff; Other General Operating Expenses such as rent, computer, equipment, supplies, etc. to support the PMU; UNDP CO Support Costs if requested by Government Implementing Partner and GEF OFP and agreed by GEF Secretariat;.*

Table 4: Total Project Budget

Summary of Funds: [43]

	Amount Year 1	Amount Year 2	Amount Year 3	Amount Year 4	Total
GEF grant administered by UNDP	***384,578***	***1,037,392***	***269,392***	***93,500***	***1,784,862***
Ministerio del Ambiente	108,519	108,519	108,519	54,261	379,818
Ministerio de Energía y Minas	571,429	571,429	571,429	285,713	2,000,000
Municipalidad Provincial de Arequipa	763,035	763,035	763,035	381,518	2,670,623
Engie Energía Perú S.A.	1,000,000	500,000	500,000	-	2,000,000
QEY Tech Perú SAC	428,571	428,571	428,571	214,287	1,500,000
Asociación de Emprendedores para el Desarrollo e impulso del vehículo Eléctrico en el Perú - AEDIVE	400,000	400,000	400,000	200,000	1,400,000
Motores Diesel Andinos, S.A. - MODASA	200,000	200,000	200,000	100,000	700,000
Agencia Francesa de Desarrollo - ADF	171,582	171,582	171,582	85,790	600,536[44]
SINOMAQ, S.A.	129,143	129,143	129,143	64,571	452,000
Universidad de Ingeniería y Tecnología - UTEC	121,500	121,500	121,500	60,750	425,250
Integra Arequipa SAC	110,857	110,857	110,857	55,429	388,000
Oficina de Cooperación Suiza - COSUDE	99,742	99,742	99,742	49,871	349,097[45]
SENATI	56,587	28,293	9,431	-	94,311
WUITO	25,286	25,286	25,286	12,642	88,500
ABB	22,000	22,000	22,000	11,000	77,000
CAVI Industrial SAC	17,714	17,714	17,714	8,858	62,000
DEUMAN	11,971	11,971	11,971	5,987	41,900
Fundación Transitemos	11,143	11,143	11,143	5,571	39,000
UNDP	55,000	60,000	60,000	25,000	200,000

[43] *Summary table should include all financing of all kinds: GEF financing, co-financing, cash, in-kind, etc…*

[44] EUR 500,000 (Exchange rate 1 USD = 0.83259 EUR, 20 April 2021, https://www1.oanda.com/lang/es/currency/converter/)

[45] CHF 320,000 (Exchange rate 1 USD = 0.91665 CHF, 20 April 2021, https://www1.oanda.com/lang/es/currency/converter/)

Enel X Peru SAC	382,000	-	-	-	382,000
Profonanpe	17,143	17,143	17,143	8,571	60,000
CITE energía	265,000	275,000	275,000	134,800	949,800
Total co-financing	**4,968,222**	**4,072,929**	**4,054,067**	**1,764,617**	**14,859,835**
TOTAL	**5,352,800**	**5,110,320**	**4,323,458**	**1,858,119**	**16,644,697**

Table 5: Summary of funds

Table 6: Budget notes

Budget note #	**Comments:** Budget note should be output based rather than input based. Even for individual consultants' outputs of the consultants must be clear. Include cost breakdown and calculation basis (e.g. daily fee and number of days/weeks, unit cost and number), as well as a total amount for the budget line.
1	11.5 person-month (USD 46,000) dedicated by PM (@USD 4,000 per month) and 8.3 person-month (USD 16,600) dedicated by Technical Assistant (@USD 2,000 per month) to activities 1.1.1, 1.1.2, 1.2.6, 1.2.7, 1.3.1, 1.3.2, 1.3.3, 1.4.1 and 1.4.2.
2	USD 60,000 for Technical Assistance contract to prepare the e-mobility strategy (output 1.2, all activities); USD 20,000 for TA contract to prepare communication and capacity-building plans; USD 110,000 for TA contract on technical, legal and fiscal regulations, including technical guidelines (1.3.1, 1.3.2, 1.3.3); USD 10,000 to carry out communication activites on institutionalization of e-mobility (1.4.1).
3	USD 50,000 to attend working groups and workshops of the global e-mobility programme and its regional platform (1.1.2). The travel budget (USD 50,000) includes 12 international trips to participate (one participant each) at 3 meetings of each of the four Working Groups @ USD 1,900, 18 regional trips to participate (2 participants each) at the kick-off meeting, 3 regional group meetings, 2 capacitation workshops, 2 marketplace events and 1 replication event, @USD 1,400 and USD 2,000 for travel contingencies. Due to COVID-19 restrictions, some activities could be held on-line.
4	USD 24,500 for organization of meetings and workshops for the e-mobility coordination space (1.1.1); USD 63,000 for training workshops (1.4.2 and 1.4.3);
5	USD 7,000 for publications on the institutionalization of e-mobility
6	4.95 person-month (USD 19,808) dedicated by PM (@USD 4,000 per month) and 9.25 person-month (USD 18,592) dedicated by Technical Assistant (@USD 2,000 per month) to activities 2.1.1, 2.1.2, 2.1.3, 2.2.1, 2.2.2, 2.2.3,, 2.3.1, 2.3.2 and 2.3.3, 2.4.1 and 2.5.1
7	USD 20,000 for technical assistance contract to support the design, implementation, monitoring and evaluation of the demonstrations (2.1.1, 2.1.2, 2.2.3, 5.1.2, 2.3.3., 5.1.3); USD 10,000 for TA to design and implement multimedia campaigns on harassment, discrimination and violence (2.1.3); USD 578,000 to provide the public transport services with 2 electric buses in each of the 2 cities and USD 100,000 to provide public transport services with 10 electric 3-wheelers in each of the 2 cities (2.2.2, 2.2.3, 5.1.2, 2.3.2, 2.3.3, 5.1.3); USD 20,902 to complete communication activities on the demonstration results (including project website design and management) (2.5.2).
8	USD 6,000 for travel to Lima and Arequipa related to the demonstrations (2.2.2, 2.2.3, 5.1.2, 2.3.2, 2.3.3, 5.1.3)
9	USD 20,250 for professional and technical training on electric vehicles (driving, maintenance, charging stations, etc.) (2.4.1).
10	USD 2,250 for publication of demonstration-related materials
11	8 person-month (USD 32,000) dedicated by PM (@USD 4,000 per month) and 9 person-month (USD 18,000) dedicated by Technical Assistant (@USD 2,000 per month) to activities 3.1.1, 3.1.2, 3.2.1, 3.2.2, 3.3.1, 3.3.2, 3.3.3, 3.4.1 and 3.4.2.
12	USD 130,000 for technical assistance contract to provide EV market analysis, scenario building, business models and financial schemes (3.1.1. 3.1.2, 3.3.1, 3.3.2, 3.4.1 and 3.4.2); USD

Budget note #	**Comments:** Budget note should be output based rather than input based. Even for individual consultants' outputs of the consultants must be clear. Include cost breakdown and calculation basis (e.g. daily fee and number of days/weeks, unit cost and number), as well as a total amount for the budget line.
	20,000 for TA contract to provide communication activities (3.2.2);
13	5.1 person-month (USD 20,400) dedicated by PM (@USD 4,000 per month) and 7.5 person-month (USD 15,000) dedicated by Technical Assistant (@USD 2,000 per month) to activities 4.1.1, 4.1.2, 4.2.1, 4.2.2, 4.3.1, 4.3.2, 4.4.1 and 4.4.2).
14	USD 90,000 for technical assistance contract for designing and implementing an EPR approach to ELV regulations (including batteries) (4.1.1, 4.2.1, 4.2.2, 4.2.3, 4.3.1, 4.3.2 and 4.4.1) as well as technical and managerial support to producers and importers (4.5.2 and 4.5.3).
15	USD 9,000 for professional training on ELV management, including batteries (4.5.1).
16	USD 1,000 in publications on ELV management (4.5.1)
17	USD 20,000 for monitoring of core indicators (activity 5.1.1) and 2.7 person-month 10,800 dedicated by PM (@USD 4,000 per month)
18	USD 13,500 for the inception workshop; USD 10,000 for M&E of demonstration in Lima (5.1.2) and Arequipa (5.1.3)
19	Contract for the terminal evaluation
20	5.7 person-month (USD 22,800) dedicated by PM (@USD 4,000 per month) to project management activities.
21	Profonanpe assumes the project administration role with an internal team and budget of USD 105,000, considering 42 months dedicated by the Project Administrator Team (@USD 2,500 per month).
22	USD 7,250 office supplies
23	USD 6,960 Furniture and other equipment for PMU office
24	USD 5,000 Computers for PMU and other hardware equipment
25	USD 8,000 Annual audits
26	USD 7,250 Other PMU office expenses

X. LEGAL CONTEXT

Option a. Where the country has signed the Standard Basic Assistance Agreement (SBAA)

233. This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of Peru and UNDP, signed on 19 January 1960. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

234. This project will be implemented by Ministerio del Ambiente (MINAM) ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

235. The designations employed and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the Secretariat of the United Nations or UNDP concerning the legal status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

XI. RISK MANAGEMENT

236. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:

a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;

b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

237. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

238. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via https://www.un.org/securitycouncil/content/un-sc-consolidated-list .

239. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

(a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").

(b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment.

240. In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

i. Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;

ii. Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;

iii. Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;

iv. Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and

v. Promptly and confidentially record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

b) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

241. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

242. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

243. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

244. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds. The Implementing Partner will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

245. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

246. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

247. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

248. Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

249. UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

250. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

251. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

252. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management Standard Clauses" are included, *mutatis mutandis*, in all sub-contracts or sub-agreements entered into further to this Project Document.

XII. MANDATORY ANNEXES

1. GEF Budget Template (available from BPPS NCE-VF)
2. Project map and geospatial coordinates of the project area
3. Multiyear Work Plan
4. Monitoring Plan
5. Social and Environmental Screening Procedure (SESP)
6. UNDP Quantum Risk Register
7. Overview of technical consultancies/subcontracts
8. Stakeholder Engagement Plan
9. Gender Analysis and Gender Action Plan
10. Procurement Plan
11. GEF focal area specific annex (Estimates of Direct and Consequential GHG Emission Reductions and Beneficiaries)
12. Additional agreements (no additional agreements in this project)
13. GEF Core indicators
14. GEF Taxonomy
15. Funding scheme to public transport operators participating in e-mobility projects

The following Annexes must completed/prepared as separate documents. Most must be completed early in the PPG phase to help inform the design of the project. They must be made available to the LPAC members. They do not need to be submitted to the GEF and do not need to be part of the project document that is signed by the relevant parties.

16. Partners Capacity Assessment Tool and HACT assessment may not be required for EAs if less than US$300,000 will be transferred to IP
17. UNDP Project Quality Assurance Report (to be completed in UNDP online corporate planning system)

Annex 1: GEF Budget Template

The budget, following the GEF template, is presented in the Table below.

Expenditure Category	Detailed Description	Component (USDeq.)							Total (USDeq.)	Responsible Entity
		Component 1	*Component 2*	*Component 3*	*Component 4*	*Sub-Total*	*M&E*	*PMC*		
		Outcome 1	*Outcome 2*	*Outcome 3*	*Outcome 4*					
Goods	Furniture	0	0	0	0	0	0	6,960	6,960	MINAM
Goods	Publications on institutionalization of e-mobility	7,000	0	0	0	7,000	0	0	7,000	MINAM
Goods	Publications on project pilots	0	2,250	0	0	2,250	0	0	2,250	MINAM
Goods	Publications on end-of-life vehicle management	0	0	0	1,000	1,000	0	0	1,000	MINAM
Goods	Computers	0	0	0	0	0	0	5,000	5,000	MINAM
Contractual Services – Company	Draft e-mobility strategy, including economic, social, space and gender impacts, and gender plan	60,000	0	0	0	60,000	0	0	60,000	MINAM
Contractual Services – Company	Design, implementation, monitoring and evaluation of demonstrations	0	20,000	0	0	20,000	20,000	0	40,000	MINAM
Contractual Services – Company	Consultancy services on technical, legal and fiscal regulations of vehicles (including technical guidelines)	110,000	0	0	0	110,000	0	0	110,000	MINAM
Contractual Services – Company	EV market analysis, scenario building, business models and financial schemes	0	0	130,000	0	130,000	0	0	130,000	MINAM
Contractual Services – Company	Organization of meetings and workshops (inception workshop)	0	0	0	0	0	3,500	0	3,500	MINAM
Contractual Services – Company	Communication activities (knowledge management, communication plan, website management and publication of materials)	10,000	20,902	20,000	0	50,902	0	0	50,902	MINAM
Contractual Services – Company	Design and implementation of multimedia campaigns on harassment, discrimitation and violence	0	10,000	0	0	10,000	0	0	10,000	MINAM
Contractual Services – Company	Provision of services with 2 e-buses in each city	0	578,000	0	0	578,000	0	0	578,000	MINAM
Contractual Services –	Provision of services with 10 electric 3-	0	100,000	0	0	100,000	0	0	100,000	MINAM

Company	wheeler taxis in each city									
Contractual Services – Company	Design and implementation of a EPR approach to ELV regulations (including batteries)	0	0	0	90,000	90,000	0	0	90,000	MINAM
Contractual Services – Company	Communication and capacity building plan	20,000	0	0	0	20,000	0	0	20,000	MINAM
Contractual Services – Company	Project Administration services (Profonanpe)	0	0	0	0	0	0	105,000	105,000	MINAM
Contractual Services – Company	Independent financial audits	0	0	0	0	0	0	8,000	8,000	MINAM
International Consultants	Terminal evaluation	0	0	0	0	0	30,000	0	30,000	MINAM
Salary and benefits / Staff costs	Project Manager	46,000	20,000	32,000	20,400	118,400	26,800	22,800	168,000	MINAM
Salary and benefits / Staff costs	Technical assistant	16,600	18,400	18,000	15,000	68,000	4,000	0	72,000	MINAM
Trainings, Workshops, Meetings	Organization of meetings and workshops	24,500	0	0	0	24,500	0	0	24,500	MINAM
Trainings, Workshops, Meetings	Workshops and training in e-mobility	63,000	0	0	0	63,000	0	0	63,000	MINAM
Trainings, Workshops, Meetings	Professional and technical training on electric vehicles	0	20,250	0	0	20,250	0	0	20,250	MINAM
Trainings, Workshops, Meetings	Professional training on ELV management of EVs and their batteries	0	0	0	9,000	9,000	0	0	9,000	MINAM
Travel	Travel to attend working groups and workshops of the global e-mob programme	50,000	0	0	0	50,000	0	0	50,000	MINAM
Travel	Travel for meetings in Lima	0	3,000	0	0	3,000	0	0	3,000	MINAM
Travel	Travel for meetings in Arequipa	0	3,000	0	0	3,000	0	0	3,000	MINAM
Office Supplies	Office supplies	0	0	0	0	0	0	7,250	7,250	MINAM
Other Operating Costs	Office operations	0	0	0	0	0	0	7,250	7,250	MINAM
Grand Total		**407,100**	**795,802**	**200,000**	**135,400**	**1,538,302**	**84,300**	**162,260**	**1,784,862**	

Annex 2: Project map and geospatial coordinates of the project area

Geospatial coordinates Lima: 12°03'00.0"S 77°03'00.0"W
Geospatial coordinates Arequipa: 16°24'00.0"S 77°32'00.0"W



Annex 3: Multi Year Work Plan

Com-ponent	Out-comes	Outputs	Activities	Year 1				Year 2				Year 3				Year 4	
				Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
Comp. 1	Outcome 1	**Output 1.1**: An intersectoral and multilevel e-mobility coordination space established, including all relevant governmental stakeholders	1.1.1: Convene and carry out periodic meetings with all key stakeholders identified														
			1.1.2: Provide technical assistance to participating governmental stakeholders for the preparation and delivery of a draft document..														
		Output 1.2: National e-mobility strategy submitted to the government and launched, including economic impacts (competitiveness, industrial development), geographical deployment (e.g. charging network) and environmental and social dimensions (gender-responsive)	1.2.1 Study of electrification options for public transport by road, with a focus on urban buses and 3-wheelers														
			1.2.2 Study of electrification options for light vehicles, with a focus on public and private fleets														
			1.2.3: Study of the charging network necessary to expand electric routes in the cities of Lima and Arequipa.														
			1.2.4 Capacity needs assessment and elaboration of a capacity-building and communications plan														
			1.2.5: Development of a gender analysis and action plan														
			1.2.6. Prepare an e-mobility strategy through participatory and multistakeholder consultations														
			1.2.7. Submission of final e-mobility strategy to the government for adoption														
		Output 1.3: Revised regulatory and fiscal framework for deployment of electric vehicles delivered for governmental approval	1.3.1: Submission to the government of national regulation reforms on vehicle approval, including EVs														
			1.3.2. Design of national regulations and standards to install public and private charging infrastructure														
			1.3.3. Reform proposals on vehicle taxes, to incentivize EV and discourage the purchase of high-carbon vehicles														
		Output 1.4: Strengthened capacities and awareness of public and private decision-makers, technical professionals	1.4.1: Implementation of the communications strategy, specifically addressing educational centers and the public to foster the use of EVs														
			1.4.2: Implementation of training activities on e-mobility policies, targeting public and private														

Com-ponent	Out-comes	Outputs	Activities	Year 1				Year 2				Year 3				Year 4	
				Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
		and consumers regarding e-mobility policy design and implementation, highlighting the social and environmental benefits of EV deployment	decision-makers and other professionals														
			Activity 1.4.3: Design of technical training modules for PT technical staff (drivers and operators) and transport and electricity professionals on EV and charging infrastructure maintenance and assembly														
Comp. 2	Outcome 2	**Output 2.1:** E-mobility pilots in public urban transport systems designed in Lima and Arequipa, integrating gender-responsive measures	2.1.1. Delivery of best practice guidelines and pilot design recommendations for e-mobility public transportation pilots														
			2.1.2: Elaboration of a design proposal for the pilots in Lima and Arequipa, following the guidelines for e-mobility pilot design														
			2.1.3 Design and implementation of multi-media campaigns against gender-based harassment, discrimination and violence, including against other vulnerable groups														
		Output 2.2: EV pilot in Lima implemented, monitored and evaluated: two electric buses and ten 3-wheelers (indicatively) in public transport lines.	2.2.1: Selection of EV operators for e-buses in Lima														
			2.2.2: Integration and operation of at least 2 e-buses and 10 electric 3-wheelers in operator's fleet in Lima														
			2.2.3. Identification and systematization of results, best practices and lessons learned														
		Output 2.3: EV pilot in Arequipa implemented, monitored and evaluated: two electric buses and ten 3-wheelers (indicatively) in public transport lines	2.3.1: Selection of EV operators for e-buses in Arequipa														
			2.3.2: Integration and operation of at least 2 e-buses and 10 electric 3-wheelers in operator's fleet in Arequipa														
			2.3.3. Identification and systematization of results, best practices and lessons learned														
		Output 2.4: Technical training delivered (including EV driving and maintenance)	2.4.1: Implementation of technical training activities on EV driving and maintenance, that also address gender issues…														
		Output 2.5: Evidence and best practices on EV pilot implementation communicated to key stakeholders and incorporated in e-mobility guidelines for replicability	2.5.1: Dissemination of project pilot results through communication materials and initiatives														
			2.5.2 Incorporation of best practices and lessons learned in e-mobility best practice guidelines														

Com-ponent	Out-comes	Outputs	Activities	Year 1				Year 2				Year 3				Year 4	
				Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
Comp. 3	Outcome 3	**Output 3.1**: Assessment of market potential in Peru carried out for the introduction of electric vehicles and electrification of road public transport systems	3.1.1: Study of scenarios on the potential of the electric vehicle market in Peru														
			3.1.2: Study of the electrification potential of road public transport systems in Lima and Arequipa														
		Output 3.2: E-mobility network established in Peru, including manufacturers, importers, technology and service providers, as well as public transport operators and other final users	3.2.1 E-mobility network engagement with the project, providing regular contacts among e-mobility stakeholders, the government and civil society														
			3.2.2: Elaboration of the network action plan (information platform and website, quarterly workshops, quarterly position papers) to accelerate the deployment of e-mobility														
		Output 3.3: Detailed financial and business models for EV commercialization and operation designed	3.3.1: Development of alternative business models for the electrification of urban public transport (buses and 3-wheelers).														
			3.3.2: Development of alternative business models for the electrification of public and private corporate fleets…														
			3.3.3: Submission to the government of alternative concession schemes to facilitate financing of EV in public transport														
		Output 3.4: EV procurement guidelines for urban public transport operators and authorities developed	3.4.1: Provision of tailored support and development of guidelines for operators of public transport services (buses and 3-wheelers) for the procurement of EVs for their fleets														
			3.4.2: Provision of tailored support to interested public authorities for promoting EVs in their own fleets and in those associated to public service concessions														
Comp. 4	Outcome 4	**Output 4.1**: Comprehensive diagnosis on national end-of-life vehicle management completed	4.1.1. Study on the description and analysis of local management conditions of ELVs and their components.														
			4.1.2. Estimate of projected ELVs components														

Com-ponent	Out-comes	Outputs	Activities	Year 1				Year 2				Year 3				Year 4	
				Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
			generation, considering internal combustion and electric vehicles.														
		Output 4.2: Proposal on end-of-life vehicle management regulation, based on the EPR approach, delivered to the government for approval	4.2.1: Research and systematization for international and regional regulation on EPR-based ELV management, with a focus on EV components and battery														
			4.2.2: Comprehensive review of existing and planned international regulations on second-life battery use														
			4.2.3: Formulation of proposals for regulation in line with national priorities and in close coordination with the identified relevant private stakeholders														
		Output 4.3: Cost-efficiency analysis and business models designed for implementation of an EPR approach to ELV management.	4.3.1 Research and systematization of successful financial and business models on ELV components and second-life battery use, based on the EPR concept														
			4.3.2 Development of commercially viable EPR business models for ELV components and EV batteries, including cost benefit analysis estimating investment needs and financing schemes														
		Output 4.4: Roadmap for EPR approach to end-of-life vehicle management submitted to the government for approval	4.4.1: Development of a road map for EPR with ELVs components, built through a participatory process and in close coordination with the Ministry of Environment														
			4.4.2: Advocacy and promotional activities to secure timely approval and enforcement of the roadmap as a planning instrument														
		Output 4.5: Awareness-raising and capacity-building activities completed, targeting government, consumers and private sector stakeholders on end-of-life vehicle management.	4.5.1: Design and delivery of a training programme for policymakers…														
			4.5.2: Provision of technical and managerial support to producers and importers…														
			4.5.3: Support to the government, industry and research sector in the design of a national campaign to raise awareness amongst consumers…														
Comp.	Outcom	**Output 5.1**: The project	5.1.1 Updating and implementation of project														

Com-ponent	Out-comes	Outputs	Activities	Year 1				Year 2				Year 3				Year 4	
				Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
5	e 5	monitoring and evaluation plan and the knowledge-management strategy are designed and implemented, providing regular reporting to the global programme	monitoring and evaluation plan														
			5.1.2. Development and implementation of monitoring, reporting and verification (MRV) plan for the demonstration in Lima														
			5.1.3. Development and implementation of monitoring, reporting and verification (MRV) plan for the demonstration in Arequipa														
		Output 5.2: Best practices and lessons learnt reports developed and shared with the global programme	5.2.1. Terminal evaluation														
			5.2.2. Final best practices and lessons learnt report prepared and shared with the Global Programme its Regional Platform.														
	PMC		Annual audits														

Annex 4: Monitoring Plan

This Monitoring Plan and the M&E Plan and Budget in Section VI of this project document will both guide monitoring and evaluation at the project level for the duration of project implementation.

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/ Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
Reduce GHG emissions from urban transport by creating the conditions to accelerate market transformation of electric mobility in Peru	***Indicator 1:*** *Number of direct project beneficiaries disaggregated by gender*	*8,000 women and 6,000 men (7 million trips during the project)*	*Women and men using the new mobility options on a regular basis*	*Travel tickets (passengers) sold in the new e-buses*	*Mid-term and end of project*	*PMU*	*Operators' ticketing records: Number of passengers using PT services of new EVs. Gender recorded through sampling*	*Assumed 240,000 and 10,500 passenger trips per year per bus and 3W, respectively. 4 buses and 20 ·W operating for at least 2 years; as a result of replication, 16 addititonal buses and 80 additional 3W operate during the last project year. Each beneficiary is assumed to make at most 500 trips during this period.*
	Indicator 2: *Number of indirect project beneficiaries*	*26,500 women and 26,500 men beneficiaries*	*Women and men living in the area served by the new mobility options*	*Population statistics*	*Mid-term and end of project*	*PMU*	*Local census: Residents in the area served by EV services*	*Assumed: 30-km corridor with 11,000 inh/km2 for buses and a 200,000 inh. district for 3W*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/ Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
	Indicator 3: *Expected metric tons of CO_2e avoided*	*178,460 t (direct) end of life of project's vehicles* *236,657 t (indirect) end of life of project's vehicles*	*Tons of direct GHG emissions avoided*	*Vehicle monitoring with GPS or other means*	*Mid-term and end of project*	*PMU*	*GPS monitoring of mileage and energy consumption of project vehicles*	*All EVs are operational in M-13.*
Outcome 1: Institutional, regulatory and fiscal frameworks established for the promotion of e-mobility, with the support of all key stakeholders, including potential EV providers and users	***Indicator 4:*** ***A national coordination space to promote the uptake of low-carbon e-mobility is established and operational***	*4 meetings/year*	*A national coordination space to promote the uptake of low-carbon e-mobility is established and operational*	*Meeting reports*	*Mid-term and end of project*	*PMU*	*Meeting reports*	*Coordination space approved and regularly convened by government* *Political risk of change of government's priorities*
	Indicator 5: *National e-mobility strategy adopted by the government*	*E-mobility strategy adopted by government*	*National e-mobility strategy endorsed by government*	*Publication of strategy*	*Mid-term and end of project*	*PMU*	*Publication of strategy*	*Political risk: government's adoption is beyond the project's capacities*
	Indicator 6: *Regulatory and fiscal schemes to incentivize the uptake of e-mobility approved by the government*	*At least one regulatory or one fiscal scheme approved by the government*	***Number of regulatory and fiscal schemes to incentivize the uptake of e-mobility***	*Adoption of scheme by government*	*Mid-term and end of project*	*PMU*	*Official publication of scheme*	*Political risk: government's adoption is beyond the project's capacities*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/ Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
	Indicator 7: *Number of public and private decision-makers trained on legal and fiscal issues of e-mobility*	*25 women, 25 men trained*	*Number of civil servants trained on legal and fiscal issues of e-mobility*	*Reports from training activities*	*Mid-term and end of project*	*PMU*	*Reports from training activities*	*Sufficient staff available in the concerned government units to be dedicated to e-mobility issues.*
Outcome 2: **EV demonstrations in urban public transport systems provide evidence of technical, financial and environmental sustainability to scale-up e-mobility**	***Indicator 8:*** *Number of kilometers served by the project's EVs*	*Buses: 216,000 in each city by end of year 3* *3W: 280,000 km in each city by end of year 3*	*Number of kilometers served by the project's 4 electric buses and 20 3-wheelers per city (Lima and Arequipa)*	*Travel tickets (passengers) sold in the new e-buses*	*Mid-term and end of project*	*Bus operators and transport authorities*	*GPS tracker installed in all project's EVs.*	*If reporting of km by operators is inadequate, the project would undertake a survey.*
	Indicator 9: *Number of women and men receiving professional training*	*30 women and 30 men*	*Number of women and men receiving professional training in e-mobility*	*Reports from training activities*	*Mid-term and end of project*	*PMU*	*Reports from training activities*	*Bus companies allowing staff to participate in training activities. Alternative: Trade unions*
	Indicator 10: *Number of EV purchases approved by PT operators in Lima and Arequipa*	*At least one purchase order approved in one city*	*PT operators in Arequipa or Lima approve the future purchase of EV*	*Questionnaire of bus operators through local public transport authorities in Arequipa and Lima*	*Mid-term and end of project*	*PMU*	*Questionnaire of bus operators through local public transport authorities in Arequipa and Lima*	*Risk of operators reluctant to make public their purchase plans. Alternative: approaching bus dealers*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/ Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
Outcome 3: Conditions established to support the inclusion of EVs in the vehicle market and to accelerate their deployment	***Indicator 11:*** *Procurement guidelines*	*Guidelines endorsed by PT authorities*	*Procurement guidelines on EV for public transport*	*Publication of guidelines*	*Mid-term and end of project*	*PMU*	*Guidelines published with government's endorsement*	*Political risk: government's endorsement is beyond the project's capacities*
	Indicator 12: *Financial mechanisms*	*Financial mechanism established*	*At least one financial mechanism to facilitate the scaling-up of e-mobility in public transport fleets*	*Publication of financial mechanism*	*Mid-term and end of project*	*PMU*	*Financial mechanism published*	*COFIDE and other financial institutions keep their current interest in supporting e-mobility*
	Indicator 13: *EVs registered*	*21 buses and 100 3-wheelers*	*Number of new EV registered in Peru in (public transport) fleets*	*Official statistics*	*Mid-term and end of project*	*PMU*	*Official statistics*	*None*
Outcome 4: Conditions created for a circular economy perspective in the electrification of mobility in Peru	***Indicator 14:*** *ELV management scheme*	*Roadmap endorsed by government*	*Diagnostic of ELV management in Peru followed by EPR-based scheme for collection, reuse and disposal of used EV batteries*	*Publication of diagnostic in the project website.* *Scheme published with government's endorsement*	*Mid-term and end of project*	*PMU*	*Diagnostic published in the project website.* *Roadmap published with government's endorsement*	*Government's and stakeholders' support to the project to collect the necessary factual information about ELV management activities.* *Political risk: government's endorsement is beyond the project's capacities*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/ Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
	Indicator 15: Professional training	*25 women, 15 men*	*Number of women and men professionally trained on end-of-life EV management*	*Reports from training activities*	*Mid-term and end of project*	*PMU*	*Reports from training activities*	*Support of companies in the sector to facilitate staff training*
Outcome 5	***Indicator 16:*** *Generation of best practices and lessons learned for the global programme*	*4 reports*	*Number of reports on best practices and lessons learned on the Peru project shared by the national coordination space with the global programme*	*Technical reports corresponding to activities 2.2.3, 2.3.3, 2.5.2 and 4.3.1*	*End of project*	*PMU*	*Global programme*	*Results of project activities are considered relevant by Global Programme managers*

Annex 5: UNDP Social and Environmental Screening Procedure (SESP)

Project Information

Project Information	
1. Project Title	Enhancing sustainability in Electric mobility for low carbon urban transport and an Extended Producer Responsibility approach in batteries and vehicle components
2. Project Number	6384
3. Location (Global/Region/Country)	Peru

Part A. Integrating Overarching Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Overarching Principles in order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the Project mainstreams the human-rights based approach
The project accelerates the adoption of electric vehicles and the improvement of public transport services in Peru, reducing greenhouse gas emissions and improving the quality of life of citizens through better mobility conditions, air quality and a more liveable urban environment. The project enhances the availability, and quality of transport services for citizens, particularly for those more dependent on public transport, such as women, low-income households, children and teenagers, and the elderly. The project will facilitate authorities to implement and enforce transparent and accountable contracts for the provision of public transport services, with decision-making processes for contract awarding, which encourage the participation of civil society and the active involvement of marginalized groups in those decisions affecting them, such as public transport service provision with electric vehicles. Participation and inclusion of all stakeholders, including marginalized groups is embedded in project design. Implementation and monitoring are being facilitated through a co-creation approach during project design and through awareness-raising and capacity-building activities during project implementation. The project will also strengthen stakeholders' and the public's involvement in monitoring and evaluation through the provision of enhanced information and mobility data, as well as information on the actual environmental performance of the passenger transport sector and its different modes.
Briefly describe in the space below how the Project is likely to improve gender equality and women's empowerment
The project will contribute to improving the quality of life for all urban dwellers in the long-run through improved air quality as well as through provision of a safer, quieter and more affordable transportation option. A significant proportion of urban women rely on public transportation, namely buses and motorcycle taxis. The project will address issues of sexual harassment in public transportation, which is a pressing issue for women that rely on this to commute in cities. Opportunities to include and prioritize women in accessing financing for technology shift will be also explored. In addition, opportunities for women empowerment will be identified, especially in form of increased participation of women in the provision of the currently male-dominated transportation services.
Briefly describe in the space below how the Project mainstreams environmental sustainability
Environmental sustainability is at the heart of project, that is geared towards promoting and supporting electric vehicles for the satisfactory achievement of the energy and clean air environment and development agenda in Peru. The reduction of GHG emissions nationally and air pollution especially in urban areas is a key element to

mainstream environmental sustainability in Peru and in the LAC region.

The proposed project is aligned to Peru's Nationally Determined Contributions (NDC, updated in 2020): It will contribute to enhance the ambition of the NDC measure T8 (promotion of electric vehicles, EV) at the national level, and T6 (national programme on sustainable urban mobility). The NDC is being supported by the identification of Nationally Appropriate Mitigation Actions (NAMA), including one on clean transport through the GEF/UNDP NAMA project for the Energy Sector (stared in 2015 and expected to be completed in 2020). The project will help to reach the NDC objective of a 5% of share of electric vehicles in the Peruvian fleet, while supporting the sustainability of urban transport and backing the achievement of Sustainable Development Goals in cities.

The proposed initiative will contribute to a longstanding environmental sustainability impact from transport, especially significant in larger cities, such as Arequipa and Lima: Air quality, which is a problem plaguing major cities in Peru.

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Describe briefly potential social and environmental risks identified in Attachment 1 – Risk Screening Checklist (based on any "Yes" responses). If no risks have been identified in Attachment 1 then note "No Risks Identified" and skip to Question 4 and Select "Low Risk". Questions 5 and 6 not required for Low Risk Projects.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5 below before proceeding to Question 6*			**QUESTION 6: What social and environmental assessment and management measures have been conducted and/or are required to address potential risks (for Risks with Moderate and High Significance)?**
Risk Description	***Impact and Probability (1-5)***	***Significance (Low, Moderate, High)***	***Comments***	***Description of assessment and management measures as reflected in the Project design. If ESIA or SESA is required note that the assessment should consider all potential impacts and risks.***
Risk 1: The project intends to work with 3-wheeler technology conversion in city demonstrations. 3-wheeler (3W) owners and drivers may have difficulties to understand the advantages of electrification. The sector plays a secondary role in urban mobility and could be marginalized by institutional stakeholders during project consultations	I = 3 P = 2	Moderate	3-wheeler taxis are well regulated in Peru, and provide an alternative for cheap and efficient transportation in some urban areas, and fares are approved by local authorities. The project intends to facilitate access to electrification for 3-wheeler operators, providing information, capacity building and factual evidence with 20 electric vehicles. The sector has a long tradition of dialog with local	The ProDoc provides a stakeholder engagement plan, in which 3-wheeler taxi representatives are included in key consultations for projects investments and activities. In order to guarantee transparency and isonomy in the financial support for technology shift, the project will guarantee a public and open call for interests for the selection of demonstrations within this category. At the project design stage, participants at the design and validation workshops suggested that 3-wheeler taxi operators participating in the pilot will decrease their

and decision-making, so that the project's outcomes could negatively impact in the living conditions of 3-wheeler workers. Principle 1, No 4: Is there a likelihood that the Project would exclude any potentially affected stakeholders, in particular marginalized groups, from fully participating in decisions that may affect them? Principle 6, no 2: Are indigenous peoples present in the Project area (including Project area of influence)? Standard 5, No.2: Would the Project possibly result in economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?			authorities to improve working conditions and fight informality. The deployment of electrification could make fuel-based 3-wheelers less attractive to users, with a negative impact on their owners. However, this effect is likely to be very limited, as the demand for 3W services is high and the lifespan of these vehicles is short (5 years), and vehicle owners can change to the new technology when renewing their vehicles. The matter of indigenous people for this project is related to the principle 1 - excluding vulnerable groups in upstream activities. This will be addressed with SESA for upstream policy. An initial research related to the territories of indigenous people in Lima and Arequipa indicates no overlaps with project area of influence for the demonstrative activities. As an e-mobility project acting in urbanized areas of Peru no traditional knowledge or practices or cultural heritage, etc. will be impacted by on-the-ground project activities.	operating costs, and enjoy better working conditions (less noise, no emissions, less vibrations), but this will not reduce the economic activity of the operators not adopting electric 3W, as the demand for these services will remain high. In order to prevent any unexpected impacts, a livelihood analysis will be undertaken during the project activity to design the pilots (D.2.1.2), as the concrete geographical areas in which the electric 3W will be operating are established and the operators in the demonstration area are fully identified. An IPP (Indigenous people plan) and FPIC (Free, prior and informed consent) are not required for the demo sites but that will be re-confirmed during the planned assessments. The SESA will include FPIC as determined appropriate and necessary for SES compliance in the course of that assessment.
Risk 2: Potential limited participation of women in public transportation sector, as well as gender-based harassment, discrimination and violence. Principle 1, No.8: Is there a risk that the Project would exacerbate conflicts among and/or the risk of violence to project-affected communities and individuals? Principle 2, No.2: Would the Project potentially reproduce discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	I = 3 P = 2	Moderate	The target of the proposed project is the public transportation sector which is predominantly male dominated. There is a risk that this trend might continue even after the project is implemented. Sexual harassment is also a key element for woman in the public transportation in LAC.	The project has developed a gender analysis and action plan to mitigate the risks of limited participation and access to opportunities. The project activities include EV training activities for drivers and others (2.4.1), where gender-equality challenges will be considered in the selection of topics and trainers and gender harassment, discrimination and violence will be addressed. The latter will also be analyzed and addressed through campaigns (2.1.3). Gender-sensitive information is embedded in the monitoring of demonstrations (EV operation and use, accessibility for women and men (2.1.2)). One of the project activities (1.2.5) is dedicated to provide an assessment on how each proposed regulation improves women's access to services, will create opportunities to women and will promote equitable treatment of women; furthermore, this activity will also assess how the proposed regulations will increase the participation of women in the transport sector, and how they will engage women directly in

				decision-making. Project output 3.4 provides guidelines for PT operators and authorities including recommendations for the adoption of gender-responsive measures in EV procurement, recruitment and training strategies of urban public transport operators. A budget of USD 129,000 has been allocated for gender-related actions.
Risk 3: Potential environmental hazards associated with waste generated during manufacture of electric vehicle batteries as well as handling batteries that have reached their end-of-life. Standard 3, No.3.2: Would the Project pose potential risks to community health and safety due to the transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)? Standard 7, No.7.2: Would the proposed Project potentially result in the generation of waste (both hazardous and non-hazardous)?	I = 3 P = 3	Moderate	Electric vehicle batteries are produced from rare earth metals some of which are not environmentally friendly. Therefore, the disposal of unrecyclable material from battery manufacturers must be handled properly. As GHG emissions during battery production are substantial, there is a need to extend the battery life through other uses (e.g. energy storage) once they are no longer fit to be used in vehicles. In addition, when batteries have reached their end-of-life, they also need proper disposal to minimize environmental impact and exposure to the society.	The potential environmental hazards of EV have been analyzed in the Problem Tree and Theory of Changed, at the beginning of the project design stage (and benefited from the analysis made by the global E-mob programme on this topic), identifying the need to establish adequate regulations for end-of-life battery management. These topics were discussed during the validation workshops, and resulted in the definition of 12 activities in component 4 of the project, based on the application of the Extended Producer Responsibility (EPR) principle to the management of batteries and vehicle components at their end-of-life; additionally, these activities aim at strengthening current waste management chains for electric vehicles. Regulation and guidance for second- and third-life use of batteries are also considered, so that batteries exhaust their whole lifespan before being sent to recycling processes. Component 4 also provides a strategic framework for mid and long term management of potential environmental hazards. The project will develop policy guidelines/regulations for re-use, collection, safe disposal/waste management and recycling of the batteries. The project will develop market mechanisms and business plans suitable to develop cooperation between vehicle importers (within an EPR approach) and third parties interested in battery reuse and recycling, while addressing the informality still present in the end-of-life vehicle (ELV) management sector.
Risk 4: Potential occupational health and safety risks associated with electric vehicle repair and decommissioning of battery banks and/or vehicle. Standard 3, No. 3.7: Does the Project pose potential risks and vulnerabilities related to occupational health and safety due to	I = 3 P = 3	Moderate	Battery Electric Vehicles (BEVs) operate at 3-phase high voltage. Consequently, they must be maintained by properly trained technicians. As expected, most of the existing crop of vehicle repair technicians were trained at a time when electric vehicles were unheard of. Therefore, retraining will be required for the technicians involved in	In accordance with the global e-mob programme, the problem tree and the theory of change, there is a need to provide adequate training to all the staff involved in the maintenance, repairs and decommissioning of EVs and their batteries. These topics were discussed during the validation workshops, and resulted in the definition of 3 activities in output 4.5, to provide training, technical support and an awareness raising campaign. Ttraining activities will build the capacity of technicians to

physical, chemical, biological, and radiological hazards during Project construction, operation, or decommissioning?			electric vehicle repair or decommissioning.	handle the repair and decommissioning of electric vehicles, and public awareness-raising campaigns will be useful to impress on the general public, who will own electric vehicles, the need for taking them to repair shops that have the appropriate equipment and training to handle such vehicles. The implementation of the Extended Producer Responsibility will provide the necessary mandates to electric vehicle dealers to take full responsibility of the EV they will commercialize, including thorough end-user awareness to avoid accidental injury due to electric shock. Potential risks associated to EVs will also be included within the different training activities envisaged in the project (1.4.2; 2.4.1) and in the future standards (1.3.2).
Risk 5: the introduction of EVs (with higher upfront costs for operators purchasing them) could result in an increase in public transport fares, reducing the mobility of vulnerable groups or forcing them to use informal transport services. This also applies to fares of 3-wheeler taxi services. Standard 5, No.2: Would the Project possibly result in economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	I = 4 P = 2	Moderate	As PT fares are already relatively high compared with purchasing power in Peru, any further raise of fares could eventually result in some low-income users looking for alternative places to live and work. The literature shows that e- buses' higher upfront costs are compensated with lower operational costs, and the same occurs for 3-wheelers. Furthermore, the purchase cost of electric buses and 3-wheelers is steadily decreasing worldwide. Public authorities' requests for fleet renewals in the past have raised similar concerns, and they have negotiated with the operators transitional conditions to attain the technological improvements without increasing fares. These transitional conditions need to be properly identified as a result of the project.	Public transport authorities have been established in most Peruvian cities to integrate PT systems and provide affordable fares. As a result of the demonstrations and through its component 2, the project will design mechanisms to compensate electric bus operators for the additional costs (which are expected to decrease in time, due to the quick global progress in battery technology) due to the new vehicles. Through its component 3, the project intends to develop financial models in order to facilitate the operators' access to electric buses and 3-wheelers, balancing their higher procurement costs and their lower operating costs. These issues have been addressed at project design, establishing a project compensation to the operators participating in the pilot to avoid a need to raise fares; the conditions will be fine-tuned in the design of the demonstrations (activity 2.1.2) and the options for upscaling will be defined during the monitoring and assessment of project demonstrations in activities 2.2.3 and 2.3.3.
Risk 6: Location of charging stations in the demonstrations, and also in the design of the networks in Lima and Arequipa, has not been decided yet and could have negative impacts on habitats if not selected appropriately.	I = 4 P = 2	Moderate	During the project, charging stations will be located within the premises (depots) of bus and 3-wheelers operators or at key high-accessibility places in the demonstration cities, in highly-urbanized areas. Charging stations are very compact in size, not requiring any significant occupation of	During project design, the stakeholders discussed this risk and considered that current national legislation on environment, habitat protection and indigenous population provide the adequate framework to avert this risk. Accordingly, the project will apply a Strategic Environmental and Social Assessment (SESA) approach; as e-mobility is a new issue and the location of charging infrastructure is highly flexible, the following

Standard 1.1: Would the Project potentially cause adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? Standard 2.2: Would the potential outcomes of the Project be sensitive or vulnerable to potential impacts of climate change?			land. Some EV chargers or the electricity distribution grid feeding them can be vulnerable to changes in climate. Best international practice is to identify a critical network with higher resilience to the impact of extreme weather events and other changes in climate.	location criteria will be integrated,: (1) identify a sufficient number of alternative locations for each charging point; (2) undertake an assessment of potential traffic generation created by the envisaged infrastructure; (3) identify the eventual needs for changes in the electricity distribution grid; (4) define mitigation measures, in case that concerns are raised from any of the relevant local stakeholders. Accordingly, project activities 1.2.3 (design of the future charging network in Arequipa and Lima), 1.2.6 (design of e-mobility strategy) and 1.3.2 (design of national regulations and standards to install charging infrastructure) are expected to adopt a SESA approach. As for the implementation of the charging stations for project demonstrations (activities 2.2.2 and 2.3.2) in Lima and Arequipa, they will be installed in the operators' premises, pointing out to a very low probability of impact in indigenous peoples territory and cultural heritage. FPIC are not required for the demo sites but that will be re-confirmed during the planned assessments to align to SES requirements. The exposure of the project's outputs to changes in climate have been assessed during project design (see risk 12 in ProDoc and the additional information on climate risks in the same section), and the project has built resilience within the design of activities 1.2.3 and 1.2.6 by including within the selection criteria the consideration of at least 20% of the future charging points in locations with low vulnerability in accordance with the relevant guidance in the National Adaptation Plan.
Risk 7: Unintended (social and/or environmental) impacts of national e-mobility strategy, regulations and standards. Principle 1, No 4: Is there a likelihood that the Project would exclude any potentially affected stakeholders, in particular marginalized groups, from fully participating in decisions that may affect them?	I = 2 P = 2	Moderate	Policies, regulations and strategies related to e-mobility may have unintended social and environmental impacts, beyond the risks identified above. Existing legislation in Peru requires undertaking a Strategic Environmental Assessment of relevant plans, policies and regulations to prevent such unintended impacts. Policies, regulations and strategies are developed in project Component 1. The unintended risks screened at project design refer to (1) the new coordination space	At the design stage, the project activities that may have such unintended impacts and the interested parties have been screened during design workshops; this has resulted in the integration within several activities in component 1 of mechanisms for stakeholders' involvement, scoping processes to identify unintended impacts and evaluation and monitoring processes to be included in the design of the expected policies, strategies and regulations. These are described in the activities related to the new deliberation body (output 1.1, coordination space, activity 1.1.2), the design of the national e-mobility strategy (activity 1.2.6) and the regulatory reforms on vehicle approval, charging infrastructure and vehicle taxes (output 1.3,

Principle 2, No.2: Would the Project potentially reproduce discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits? Principle 3, Standard 1, Question 1.1: Would the Project potentially cause adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? Standard 5, No.2: Would the Project possibly result in economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?.			(output 1.1) not advancing the participation of affected stakeholders in policy decision-making or even establishing additional barriers to such participation; (2) the national e-mobility strategy (activity 1.2.6) and regulatory reforms (output 1.3) and the subsequent expansion of EVs not improving mobility (and associated livelihood) conditions or creating unintended impacts on habitats.	activities 1.3.1, 1.3.2 and 1.3.3).
	QUESTION 4: What is the overall Project risk categorization?			
	Select one (see SESP for guidance)			**Comments**
	Low Risk	☐		
	Moderate Risk	√		The Project is considered to be low-range Moderate risk as the mitigation strategies were inserted at the project design stage. The risk considered to be more substantial – hazardous waste- is being addressed as a core component with specific budget, results and indicators embedded in project logframe.
	High Risk	☐		
	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are relevant?			
	Check all that apply			**Comments**
	Principle 1: Human Rights	√		Comprehensive Stakeholder Engagement Plan
	Principle 2: Gender Equality and Women's Empowerment	√		Gender Analysis and Action Plan
	1. Biodiversity Conservation and Natural Resource Management	☐		

	2. Climate Change Mitigation and Adaptation	☐	
	3. Community Health, Safety and Working Conditions	√	Extended Producer Responsibility (EPR) approach included as a component of the project.
	4. Cultural Heritage	☐	
	5. Displacement and Resettlement	√	
	6. Indigenous Peoples	√	An IPP and FPIC are not required for the demo sites but that will be re-confirmed during the planned assessments. The SESA will include FPIC as determined appropriate and necessary for SES compliance in the course of that assessment.
	7. Pollution Prevention and Resource Efficiency	√	Extended Producer Responsibility (EPR) approach included as a component of the project.

Final Sign Off

Signature	*Date*	*Description*
QA Assessor		UNDP staff member responsible for the Project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver		UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair		UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks		
Principles 1: Human Rights		**Answer (Yes/No)**
1.	Could the Project lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	No
2.	Is there a likelihood that the Project would have inequitable or discriminatory adverse impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups? [46]	No
3.	Could the Project potentially restrict availability, quality of and access to resources or basic services, in particular to marginalized individuals or groups?	No
4.	Is there a likelihood that the Project would exclude any potentially affected stakeholders, in particular marginalized groups, from fully participating in decisions that may affect them?	Yes
5.	Is there a risk that duty-bearers do not have the capacity to meet their obligations in the Project?	No
6.	Is there a risk that rights-holders do not have the capacity to claim their rights?	No
7.	Have local communities or individuals, given the opportunity, raised human rights concerns regarding the Project during the stakeholder engagement process?	No
8.	Is there a risk that the Project would exacerbate conflicts among and/or the risk of violence to project-affected communities and individuals?	Yes
Principle 2: Gender Equality and Women's Empowerment		
1.	Is there a likelihood that the proposed Project would have adverse impacts on gender equality and/or the situation of women and girls?	No
2.	Would the Project potentially reproduce discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	Yes
3.	Have women's groups/leaders raised gender equality concerns regarding the Project during the stakeholder engagement process and has this been included in the overall Project proposal and in the risk assessment?	Yes
4.	Would the Project potentially limit women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	No
Principle 3: Environmental Sustainability: Screening questions regarding environmental risks are encompassed by the specific Standard-related questions below		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
1.1	Would the Project potentially cause adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	Yes

[46] Prohibited grounds of discrimination include race, ethnicity, gender, age, language, disability, sexual orientation, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender people and transsexuals.

1.2	Are any Project activities proposed within or adjacent to critical habitats and/or environmentally sensitive areas, including legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	No
1.3	Does the Project involve changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	No
1.4	Would Project activities pose risks to endangered species?	No
1.5	Would the Project pose a risk of introducing invasive alien species?	No
1.6	Does the Project involve harvesting of natural forests, plantation development, or reforestation?	No
1.7	Does the Project involve the production and/or harvesting of fish populations or other aquatic species?	No
1.8	Does the Project involve significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.9	Does the Project involve utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)	No
1.10	Would the Project generate potential adverse transboundary or global environmental concerns?	No
1.11	Would the Project result in secondary or consequential development activities which could lead to adverse social and environmental effects, or would it generate cumulative impacts with other known existing or planned activities in the area? *For example, a new road through forested lands will generate direct environmental and social impacts (e.g. felling of trees, earthworks, potential relocation of inhabitants). The new road may also facilitate encroachment on lands by illegal settlers or generate unplanned commercial development along the route, potentially in sensitive areas. These are indirect, secondary, or induced impacts that need to be considered. Also, if similar developments in the same forested area are planned, then cumulative impacts of multiple activities (even if not part of the same Project) need to be considered.*	No
Standard 2: Climate Change Mitigation and Adaptation		
2.1	Will the proposed Project result in significant[47] greenhouse gas emissions or may exacerbate climate change?	No
2.2	Would the potential outcomes of the Project be sensitive or vulnerable to potential impacts of climate change?	Yes
2.3	Is the proposed Project likely to directly or indirectly increase social and environmental vulnerability to climate change now or in the future (also known as maladaptive practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	No
Standard 3: Community Health, Safety and Working Conditions		
3.1	Would elements of Project construction, operation, or decommissioning pose potential safety risks to local communities?	No
3.2	Would the Project pose potential risks to community health and safety due to the transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	Yes
3.3	Does the Project involve large-scale infrastructure development (e.g. dams, roads, buildings)?	No

[47] In regards to CO_2, 'significant emissions' corresponds generally to more than 25,000 tons per year (from both direct and indirect sources). [The Guidance Note on Climate Change Mitigation and Adaptation provides additional information on GHG emissions.]

3.4	Would failure of structural elements of the Project pose risks to communities? (e.g. collapse of buildings or infrastructure)	No
3.5	Would the proposed Project be susceptible to or lead to increased vulnerability to earthquakes, subsidence, landslides, erosion, flooding or extreme climatic conditions?	No
3.6	Would the Project result in potential increased health risks (e.g. from water-borne or other vector-borne diseases or communicable infections such as HIV/AIDS)?	No
3.7	Does the Project pose potential risks and vulnerabilities related to occupational health and safety due to physical, chemical, biological, and radiological hazards during Project construction, operation, or decommissioning?	Yes
3.8	Does the Project involve support for employment or livelihoods that may fail to comply with national and international labor standards (i.e. principles and standards of ILO fundamental conventions)?	No
3.9	Does the Project engage security personnel that may pose a potential risk to health and safety of communities and/or individuals (e.g. due to a lack of adequate training or accountability)?	No
Standard 4: Cultural Heritage		
4.1	Will the proposed Project result in interventions that would potentially adversely impact sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: Projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	No
4.2	Does the Project propose utilizing tangible and/or intangible forms of cultural heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
5.1	Would the Project potentially involve temporary or permanent and full or partial physical displacement?	No
5.2	Would the Project possibly result in economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	Yes
5.3	Is there a risk that the Project would lead to forced evictions?[48]	No
5.4	Would the proposed Project possibly affect land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No
Standard 6: Indigenous Peoples		
6.1	Are indigenous peoples present in the Project area (including Project area of influence)?	Yes
6.2	Is it likely that the Project or portions of the Project will be located on lands and territories claimed by indigenous peoples?	No
6.3	Would the proposed Project potentially affect the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the Project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to the screening question 6.3 is "yes" the potential risk impacts are considered potentially severe and/or critical and the Project would be categorized as either Moderate or High Risk.*	No

[48] Forced evictions include acts and/or omissions involving the coerced or involuntary displacement of individuals, groups, or communities from homes and/or lands and common property resources that were occupied or depended upon, thus eliminating the ability of an individual, group, or community to reside or work in a particular dwelling, residence, or location without the provision of, and access to, appropriate forms of legal or other protections.

6.4	Has there been an absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	No
6.5	Does the proposed Project involve the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6	Is there a potential for forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources?	No
6.7	Would the Project adversely affect the development priorities of indigenous peoples as defined by them?	No
6.8	Would the Project potentially affect the physical and cultural survival of indigenous peoples?	No
6.9	Would the Project potentially affect the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices?	No
Standard 7: Pollution Prevention and Resource Efficiency		
7.1	Would the Project potentially result in the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	No
7.2	Would the proposed Project potentially result in the generation of waste (both hazardous and non-hazardous)?	Yes
7.3	Will the proposed Project potentially involve the manufacture, trade, release, and/or use of hazardous chemicals and/or materials? Does the Project propose use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Stockholm Conventions on Persistent Organic Pollutants or the Montreal Protocol*	No
7.4	Will the proposed Project involve the application of pesticides that may have a negative effect on the environment or human health?	No
7.5	Does the Project include activities that require significant consumption of raw materials, energy, and/or water?	No

Annex 6: UNDP Risk Register

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
1	Inability to properly manage EVs (especially batteries) at their end-of-life due to weak ELV management control and insufficient regulations; aggravated if Peru allows import of 2-hand EVs with low remaining battery life.	Environmental Moderate P = 4 I = 2	Probability: High, as ELV management is not well developed, and developed countries are starting to export 2-hand EVs with low battery capacity. Impact: Low environmental hazard, as the number of EV in Peru still very low and at the beginning of their lifespan, even if some EV are imported.	New regulations developed by the project. Responsibility required from producers and importers of EVs used in the project demonstrations	MINAM
2	Traditional gender barriers in the transport sector remain, and women are marginalized in accessing new e-mobility jobs.	Social Substantial P = 3 I = 4	Probability: Medium: women share in transport jobs is very low, but their involvement in innovative technologies, like EVs is higher. Impact: Very high, as gender objectives will not materialize and EV technologies are likely to dominate future road transport.	Specific professional training activities are envisaged within the project, targeting women	MTC/ MIMP
3	Higher public transport fares due to additional EV costs, jeopardizing mobility of low-income groups.	Social Substantial P = 2 I = 5	Probability: Low, as EV costs are rapidly decreasing compared to ICE vehicles Impact: Very high, as current fares are already hardly affordable to some social groups; social opposition to EVs would be likely to follow.	The project includes development of financial and fiscal measures to reduce EV costs and their impact on fares	MTC/ MIMP/Local transport authorities
4	PT operators buying EVs cannot access financing with the same conditions than for ICE vehicles	Financial Substantial P = 4 I = 4	Probability: High, as EV are more expensive and there is not much experience in the country on their lifespan. Impact: Very high, as most PT operators have low financial capacity; EVs would not be introduce very slowly in Peru.	The project develops incentives to reduce financial needs, and awareness raising activities on EV technology; financial partners are included in the project.	MEF/ MEM/ MINAM
5	Electric buses and 3-wheelers unable to cope with current operational requirements (range)	Operational Low P = 1 I = 4	Probability: Low, as EV autonomy is rapidly increasing and EV are intended to be used in the most appropriate routes and services. Impact: High, as this would reinforce prevention against EV deployment. The project would fail to demonstrate the technical feasibility of EVs.	The project is selecting well-proven EV models through a public RfQ. Project partnership include all relevant technical stakeholders.	ATU/ Local authorities
6	Charging infrastructure coverage insufficient for convenient EV rechaging	Operational Moderate P = 2 I = 4	Probability: low, as the project focuses on fleets that will recharge at depots (buses) or at owners' households (3-wheelers) with low technical requirements for charging. Impact: High, as EV would provide a more limited service. The project would fail to fully demonstrate the technical feasibility of EVs.	The project will address this within the E-mobility strategy and action plan. Electricity providers are included as partners.	MEM

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
7	Political instability in Peru jeopardizes the continuity of policies and the approval of regulations and policies.	Political High P = 4 I = 5	Probability: very high, as recent developments in Peru have increased political polarization. Impact: critical, as the sustainability of project results would be compromised. Adoption an implementation of regulations would fail.	The project establishes a coordination space and a national network to institutionalize and to facilitate wide political and social consensus on e-mobility	MINAM
8	Current concession and authorization regimes for PT and 3-wheeler taxis jeopardizes the use of more costly EVs (e.g. leasing options not accepted).	Regulatory Substantial P = 2 I = 5	Probability: Low, as PT authorities are already aware and try to encourage the use of better vehicles. Impact: Very high, as operators would be unable to use EVs. The project demonstrations would not materialize and business models could not be implemented.	The project is building upon the experience of previous pilots; PT authorities are project partners.	MTC/ATU/ Local government
9	Cheaper so-called "clean technologies" like CNG promoted by the government, in spite of their negative impact on GHG emissions	Strategic Moderate P = 2 I = 4	Probability: Low. Although CNG (and LPG) technologies have been widely promoted in Peru, the NAMA Energy project has provided evidence on their high emissions, and costs compared to benefits. Impact: High, as most stakeholders and consumers would not support a more expensive technology if CNG is presented as a "green" option. Introduction of EVs in public transport would not materialize.	The project includes extensive activities on networking and awareness-raising. The e-mobility strategy is expected to address this challenge, establishing an adequate transitioning period.	MINAM
10	The deployment of clean electric 3-wheelers jeopardizes policies to replace them by conventional public transport	Strategic Moderate P = 2 I = 3	Probability: Low, as the advantages (safety, comfort) of buses is overwhelming. Impact: Low, as there are no consistent plans for replacement of 3-wheeler taxi services by regular bus services. The project would not receive support from governments to work on 3-wheelers.	The project will develop guidelines at the end of demonstrations, with PT authorities. Electric 3-wheelers will be considered as a transitional option until safer PT modes can be deployed in those districts.	MTC
11	The competition to regular public transportation by informal operators put concessions at risk of financial failure	Regulatory Low P = 3 I = 1	Probability: Medium, as regulations are not being sufficiently enforced by public authorities. Impact: Low, as the introduction of electric buses with project support is not increasing the financial burden on formal operators.	The project will monitor the pilot corridors making use of e-buses and monitor informal supply and demand, so that authorities can establish an enforcement plan.	Project team. ATU (Lima). SIT (Arequipa)
12	Climate risk: More frequent extreme weather events due to climate change lead to disruptions in the electricity system and to difficulties to charge EVs; therefore, reliance on the electric grid increases the vulnerability of urban transport systems to climate change	Environmental Low P = 2 I = 1	Probability: Low, as Peru is already planning and implementing the adaptation of the electric grid to climate change (NDC, 2016 and 2020 update). Impact: Low, as the resiliency of the electricity grid to changes in climate is similar to that of the fuel distribution system. Also, the effects of mobility constrains can be to a good extent reduced by alternative transport means or mobility demand measures, such as teleworking and teleservices.	MINAM will integrate electrification trends within its adaptation planning process for the transportation system in Peru. The project team will identify alternative charging options in case of electricity disruption in the usual charging points used by the project's EVs.	MINAM

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
13	The COVID pandemic affects project implementation due to lockdown and social distancing measures, and to economic downturn.	Economic/ Health Moderate P = 3 I = 3	Probability: Medium, Peru was severely hit by the pandemic in Spring 2019 and the government is actively improving its capacity of response towards new crisis. Mobility has recovered since then, although public transport ridereship has not reached pre-COVID levels yet. Impact: Medium. Project activities can easily be undertaken under lockdown or social-distancing conditions; public transport authorities and operators are committed in recovering pre-COVID ridership levels, but they need to implement additional measures to provide a safe trip and regain the trust of their customers.	The pandemic impact in Peru has been high until September 2020. COVID prevention policies include mobility as a key area for action, and the project will include COVID lessons learned in other countries. The following mitigation measures will be implemented If the pandemic results in continued application of lockdown and social distancing measures: Components 1, 3 and 4: Physical meetings will be replaced by virtual meetings. Online tools (such as clouds for document preparation) will be used to facilitate the development of draft policies and regulations. Travel to activities of the global programme will be held through means of 'virtual missions' if travel restrictions are established. Component 2: Instructions will be provided for the operation of the vehicles used in the pilot actions, in line with those issued by public transport authorities. Targets for the relevant indicators will be revised accordingly. Additional efforts will be made by PMU to identify potential COVID-contributions of e-mobility (e.g. cleaner air).	Project team by end Y1

Annex 7: Overview of Project Staff and Technical Consultancies

Consultant	Time Input	Tasks, Inputs and Outputs
For Project Management / Monitoring & Evaluation		
Local / National contracting		
Project Manager *Rate: USD4,000/month*	*M1 Full time during 3.5 years (42 months)*	*The Project Manager will be appointed by the project implementing partner. The PM will be responsible for the overall management of the Project, including the mobilisation of all project inputs, supervision over project staff, consultants and sub-contractors.* *Duties and Responsibilities* • *Manage the overall conduct of the project in close coordination with the National Project Director.* • *Plan the activities of the project and monitor progress against the approved workplan.* • *Execute activities by managing personnel, goods and services, training and low-value grants, including drafting terms of reference and work specifications, and overseeing all contractors' work.* • *Monitor events as determined in the project monitoring plan, and update the plan as required.* • *Provide support for completion of assessments required by UNDP, spot checks and audits.* • *Manage requests for the provision of UNDP financial resources through funding advances, direct payments or reimbursement using the FACE form.* • *Monitor financial resources and accounting to ensure the accuracy and reliability of financial reports.* • *Monitor progress, watch for plan deviations and make course corrections when needed within project board-agreed tolerances to achieve results.* • *Ensure that changes are controlled and problems addressed.* • *Perform regular progress reporting to the National Project Director and the Project Board as agreed with the board, including measures to address challenges and opportunities.* • *Prepare and submit financial reports to UNDP on a quarterly basis.* • *Manage and monitor the project risks – including social and environmental risks - initially identified and submit new risks to the Project Board for consideration and decision on possible actions if required; update the status of these risks by maintaining the project risks log;* • *Capture lessons learned during project implementation.* • *Prepare revisions to the multi-year workplan, as needed, as well as annual and quarterly plans if required.* • *Prepare the inception report no later than one month after the inception workshop.* • *Ensure that the indicators included in the project results framework are monitored annually in advance of the GEF PIR submission deadline so that progress can be reported in the GEF PIR.* • *Prepare the GEF PIR;* • *Assess major and minor amendments to the project within the parameters set by UNDP-GEF;* • *Monitor implementation plans including the gender action plan, stakeholder engagement plan, and any environmental and*

Consultant	Time Input	Tasks, Inputs and Outputs
		social management plans; • *Monitor and track progress against the GEF Core indicators.* • *Support the Terminal Evaluation process.* • *Provide regular liaise with the Global Program, answer to its requests (data, updates) and assure active participation from Peru* • *To ensure technical quality, both in terms of TORs and deliverables of activities* • *To facilitate access to UNDP, in its oversight role, of TORS and deliverables of key activities, both in terms of ad hoc requests and standard processes, e.g., PIRs*
Technical assistant *Rate:* *USD 2,000/month*	*Full time since M-7 (36 months)*	*Under the overall supervision and guidance of Project Manager, the Technical assistant will have the responsibility for leading knowledge management outputs and designing and implementing the project communication and capacity building activities across all project components. It will also be responsible for the implementation of project safeguards, gender action plan and stakeholder engagement plan. Specific responsibilities will include:* • *Oversee and monitoring the implementation of the gender action plan.* • *Oversee and monitoring the implementation of the stakeholder engagement plan and reaches out to project's stakeholders.* • *Convene, prepare documentation and provide minutes and reports for project's meetings and workshops.* • *Collect factual information and prepares briefings for the PM supporting the implementation of the project's technical actions.* • *Develop a project communications strategy / plan, incorporate it with the annual work plans and update it annually in consultation with project stakeholders; coordinate its implementation* • *Coordinate the implementation of knowledge management outputs of the project;* • *Coordinate and oversee the implementation of public awareness activities across all project components;* • *Coordinate and oversee the implementation of training and capacity-building activities across all project components;* • *Facilitate the design and maintenance of the project website/webpages and ensure it is up-to-date and dynamic;* • *Facilitate learning and sharing of knowledge and experiences relevant to the project;* • *Develop the implementation of the stakeholder engagement plan, and the gender action plan, and oversees the implementation and monitoring of their activities;O* • *Oversee and guide the design of surveys/ assessments commissioned for monitoring and evaluating project results;* • *Oversee and coordinate the activities of the project working groups.* • *To ensure technical quality, both in terms of TORs and deliverables of activities* • *To facilitate access to UNDP, in its oversight role, of TORS and deliverables of key activities, both in terms of ad hoc requests and standard processes, e.g., PIRs*

Consultant	Time Input	Tasks, Inputs and Outputs
International / Regional and global contracting		
Individual consultant	*M-37*	*Terminal evaluation*
		The Terminal Evaluation will assess the key financial aspects of the project, including the extent of co-financing planned and realized. The evaluator will receive assistance from the Country Office (CO) and Project Team to obtain financial data and other factual evidence, which will be included in the terminal evaluation report. In particular, the Terminal evaluation will include the following: *MAINSTREAMING. The evaluation will assess the extent to which the project was successfully mainstreamed with other UNDP priorities, including poverty alleviation, improved governance, the prevention and recovery from natural disasters, and gender.* *IMPACT. The evaluation will assess the extent to which the project is achieving impacts or progressing towards the achievement of impacts.* *CONCLUSIONS, RECOMMENDATIONS & LESSONS. The evaluation report must include a chapter providing a set of conclusions, recommendations and lessons.* *The principal responsibility for managing this evaluation resides with the UNDP CO in Peru. The Project Team will be responsible for liaising with the Evaluator to set up stakeholder interviews, arrange field visits, coordinate with the Government, etc.* *The Terminal evaluator must present the following qualifications:* • *Master's or equivalent degree in urban transport, urban studies, civil engineering, environment or related field* • *Minimum 10 years of progressive experience in urban transport and mobility planning and development, urban planning and development, environment and in addition experience related to climate change mitigation projects* • *Knowledge of UNDP and GEF evaluation procedures* • *Previous experience with results□based monitoring and evaluation methodologies* • *At least 5 similar evaluation/review tasks in urban transport projects completed* • *Excellent English and Spanish are required* *Assets would include:* • *Experience of implementing GEF funded or relevant/ similar donor funded transport projects* • *Experience in the Latin American region*
For Technical Assistance		
Outcome 1		
Local / National contracting		
National company	*8 months*	*Consultancy services to deliver a draft e-mobility strategy, including economic, social, space and gender impacts, as well as a gender analysis and action plan:* *- Identification of social and territorial impacts of e-mobility.* *- Identification of electrification options for public transport by road and for light vehicles.*

Consultant	Time Input	Tasks, Inputs and Outputs
		- Preparation of the draft e-mobility strategy. *- Guidelines for the design of a national charging infrastructure network.* *- Gender analysis and gender action plan of the e-mobility strategy.*
National company	*11 months*	*Consultancy services on technical, legal and fiscal regulations of vehicles:* *- Analysis of national regulations on vehicle approval and preparation of proposals for the inclusion of electric vehicle and possible barriers to high-carbon vehicles.* *- Development of regulations and standards for installing public and private charging infrastructure, based on existing and proposed legislation and standardization activities.* *- Analysis of the fiscal framework for vehicles (all categories), and development of proposals to incentivize EV and discourage the purchase of high-carbon vehicles.*
National company	*7 meetings in 42 months*	*Organization of meetings and workshops: Services for supporting meetings of the coordination space.* *- Provision of services for the organization of the meetings of the coordination space on e-mobility (8 meetings in total).* *- Organization of capacity-building activities and workshops (7) on e-mobility, transport and energy, addressed to civil servants and to specialists in the transport and energy sectors.*
National company	*27 months*	*Communication activities (knowledge management, communication plan, website management and publication of materials) (also contributing to Outcomes 2 and 3).* *The contractor will be responsible of the implementation of the communication plans and campaigns for the key project activities, the design and publication of training and dissemination materials, and the design, implementation and management of the project website. This includes:* *- The design and publication of capacity-buliding materials on e-mobility addressed to civil servants and professionals.* *- The communication plan for the project demonstrations.* *- The design and publication of the communication and campaign materials for the project demonstrations.* *- The design and publication of training materials on EV operations.* *- Support to the project networking activities, including the design, implementation and management of the project website.* *- Preparation of a campaign addressed to the general public on the advantages of electric vehiicles, to be implemeated by the government and the private sector.*
National company	*4 months*	*Design of communication plan.* *The contractor will be responsible of the preparation of a capacity-building plan as well as the communication plan of the project.* *This includes:* *- The assessment of capacity-buliding needs on e-mobility, with a focus on the urban public transport sector, targeting national and local civil servants and professionals.* *- The capacity-building plan for the project, including budgeting and programming of all training activities.*
For Technical Assistance		
Outcome 2		

Consultant	Time Input	Tasks, Inputs and Outputs
Local / National contracting		
National company	*5 months*	*Design and implementation of demonstrations (can be combined with monitoring and evaluation, Outcome 5):* *- Preparation of best-practice guidelines on EV in public transport (buses and mototaxis).* *- Preparation of recommendations for the design of the demonstrations.* *- Analysis of the potential impact electrification may have on the living and working conditions of 3-wheeler owners and drivers* *- Detailed design of the implementation and management of the pilots in Arequipa and Lima, including a specific analysis to select locations for charging stations without negative impacts in Lima and Arequipa.*
National company	*12 months*	*Design and implementation of multi-media campaigns against gender-based harassment, discrimination and violence, including against other vulnerable groups* *- Design of the multimedia campaign through a participatory process including women and other vulnerable groups, as well as public transport and 3-wheeler professionals.* *- Analysis of public transport users' profiles and selection of communication channels in Lima and Arequipa.* *- Monitoring and assessment of impact on public transport users and other citizens.*
National company	*5 workshops*	*Professional training (on PT jobs, focus on drivers and maintenance) and also on ELV management (component 4)* *The contractor will organize at least 3 hands-on training workshops addressed to future drivers and maintenance staff of electric buses and 3 wheelers, making use of the training materials provided by the project. This includes the availability of a convenient training facility, trainers and vehicles.* *The contractor will design a training programme for policy makers and future professionals on ELV management, and deliver at least two professional y workshops addressed to future staff working in ELV management of EV and batteries.This includes the availability of a convenient training facility, trainers and vehicles.*
National company	*At least 24 months*	*Provision of services with 2 e-buses in each city.* *The contractor(s) will provide regular urban transport service with a minimum of 56,000 km travelled per year with each electric bus within a 24-month period with the following characteristics:* *- The service must be provided with new electric buses, registered at the name of the contractor in the second half of 2021 and provided with a GPS tracking system. The contractor will provide evidence of availability of the buses at the time the services will be expected to start.* *- Contracts will be awarded for 2 buses in Arequipa and 2 buses in Lima.* *-The contractor(s) will hold an authorization to operate a structural high-demand bus line in the metropolitan areas of Arequipa or Lima or will provide evidence of an agreement with an authorized bus operator to run the buses on its behalf.* *- The contractor(s) will guarantee that the operators providing the demonstration services will prepare and implement a gender policy, which will be monitored during the project lifespan.* *- The contractor will be compensated with USD 1.33 per km travelled in regular transport services, up to a maximum of 150,000 USD* *- Monthly payments will be provided, based on the verified km served in regular transport services, based on a GPS tracking system. The contractor will grant access to the GPS tracking system to the PMU.*

Consultant	Time Input	Tasks, Inputs and Outputs
		- Minimum technical specification of the electric bus: *Bus length: 12 m.* *Maximum passenger capacity (seating and standing): 80 passengers.* *Maximum motor power: 2x150 kW* *Maximum climb gradient: 18%* *Battery capacity: 350 kWh* *GPS tracking system including information of km travelled, number of passengers.*
National company	*At least 24 months*	*Provision of services with 10 electric 3-wheeler taxis in each city.* *The contractor(s) will provide authorized taxi service with an electric 3-wheeler for a minimum of 14,000 km travelled per year with each vehicle within a 24-month period with the following characteristics:* *- The service must be provided with new electric 3-wheelers, registered at the name of the contractor in the second half of 2021 and provided with a GPS tracking system. The contractor will provide evidence of availability of the buses at the time the services will be expected to start.* *- Contracts will be awarded for 20 mototaxis in Arequipa or Lima.* *-The contractors will hold an authorization to operate mototaxis in the metropolitan areas of Arequipa or Lima or will provide evidence of an agreement with an authorized mototaxi operator or association to run the 3-wheelers on their behalf.* *- The contractors will show evidence of the availability of the required electric 3 wheeler at the time of starting the demonstration.* *- Minimum in-service mileage travelled: 14,000 km per year.* *- The contractor will be compensated with USD 0.18 per km travelled in regular transport services, up to a maximum of 5,000 USD.* *- Monthly payments will be provided, based on the verified km served in regular transport services, based on a GPS tracking system. The contractor will grant access to the GPS tracking system to the PMU.* *- Minimum technical specifications of the electric 3-wheelers:* *Minimum battery capacity: 4.5 kWh*
For Technical Assistance		
Outcome 3		
Local / National contracting		
National company	*13 months*	*EV market analysis, scenario building, business models and financial schemes.* *The consultant will undertake a comprehensive analysis of the expected trends of the EV market in Peru, including:* *- Development of scenarios for the EV market in Peru.* *- Detailed study of the electrification potential of road public transport systems in Peruvian cities.* *- Development of business models considering all the relevant stakeholders in the value chain of electric buses, 3-wheelers in uirban public transport systems.* *Development of business models considering all the relevant stakeholders in the value chain of light-duty vehicles in private and*

Consultant	Time Input	Tasks, Inputs and Outputs
		public fleets. *- Development of detailed financial schemes (including potential financial institutions and governmental incentives) to facilitte the electrification of buses in urban public transport.* *- Development of detailed financial schemes (including potential financial institutions and governmental incentives) to facilitte the electrification of large private and public light-duty vehicle fleets.* *- Development of recommendations and guidance to public transport operators (buses and 3-wheelers) to electrify their fleets.*
For Technical Assistance		
Outcome 4		
Local / National contracting		
National company	*9 months*	*Design and implementation of a EPR approach to ELV regulations (including batteries).* *This technical assistance will cover the various aspects of ELV management in Peru, with a view to introduce the EPR concept and to include electric vehicles and their components in the system. It will provide:* *- An analysis of current ELV management practice in Peru.* *- An assessment of ELV management needs in Peru and the impact of EV.* *- A review of international best practice and regulation on ELV management for EV-* *- Draft regulations on ELV management for electric vehicles.* *- Development of business models on ELV management and second-life battery use.* *- Proposals to develop commercially-viable EPR business models in this sector.* *- A roadmap for EPR implementation in ELV management of EVs.* *- Support to project activities to secure approval and enforcement of the ELV strategy by the government.* *- Technical and managerial support to producers and importers in order to be involved in a future EPR system.*
For Technical Assistance		
Outcome 5		
Local / National contracting		
National company	*25 months*	*Monitoring and evaluation of demonstrations (can be combined with design and implementation, Outcome 2)* *- MRV plan for the demonstrations in Arequipa and Lima.* *- Assessment of the potential impact of electrification on fares and options to avoid it.* *- Evaluation reports of the demonstrations in Arequipa and Lima.* *- Knowledge management of the demonstration results.*
For Technical Assistance		
Project Management		
Local / National contracting		
Project administration services	*Full time since M-1 (42 months)*	*Under the guidance and supervision of the Project Manager, the Project administration services provided by Profonanpe will have the following specific responsibilities:* • *Keep records of project funds and expenditures, and ensure all project-related financial documentation are well maintained*

Consultant	Time Input	Tasks, Inputs and Outputs
Rate: *USD 2,500 per month*		*and readily available when required by the PM;* • *Review project expenditures and ensure that project funds are used in compliance with the Project Document and GoI financial rules and procedures;* • *Validate and certify FACE forms before submission to UNDP;* • *Provide necessary financial information as and when required for project management decisions;* • *Provide necessary financial information during project audit(s);* • *Review annual budgets and project expenditure reports, and notify the PM if there are any discrepancies or issues;* • *Consolidate financial progress reports submitted by the responsible parties for implementation of project activities;* • *Liaise and follow up with the responsible parties for implementation of project activities in matters related to project funds and financial progress reports.* • *Monitor project progress and participate in the production of progress reports ensuring that they meet the necessary reporting requirements and standards;* • *Ensure project's M&E meets the requirements of the Government, the UNDP Country Office, and UNDP-GEF; develop project-specific M&E tools as necessary;* • *Oversee and ensure the implementation of the project's M&E plan, including periodic appraisal of the Project's Theory of Change and Results Framework with reference to actual and potential project progress and results;* • *Facilitate the terminal evaluation of the project; including management responses;* • *Facilitate annual reviews of the project and produce analytical reports from these annual reviews, including learning and other knowledge management products;* • *Support project site M&E and learning missions;* • *Ensure efficient management of procurement processes for the project through proper monitoring and control of procurement processes, including organization of RFQ, ITB or RFP, receipt and evaluation of quotations, bids or proposals, negotiation of certain conditions of contracts in full compliance with UNDP and GEF rules and regulations.*

Annex 8: Stakeholder Engagement Plan

Introduction

The Stakeholder Engagement Plan (SEP) is designed to ensure effective engagement of all relevant stakeholders throughout the e-mob project lifecycle in Peru. This plan builds upon the interviews and workshops conducted during project preparation. The e-mob project in Peru will aim at maintaining fluid and two-way dialogue with the relevant national and local government institutions and agencies, the private sector, and civil society in Peru for national activities, as well as with international NGOs, the international community and other participating countries at the global programme level.

The process of identifying the private sector actors that are co-financiers of the project consisted of two stages: first, an initial mapping was carried out by UNDP in conjunction with the Ministry of the Environment. This was undertaken through a selection of key stakeholders of the GEF5 project under implementation "Appropriate National Mitigation Actions (NAMA) in the power generation sector and its final use in Peru", mainly the Electric Transportation NAMA which already had identified the private and public sector actors working with E-mobility in Peru. Additionally, workshops were held to socialize the project with representatives of manufacturers and marketers of public transport vehicles (buses, 3 wheelers) as well as transport infrastructure enterprises. This mapping stage considered as main elements for the selection and analysis of stakeholders:

- Identification of actors that could benefit and be disadvantaged with the development of project's activities.
- Level of knowledge / information that the actors have on e-mobility.
- Type of relationships that preponderate between actors (collaboration, conflict, etc.), and their level of interest
- The likely benefits for each actor.

Finally, the second phase consisted of verifying the capacities of the actors identified through the workshops mentioned above as well as in bilateral meetings and interviews that were carried out with them. The possibility is left open that later they may be responsible parties to implement project activities in which case they will be involved and selected in accordance with UNDP policies and rules.

Definitions[49]

Consultation: Consultation involves information exchanges among the government, the Implementing Agency, the project executing agencies, and other stakeholders. Although decision making authority rests with the government, the Implementing Agencies, and the project executing agencies, periodic consultations throughout the project cycle help managers make informed choices about project activities. More important, it provides opportunities for communities and local groups to contribute to project design, implementation, and evaluation.

Public Involvement: Public involvement consists of three related, and often overlapping, processes: information dissemination, consultation, and stakeholder participation. Stakeholders are the individuals, groups or institutions which have an interest or "stake" in the outcome of a GEF-financed project or are potentially affected by it. Stakeholders include the recipient country government; project executing agencies; groups contracted to carry out project activities and/or consulted at various stages of the project; project beneficiaries; groups of people who may be affected by project activities; and other groups in the civil society which may have an interest in the project.

Stakeholder participation: Where stakeholders collaboratively engage in the identification of project concepts and objectives, selection of sites, design and implementation of activities, and monitoring and evaluation of project outcomes. Developing strategies for incorporating stakeholder participation throughout the project cycle is particularly necessary in projects which have impacts on the incomes and livelihoods of local groups, especially disadvantaged populations in and around project sites (e.g., indigenous peoples, women, poor households).

[49] From GEF (2018). Guidelines on the Implementation of the Policy on Stakeholder engagement. Retrieved from https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=1&cad=rja&uact=8&ved=2ahUKEwim9PnSm5TmAhWD34UKHZScBRkQFjAAegQIBRAC&url=https%3A%2F%2Fwww.thegef.org%2Fsites%2Fdefault%2Ffiles%2Fdocuments%2FStakeholder_Engagement_Guidelines.pdf&usg=AOvVaw0eP2ZYZbiiUrD-sKV5Kqs_

Legal requirements for public consultation in Peru

In 2011, the Law on Prior Consultation (*Ley de consulta previa*) was approved, followed by its Regulation in 2012. It refers to consultation to indigenous communities for those legislative or administrative decisions that directly impact on them. This law is not relevant in this case, as the project is not directly affecting indigenous communities.
Citizens' participation and access to environmental public information is regulated since 2009 (*Decreto Supremo N° 002-2009-MINAM que aprueba el reglamento sobre transparencia, acceso a la información pública ambiental y participación y consulta ciudadana en asuntos públicos*). It only covers the environmental impact assessment procedure; in accordance with Peruvian law, the contents of the e-mob project are not subject the environmental impact assessment process. However, it provides general principals on public participation in the development of public policies, which serve as a reference for this project.

GEF guidelines[50]

All GEF funded projects are required to meet best international practice and specifically the requirements for stakeholder engagement and public consultations. The project stakeholder engagement activities should be robust, and enough disclosure on information should be made in order to promote better awareness and understanding of its strategies, policies and operations. During disclosure, the project is required to: (1) Identify people or communities that are or could be affected by the project as well as other interested parties; (2) ensure that such stakeholders are appropriately engaged on environmental and social issues that could potentially affect them, through a process of information disclosure and meaningful consultation; and (3) maintain a constructive relationship with stakeholders on an on-going basis through meaningful engagement during project implementation. The stakeholder consultations are an on-going process taking place during the project life and during this process it is necessary to ensure that stakeholders are informed about environmental and social consequences of the project implementation and ensure the opportunity for feedback.

Identification of stakeholders for engagement and methods of communication

In order to ensure inclusive participation and consultation, the following stakeholders have been identified for consultation. The list includes the identified social groups that are associated with the project in different ways: those directly or indirectly by the outcomes of project implementation; those directly or indirectly participating in the project, and those with a capacity to influence and decide on project implementation and outcomes.

- National government. MINAM and MINEM have prioritized e-mobility within their policies to reduce climate change emissions, improve air quality in cities and increase energy efficiency in the transport sector. MTC recognizes the contribution EVs can make in its effort to improve public transport in cities. Other ministries identify some opportunities, such as national industry expansion (PRODUCE and *Instituto Tecnológico del Perú*, *ITP*) or urban development regulations of interest for the MVCS. The project will support the coordination among these ministries and the establishment of a national strategy.
- Local governments and public transport authorities. Local governments in the participating cities have identified public transport as a key political priority; this is being translated into updated urban mobility plans, revised concession contracts and competences being transferred to public transport authorities, in cooperation with MTC. The project will work with the pilot cities in the integration of EV within their sustainable mobility plans and public transport provision contracts.
- Bus operators. Following a successful transition, bus networks are in the process of being or have already been restructured, and an increasing number of high-demand bus lines are being served by operators with large fleets and stronger financial capacities. These operators are the key partners in the project demonstrations and also a priority target for capacity building and awareness-raising project actions. They are expected to be interested in knowing EV as an emerging technology, although initially, their interest in using EVs can be low, as it requires a higher upfront investment with long-term returns.
- 3-wheelers associations authorized by local governments provide key support and services to their associates; they are expected to play a key partner role in the project, facilitating the access of individual 3-wheeler taxi

[50] From GEF (2018). Guidelines on the Implementation of the Policy on Stakeholder engagement.

operators to information, technical support and financial opportunities to replace their gasoline (or gas) vehicles by electric ones.

- The financial sector is a key project partner to facilitate the access of operators to EVs. As EV provide a substantial reduction in operating costs in exchange of higher upfront capital costs, it offers a significant business opportunity to the financial sector to develop appropriate loans and leasing schemes. COFIDE will participate in the project, and private banks will also be invited to project activities. COFIDE has already experience in financing bus fleet renewal (through its COFIGAS programme) and is currently designing a new bus fleet renewal instrument with the Inter-American Development Bank (IDB).
- EV manufacturers, dealers and importers are essential for the deployment of EV in Peru. The project includes as partners local bus and 3-wheeler manufacturers, as well as national branches and importers of global EV producers. For national manufacturers, the expansion of EV-technologies provides an opportunity to compete with incumbent foreign industries in a still fast-developing technology (some companies (QEV, MODASA) are considering to initiate e-bus manufacturing). For global EV forerunners, the project provides an opportunity to enter a fast-growing market, in accordance with public transport fleet renewal forecasts.
- For players in the electricity sector, the project provides an opportunity to increase electricity demand at a time in which most of the country is served by an integrated system, with a strong and still growing share of renewables and significant spare capacity. Furthermore, the project will support the consolidation of new market opportunities, such as infrastructure charging or electric bus leasing concepts based on energy provision.
- Sectoral associations and special-interest groups such as AAP and AEDIVE provide to the project strong knowledge on the automotive market and access to a thick network of market players.
- University departments and other academic institutions are keen in participating in the project, providing their knowledge and looking for upcoming challenges to accommodate e-mobility technologies within their curricula and their research agendas.
- Some grassroots associations and civil society stakeholders have also been active in urban mobility from different sensitivities: the environment, urban living conditions, equity in mobility, safety and security.

Stakeholders to be affected, directly or indirectly, by the outcomes of the Project implementation	**Stakeholders that participate in the Project implementation**	**Stakeholders being able to influence and decide on the Project implementation or use project outcome for decision making**
PT bus operators and staff 3-wheeler taxi operators Vehicle dealers and importers Local population in two pilot cities, within the area of influence of the demonstrations Persons receiving training in e-mobility Citizens in the pilot cities	Project staff National government (MINAM, MEM, MTC) Members of Project Steering Committee PT operators participating in the demonstrations 3-wheeler taxi operators participating in the demonstration Local public transport authorities EV suppliers (dealers, importers, maintenance…) EV charging infrastructure suppliers Financial partners Academia Civil society organizations and networks	National government (MINAM, MEM, MTC). National government (MEF, MVCS, MIMP). Local government (municipal, provincial, metropolitan and departmental levels) in pilot cities Public transport authorities in pilot cities Energy regulator (OSINERGMIN)

Table A8.1: Identification of stakeholders

Summary of any previous stakeholder engagement activities

This Stakeholder Engagement Plan (SEP) builds upon the interviews and workshops conducted during project preparation. All the workshops and most of the interviews took place during the two missions (November 2019

and February 2020) carried out by the international consultant. The national consultant participated in all these activities and carried out additional interviews with those stakeholders that were not available during the missions.
Meetings during the mission in November 2019 included the national government (MINAM, MINEM, MTC MVCS, PRODUCE, Osinergmin), the industrial sector (MODAS, QEV Tech, AEDIVE, BYD, Asociación Automotriz), the Municipality of Lima, the Public Transport Authority of Lima (ATU), the electricity utilities (Engie, Enel X), and one of the bus operators in Lima (ALLIN Group). The mission included a co-design workshop focusing on the project's components and another workshop on social, environmental and gender issues.
Meetings during the mission in November 2020 focused on the national government stakeholders. Two final validation workshops were held, one focusing on social, environmental and gender safeguards and one on the final validation of the project.
Outside the two missions' dates, the national consultant held bilateral meetings with financial institutions (IADB, CAF, COFIDE), and with the local stakeholders in Arequipa (municipality, public transport authority (SIT) and bus operators).
All the meetings followed a similar methodology: the MINAM representative provided an overview of the project and its consistency with the national government's policy and priorities. This was followed by a detailed presentation of those aspects of the project closer to the interests of the participants. A discussion followed, aiming at identifying co-financing opportunities and to assess the consistency of the project's approach and the prospects for the involvement of the participants.
The workshops followed UNDP's general guidance of social and environmental safeguards, including presentation of other projects in the same area and their social, environmental and gender impacts, detailed discussion on potential project's impacts and risks and discussion on mitigation measures, if relevant. Workshop participants also considered how to better integrate the results of the discussion within the project's structure and contents. The project's Social and Environmental Screening Process (SESP) was distributed among participants to validate its contents.
During the general validation workshops, the participants were distributed in different groups addressing their main shared interests and discussing those project components more relevant for them. This was followed by a general discussion. A summary of the draft project document was distributed in advance among the participants, to facilitate the discussion.
During its implementation, the project is expected to maintain a fluid two-way dialogue with the relevant national and local government institutions and agencies, the private sector, and civil society in the country, as well as with international institutions and the countries participating in the UNEP/GEF e-mobility global programme. The SEP provides the framework to make it possible.

The identification of the stakeholders potentially interested in the project built upon activities in previous and on-going projects, and the knowledge of the various ministries involved in their respective areas of competence.

The list of stakeholders invited to participate in the activities during project preparation is provided in the Table below.

Stakeholder name	Stakeholder group
MINAM, MINEM, MEF, MTC, PRODUCE, MVCS, FONAM,	National government
Municipalidad Metropolitana de Lima Municipalidad de Arequipa	Local Government
Sistema Integrado de Transporte Arequipa (SIT) Autoridad de Transporte Urbano para Lima y Callao (ATU)	Public Transport Authorities
Transportes Cruz del Sur, Perú Masivo, Lima Vías Express, Transvial Lima, Perú Bus Internacional, Allin Group, Consorcio Transporte Arequipa, Consorcio Nueva Alternativa, Expreso Próceres Internacional, Consorcio Empresarial Futuro Express, Consorcio Santa Catalina, COTRANSCAR, Transporte Ecológico Vial,	Bus operators
Confederación Nacional de Mototaxis del Perú	3-wheeler taxi associations
COFIDE, IADB, CAF,	Financial sector

Stakeholder name	Stakeholder group
SENATI, BYD, SINOMAQ, Modasa, DEUMAN, Mitsubishi, ABB, Laboratorio Diesel Senatinos, Euromotors, Samcorp,	EV & charging infrastructure manufacturers, car dealers
Engie, CITE Energía, Enel, Enel X, Enel LATAM,	Electricity utilities
CAVI, APVEA, AEDIVE, EPEI, Cámara Peruana de Energías Renovables, Global Sustainable Electricity Partnership	Sectoral associations and groups
UTEC, Pontificia Universidad Católica del Perú, Universidad Nacional de Ingeniería,	Academia
Fundación Transitemos, Lima Cómo Vamos, CIDATT, Global Green Growth Institute, Sustainable Earth, Agencia Francesa de Desarrollo Swisscontact Agencia Suiza para la Cooperación (COSUDE) GiZ	NGOs and civil society associations
QEV Technologies, Eco Energy, Green Energy, WSP,	Consultancy firms

Table A8.2: List of stakeholders

Stakeholder Concerns Analysis

During its planning stage, the project has organized two design workshops in March 2019 and November 2019, as well as a final validation workshop in February 2020. Interviews with all the stakeholders identified above have taken place in order to identify their concerns and priorities and to integrate their know-how. As a result, the project is taking appropriate responsive measures throughout its lifespan to consolidate and maintain the wide current support built up during the planning stage. As a result of the project design activities, the interests and concerns of the key stakeholder groups have been identified and are presented in the table below.

Stakeholder group	Key expectations	Key concerns	Recommendations for project implementation
National government	Project support to policies on climate change mitigation, improvement of air quality in cities energy efficiency and sustainable urban transport. National e-mobility strategy delivered by the project Quicker and ordered deployment of EVs in Peru	Fiscal costs of policies incentivizing public transport. Legal framework need to be updated. Impacts on the electricity sector (infrastructure, tariffs).	An institutionalized multi stakeholder, intersectoral and multilevel coordination space is needed for the various activities developed by different governmental services on e-mobility
Local governments and PT authorities	Improvement in service quality of buses and 3-wheeler taxis Improved air quality Coordinated strategy and legislation on e-mobility provided by national government	Affordability of EVs and impact on fares Public's acceptance of EVs Compatibility of EVs with on-going urban mobility strategies	Involvement of local governments (transport authorities) in PSC Regular coordination of project activities at local level
Vulnerable social groups	Affordable mobility services, better tailored to their particular needs and priorities	E-mobility could increase fares or could focus on affluent users	Involvement of vulnerable groups in the design of local demonstrations
Bus operators	Support to access EV technologies at reasonable costs, in order to benefit from	Affordability of EVs and impact on operational costs Impacts of staff training	RfQ for selection and training of operators participating in the pilot

Stakeholder group	Key expectations	Key concerns	Recommendations for project implementation
	lower fuel consumption and maintenance. Capacity building on EV technology, its potential and limitations	Additional investment needs on charging infrastructure and prevention towards EV safety and reliability	Extensive access to pilot results and dissemination of lessons learned among operators Technological-risk reduced through demonstrations
3-wheeler taxi associations	Support to access EV technologies at reasonable costs, in order to benefit from lower fuel consumption and maintenance. Capacity building on EV technology, its potential and limitations	Uncertainty on the future of 3-wheelers as mass transport expand and improve Affordability of EVs and impact on operational costs Training needs Additional investment needs on charging infrastructure	RfQ for selection of associations and individuals participating in the pilot Extensive access to pilot results and dissemination of lessons learned among operators
Financial sector	New financial schemes and business opportunities associated to e-mobility	EVs more expensive, with uncertainties and limited experience in key issues such as reliability or lifespan	New business models, including its financial viability developed in consultation with sector, Technological-risk reduced through demonstrations
EV manufacturers	Strong support to the expansion of EVs in Peru, as a result of project activities	Negative image of EVs if demonstrations or other project activities are poorly managed	Demonstrations and project dissemination activities carefully designed
Electricity utilities	Increased electricity consumption outside peak demand period. New business opportunities linked to charging infrastructure and services	EV market developing without clear regulations, and imposing additional burdens on utilities	Charging infrastructure and actions in the electricity sector to be included in the national e-mobility strategy within the project
Sectoral associations and groups	Clear EV regulations established	Anarchic market competition of ICE vehicles and EV due to lack of or inappropriate regulations.	Stakeholders' involvement in the development of the projects' regulatory proposals
Academia	New research and educational opportunities linked to EV	Insufficient research resources and limited ability of the project to influence on this	Networking with other projects and research partners within the e-mob global programme
NGOs and civil society associations	Implementation of sustainable mobility practices in Peruvian cities. Improvement of urban mobility conditions of vulnerable groups and women	EV focus could compete for scarce resources and efforts with other priorities in sustainable mobility needs (bus infrastructure, car restrictions)	The project approach intends to mobilize additional resources and efforts from private stakeholders, rather than competing for the existing ones

Table A8.3: Key interests and concerns of stakeholders

The project will engage or communicate to the identified stakeholders as outlined below.

Stakeholder group	Means of engagement	Rules for communication
National government	PSC Regular meetings with PMU	Communication to be done by persons authorised to communicate. In accordance with administrative procedure requirements.
Local governments and PT authorities	PSC Regular meetings with PMU	Communication to be done by persons authorised to communicate In accordance with administrative procedure requirements.
Vulnerable social groups	Regular consultation meetings at national and local (demonstration cities) level	Internal project communication rules to be defined according to communication strategy.
Bus operators	Exchange of correspondence, meetings, training courses, demonstration design supervision. Data collection templates and procedures	In accordance with external project communication procedures[51].
3-wheeler taxi associations	Exchange of correspondence, meetings, training courses, demonstration design supervision. Data collection templates and procedures	In accordance with external project communication procedures.
Financial sector	Exchange of correspondence, meetings	In accordance with external project communication procedures.
EV manufacturers	Exchange of correspondence, meetings	In accordance with external project communication procedures.
Electricity utilities	Exchange of correspondence, meetings	In accordance with external project communication procedures.
Sectoral associations and groups	Exchange of correspondence, meetings, training activities	In accordance with external project communication procedures.
Academia	Exchange of correspondence, meetings, training activities	In accordance with external project communication procedures.
NGOs and civil society associations	Exchange of correspondence, meetings, training activities	During public meetings and on-demand

Table A8.4: Engagement and communication approach

Making Information Available

The project will have a knowledge management transversal approach, which will ensure to make information available to stakeholders and to the public, so that they can know the environmental and social risks and impacts associated with the project, as well as the opportunities it provides. Project data will enable them to take better-informed decisions in urban transport and e-mobility.

On an ongoing basis, the project will provide disclosure and consultation on the project's environmental and socio-economic performance to all stakeholders thorough project briefs or annual reporting. The project will also provide:

• An update on the project achievements and its contributions to enhancing transparency.

[51] To be defined at the beginning of the Project.

• An overview of the stakeholder engagement process and how affected parties can participate and provide feedback through meeting or other channels.
• Project impacts and how the government is using the project data to enhance the mobility of urban population and to reduce GHG emissions in Peru.

Monitoring and Reporting

Monitoring is an integral component of project management as it tracks and assesses progress towards achieving tangible development results associated with the project being implemented. It is an essential management tool which provides an opportunity to know whether results are being achieved as planned, what corrective action are needed to ensure delivery of the intended results and how they are making positive development contributions. This helps to detect problems earlier and coming up with appropriate measures to address them. Therefore, monitoring usually provides data used for analysis and synthesis prior to reporting for decision making. Monitoring and reporting responsibility and period are presented in the Table below.

	Parameter	Monitoring and reporting responsibility	Reporting period
1	Number of government agencies, civil society organizations, private sector and other stakeholder groups involved in the project implementation phase	PMU	Annual
2	Number of persons (sex disaggregated) involved in project implementation	PMU	Annual
3	Number of engagement activities (e.g. meetings, workshops, consultations) with stakeholders during project implementation	PMU	Annual
4	Satisfaction rate provided by stakeholders on how their concerns are integrated in the project	Academia partners	Annual
5	Grievances handling mechanism (how grievances are received and results communicated to all stakeholders)	PMU, UNDP Social and Environmental Compliance Unit (SECU)	Annual

Table A8.5: Monitoring and reporting

Stakeholder management programme

Stakeholder engagement actions are identified in the table below, including materials, responsibility, location and dates.

Stakeholder group	Engagement method	Materials to use	Location	Responsible	Date
National government	Briefings on project implementation status. Training meetings on regulatory and data collection issues	Project reports Project surveys and data bases	Governmental offices	PMU	Quarterly
Local governments and PT authorities	Scheduled meetings on local demonstrations	Presentations, reports	Project office, local governments' offices	PMU	Quarterly (until end of demonstration)
Vulnerable social groups	Consultation meetings Targeted group meetings, if necessary	Website, presentations, leaflets	Local civic centers	PMU	Quarterly (until end of demonstration)
Bus operators	Scheduled meetings on local	Presentations,	Project office	PMU	As required

Stakeholder group	Engagement method	Materials to use	Location	Responsible	Date
	demonstrations and business models	reports			
3-wheeler taxi associations	Scheduled meetings on local demonstrations and business models	Presentations, reports	Project office	PMU	As required
Financial sector	Scheduled meetings on EV financing options for demonstrations and business models	Presentations, reports	Project office	PMU	As required
EV manufacturers	Scheduled meetings on EV options for demonstrations and business models	Presentations, reports	Project office	PMU	As required
Electricity utilities	Scheduled meetings on energy supply options for demonstrations and business models	Presentations, reports	Project office	PMU	As required
Sectoral associations and groups	Scheduled meetings; workshops	Presentations, reports	Project office	PMU	As required
Academia	Design workshops, evaluation workshops	Presentations, reports	Project office	PMU	As required
NGOs and civil society associations	Workshops: design, evaluation and dissemination	Presentations, reports	Project office	PMU	As required

Table A8.6: Stakeholder management programme

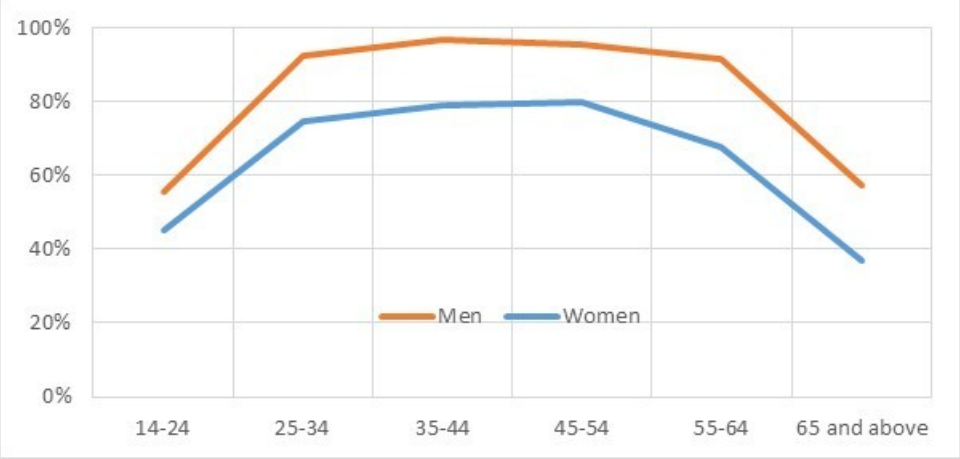
Annex 9: Gender Analysis and Gender Action Plan

1. Gender Analysis

1. Context of the legal status of women in Peru	According to the 2017 national census, Peru's female population totals 14,931,100 women, comprising 50,8% out of the country's total of 31,237,385 (INEI, 2018a). 79.4% of the total female population currently lives in urban areas, as a consequence of centralization and rural-urban migration patterns. An example of this is the fact that only Lima, the capital city, concentrates 41% of the urban female population and 29% of the national total. In contrast, a third of the total female population lives in rural areas in the Andean and Amazon regions. This trend is further exemplified by the fact that, while in urban areas there are more women than men in all age groups above 15 years of age, in rural areas there are more men than women in the majority of age groups, except those over 65 years of age. These contrasts can be explained by the emigration of women towards urban areas as they search for better living conditions, a trend that may be further accentuated as urbanization continues to increase in the country. Only between 2007 and 2017, the urban population in Peru grew by 17.3% (INEI 2018). In this context, the pervasiveness of gender inequality and gender-based violence remain serious challenges in Peru. In 2018, the World Economic Forum scored Peru with a 0.72 Gender Gap Index, which measures gaps between women and men in economic participation, educational attainment, health and survival, and political empowerment (with a score of 1 representing the equality benchmark). Peru ranked 52nd out of 149 countries in the overall index, lagging behind other South American countries such as Bolivia, Argentina, Colombia and Ecuador (WEF 2018). Yet, in the economic participation and opportunity sub index, Peru fell even further behind, occupying the 94th place worldwide. Furthermore, according to INEI's 2018 Gender Gap Report, Peru's Gender Inequality Index is 0.385, which indicates a combined percentage loss of 38.5% on improvements made in reproductive health, empowerment and participation in the labour force due to gender inequalities. While a positive improvements regarding gender inequality have been observed in Peru over the past decades (Peru's GDI score was 0.527 in 2000, compared to 0.385 in 2018), statistics confirm that significant gaps persist between women and men in a variety of key development indicators, where women continue to be substantially disadvantaged in access to health, education, labour and political participation, among others. On average, in urban areas women earn 66.6% of men's income, a gap that increases for the age group of women and men over 45 years old, where women earn 63% of men's income. As for employment, in urban areas 80% of men and 62% of women of working age are employed, while in rural areas these figures change to 89% for men and 73% for women (INEI 2018b). This last statistic must be read considering that women continue to emigrate from rural to urban areas, with the latter being where most of Peru's female population currently resides. Furthermore, these gender gaps also intersect with other identities that amplify inequalities, where the greatest barriers and discrimination are faced by women living in poverty, rural and marginal peri-urban women, indigenous and afroperuvian women, women with disabilities and LGBTQ women. Peru is also facing a critical situation regarding violence against women, ranked as the second country in the region with the highest levels of sexual violence. Reported cases of sexual and domestic violence have increased by 26.2% and 79.3% respectively between 2012 and 2018. In 2018, 63.2% of women aged 15 to 49 years old reported having been victims of intimate partner violence at some point during their lifetime (INEI 2019). It is important to note that while intimate partner violence against women occurs across all socioeconomic levels, the data shows a higher prevalence among lower socioeconomic groups. The pervasiveness of gender violence in Peru resulted in 166 cases of femicide registered in 2019, the highest number in 10 years. Lima, the capital city, continuously registers the highest number of femicides in the country. Women also experience high levels of violence in public spaces, with 7 out of 10 women between 18 and 29 years having been victims of street sexual harassment, and 9 out of 10 in Lima and Callao (Vallejo & Rivarola, 2013). The physical, psychological, economic and sexual violence experienced by women ultimately affect their opportunities and development capacities,

	increase their vulnerability to various social and environmental risks (such as climate change), and further deepen gender gaps[52]. In light of these challenges, the Peruvian Government has recognized the importance of addressing gender gaps to achieve the country's development goals. Peru is a party of the 1979 Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) and its Facultative Protocol since June 5th, 1982. In line with these commitments and following the CEDAW's reports and recommendations, Peru has enacted several laws and regulations that aim to combat gender-based violence and inequalities, particularly with the benchmark creation of the Ministry of Women and Vulnerable Populations in 2000. In 2007, the National Policy on Gender and Equal Opportunities between Men and Women was established as a national policy of compulsory enforcement by all national entities. That same year, the Law 28983 on equal opportunities for women and men was passed, which mandates that the State has the obligation to implement public policies under a gender approach. Subsequent policies and plans have sought to address structural gender discrimination, such as the National Policy on Gender Equality (Supreme Decree N°008-2019-MIMP), Law N°30709 for equal remuneration between working men and women enacted in 2017, and the Ministry of Employment's 2018-2021 Sectoral Plan for Equality and Non-discrimination in employment (Ministerial Resolution N°061-2018-TR). With regards to combatting gender violence, Peru enacted Law N° 26260 in 1993 for the Protection against Domestic Violence, which recognizes intra-family violence as an infringement on fundamental rights. In 2013, an amendment was made to the Criminal Code and the Penal Enforcement Code to prevent, punish and eradicate femicide, punishable with up to life sentence. In November 2015 Law N° 30364 was passed to eradicate violence against women, establishing mechanisms, measures and integral policies to prevent, provide care and protection to victims, and to prosecute, sanction and reeducate sentenced offenders. The 2016-2021 Plan against gender violence is also currently in effect and under implementation, under the purview of the Ministry of Women and Vulnerable Populations. While these constitute important advancements in Peru's legal framework for the protection of women's rights, deeply entrenched and institutionalized traditional and patriarchal sociocultural norms still challenge further progress and the full respect and protection of women's reproductive and sexual rights. On several occasions, the CEDAW has expressed concern over discriminatory practices on the part of judicial authorities, prosecutors and police officers that dismiss women victims of violence and thus prevent their access to justice. Furthermore, the incorporation of the gender approach in Peru's National School Curriculum in 2016 to promote tolerance, non-discrimination and gender equality, received significant backlash from conservative groups and congressmen, who made bids to exclude what they consider to be a "gender ideology" that confuses children's gender identity and encourages homosexuality. These sociocultural norms represent substantial challenges to fully incorporate gender-responsive policies across sectors and thus ensure the fulfilment of women's rights in Peru. It is important to mention that, in the case of the environmental sector, the 2015 Gender and Climate Change Action Plan prepared by the Ministry of Environment constitutes a key strategy for gender mainstreaming in environmental policy, as it recognizes the differentiated impacts of climate change on women and men, and the fact that climate change mitigation and adaptation measures often do not take into account the need to address women and men's differentiated needs and ensure their full participation, thus risking to create or reinforce already existing gender inequalities. The Plan specifically addresses gender gaps in the energy sector, highlighting the low enrollment of women in energy-related technical or postgraduate degree courses, and consequently the low representation of women in managerial positions in the energy sector. Furthermore, it highlights barriers that women face in their use of public transport services due to costs, unreliability of the service and because of the

[52] Violence against women has increased during the COVID-19 quarantine, and in mid-June, the Press reported 21 femicides, and 25 serious live-threatening domestic events in Peru, as well as 1,000 women reported as missing and 342 teenagers and girls raped.
https://www.elperiodico.com/es/internacional/20200615/crecen-los-feminicidios-y-violaciones-en-peru-durante-el-confinamiento-8000538

	probability of being sexually harassed. Thus, the Plan pushes for the need to involve women in the energy production, distribution and consumption chains, at the operational and managerial level.
2. Division of labour	Over the past decade, an important increase in the number of women participating in the labor market has been observed in Peru, with important implications for household wellbeing, the reduction of poverty and the closure of development gaps. While in 2007 the female labor in Peru force totaled 6,553,000 women, by 2017 this number had risen to 7,766,900. Yet, the rate of female participation is still lower than male participation, while important gaps persist between urban and rural areas and by educational level, age group, ethnicity, and others. Indeed, according to the INEI (2018a) data shows that the participation of women in the formal economy is always lower than that of men, regardless of the age. For men and women over 14 years old, the participation in formal employment is 80.7% and 64.0% respectively. This trend is observed in the following graph:  *Source: INEI. Indicadores de género (https://www.inei.gob.pe/estadisticas/indice-tematico/brechas-de-genero-7913/). Year: 2018* This trend is further supported by other persistent gender gaps in the labor market: the unemployment and underemployment rates are higher for women than for men (4.4% vs. 3.5% and 55.2% vs. 34.3% respectively). The shares of women working in unpaid family businesses and households are also substantially higher than for men (15.3% vs. 5.5% and 4.9% vs. 0.2% respectively). These gaps evidence the persistence of the traditional sexual division of labor, in which women engage in unpaid work while men work in the formal economy. Indeed, women not only dedicate time to economic activities to generate income, but also a significant amount of time for childcare and unpaid domestic work. It is estimated that women in Peru work over 75 hours per week, 9 hours and 15 minutes more than men, and dedicate more time than men to domestic work. While women dedicate 39 hours and 28 minutes per week to unpaid work, men dedicate 15 hours and 54 minutes. In contrast, men dedicate more time to paid work, 14 hours and 19 minutes more than women. The disproportionate burden of domestic work on women is persistent across all age groups. It is indeed telling that the age group for men that dedicates the most time to domestic work per week – 20.27 hours per week for men over 65 years old – is still lower than the age group for women that dedicates the least amount of time to domestic work per week – 21.55 hours for women between 12 and 17 years old (INEI 2018). As a result, the sexual division of labor and the disproportionate burden that women bear for childcare represents an important challenge for their economic empowerment and autonomy. An analysis of maternity and insertion in the labor force for women sheds light on increased barriers they may face due to tensions between responsibilities related to their domestic and income-generating activities. In 2018, INEI reported that during the last 11 years, the participation of women with children under 6 years old in the labor force, the economically active population and the employed population has decreased, while in contrast the participation of women without children under 6 years old has increased. Likewise, the employment rate for women with children under 6 years old was 63.6%, while for women without children under 6 years old the rate rose to 69.1% (INEI 2018). Important gender gaps are also reflected in female participation with regards to the sector of employment and

	occupational group. 67.6% of women work in services (41.1%) and commerce (26.5%), compared to 24% and 12.8% for men, respectively. One of the largest gaps is observed in the transport and communications sector in Peru, where female participation is very low compared to men's (2.1% vs. 13.6%), suggesting the pervasiveness of gender barriers to access jobs in this sector (INEI 2018). Data for 2008 also showed that, with regards to independent workers, 98.4% of those who carried out their activity in a vehicle to transport passengers were men, while only 1.6% were women. As the Gender and Climate Change Action Plan suggests, low enrollment rates of women in technical or professional degree courses in these fields may explain this gap. The fact that these sectors are significantly male dominated may also reinforce gender stereotypes that discourage women from engaging in these activities This context highlights the need for the proposed project to support greater female participation in the sectors related to electric mobility, providing them the necessary tools to access new jobs associated with the electric vehicles that will be piloted. Gender stereotypes and other barriers that stem from the sexual division of labor must be taken into consideration to ensure the effectiveness of the interventions, and to avoid reinforcing already existing gender gaps.
3. **Access and control over resources and services**	The access and quality of public transportation is a significant factor that impacts women's economic, social and political autonomy. Women not only use public transportation to go to work, but also to carry out activities related to childcare and domestic work. A low quality of service, high costs and the lack of safety may discourage citizens from using public transport, and women face higher barriers given their condition of vulnerability due to the pervasiveness of gender violence. Indeed, women in Peru report often facing sexual harassment in public transport. In Metropolitan Lima, 29% of women (vs. 10% of men) declared to have been harassed while using public transport (Lima Cómo Vamos, 2019). Harassment includes inappropriate staring, vulgar gestures, whistles, nonconsensual touching and indecent exposure (Vallejo & Rivarola, 2013). According to a study by the IADB in 2016, the perception of insecurity in public transport in Lima is among the highest in the region, with a high incidence of sexual violence and victimization and yet very little reporting of these crimes (around 70% of victims do not report them to the police). The study also showed that the perception of insecurity is higher among women who do not regularly use public transport, which indicates that insecurity may negatively impact and limit women's choices in mobility. In fact, 53% of surveyed women reported having changed the transportation means to be safer, and 16% do not travel at night anymore. Still, public transportation is the main transport means for women in Lima, with 74% of surveyed women heads of household using it for their most frequent travel (IADB 2016). In August 2019, the Ministry of Transport and Communications requested all bus operators to include a warning notice inside the bus with legal penalties for improper behavior. In 2015, Law N° 30314 requested regional, provincial and local governments to issue ordinances with financial penalties for sexual harassment in public spaces. Since 2018 (D.L. n° 1410), sexual harassment in any space is an offence included in the penal code. The availability of adequate space for comfortable travel in public transportation and the compliance with schedules have been identified as important factors that can reduce the likelihood of victimization. This stresses the importance of gender-sensitive public transport design to reduce gender gaps related to mobility, which in turn are key reduce further socioeconomic gender gaps. As for employment, the gender gap identified in the participation of women and men in the transport sector, and specifically as service providers for the transportation of passengers (2% women vs. 98% men) also indicates the lack of access for women to employment opportunities in public transportation. There is no data regarding the participation of women in maintenance of public transport services, yet it is likely that the gender gap persists. The project aims at empowering women participation using as a leverage the introduction of the new electric vehicles in various companies' fleets, which will be tested during the project. This will include the improvement of public transport quality as a way to improve mobility conditions for women, taking into consideration gender-responsive design in electric buses.
4. Participation in decision-making	In September 2017, the National Institute of Statistics (INEI) published the study "Perú: Brechas de género 2017", which included the government, and the public and private sectors. **Women elected for national congress in Peru, 1956-2021**

	 ***Source: INEI, 2017*** As the figure above shows, the number of women elected as MP has consistently increased in the last two decades. However, the share of women has slightly decreased during the last two congressional periods, compared to the 29.2% peak reached in 2006-2011, and remains below the share established in national legislation. The current cabinet of ministers consists of 19 members, including its president. Ten 10 women are members of the cabinet. The presence of women in local councils (*regidoras electas*) remains below 30% (28.5% in 2015-2018). The percentage of women in high governmental positions in autonomous organisms, decentralized services, autonomous entities and public companies does not exceed 35%, except in the social sector, where it reaches 53%. As transport is a sector with high male participation and low participation of women, it can be expected that the gender gap will increase in decision-making positions. However, there is no information available on the characteristics of the management positions in public transport companies in Lima or other cities in the country.
5. **Education and knowledge**	In terms of access to education, the levels achieved by people over 15 years old are very similar for men and women: 20.3 and 10.0 years, respectively. Although consistently decreasing over time, the level of illiteracy remains thrice higher for women over 15 than for men: 9% vs. 3%. About 35% of women and 43% of men have completed secondary education and 15% of women and 18% of men have completed a university (or postgraduate) degree. As the project has components related to technology and mobility, strong interaction can be expected with the university, and with professionals linked to different branches of engineering. In this sector, the percentage of male students is thrice that of women. However, the percentage of women enrolled in higher education studies is higher (32.4% of women and 29.8% of men). In addition to the project's connection with the scientific-technological area, the incorporation of electric mobility requires that drivers and vehicle maintenance specialists acquire new knowledge. In the case of drivers, it is worth noticing that only 0.8% of the professional bus driver licenses (A-IIIa) in Peru are held by women. The project aims at interacting with different men and women linked to the professional, technical and transport sectors with experience in managing and maintaining vehicle fleets, in order to optimize the use of new technologies and to improve the quality of the service provided.

2. Gender Action Plan

2.1 Relevant background

There is limited information in Metropolitan Lima and other Peruvian urban areas about gender gaps related to transportation and mobility, which may indicate limited awareness and interest in gender issues by the transport sector. A survey made by CPI[53] in 2016 among students and employees showed that the share of public transport was higher among women (86.4%) than among men (78.4%), confirming the importance of public transport plays for women's urban mobility. Meanwhile, the share of private car use was found to be higher for men (19.4%) than for women (8.9%).

Female participation in transport sector jobs is very low (1.8% of total employed women) compared to men (13.7%). There are no details available on employment in the bus companies serving Lima and other cities, but there is no reason to think that they do not follow the same low-female-employment pattern of the transport sector as a whole. Therefore, it seems adequate to talk of both, a "mobility gap" and a "job gap" concerning gender mobility challenges in Metropolitan Lima and in other Peruvian cities:

- The mobility gap is related to the low quality of service of public transport compared to car use. Circulation and parking restrictions to discourage car use are rare. Buses do not have reserved lanes in most corridors, and cannot compete with the travel times of private cars or relatively cheap taxi services. This gap in the quality of mobility conditions between cars and public transport is disproportionately suffered by women: no gendered information is available on car ownership but, considering car use data in Lima (above), it could be inferred that women's access to private cars is at least 10 percentage points lower than men's. The gender gap in mobility is likely to further increase with urban growth, so that many women will face increasing distances and travel times. The gender mobility gap is sustained by widespread acceptance of the privileges of car users, in spite of their social and environmental consequences.
  The perception of insecurity by women travelling in public transport in Lima is reported as one of the highest in Latin America[54]: between 75% and 80% of women reported feeling unsafe while travelling alone in Lima at night, and 64% perceived public transport as unsafe or vey unsafe.
- The job gap can be linked to a conservative management culture in all public transport operators (PTOs). On the one side this conservative culture has been unable to attract additional passengers, even in a context of population growth; on the other hand, the resistance to change can also explain the low share of employed women in the sector.

The design and operational conditions of the public transport system are also poorly suited to cope with many women's needs. Operations have been designed historically to cope with the needs of daily home-to-work trips at peak hours. There is no information on the share (%) of mobility demand in Metropolitan Lima or other cities related childcare and domestic work (household shopping, socializing children or providing support to dependent adults), but there is wide evidence that these trips are frequently chained by users, and are poorly served by existing bus systems[55].

2.2. Implementation Plan

Implementation of the gender action plan has been embedded within the project structure as follows:

- **The integration of gender issues within public policies addressing e-mobility will be developed through component 1, output 1.2.**
  - The e-mobility strategy provides the adequate framework for a comprehensive consideration of the opportunities and challenges associated to the deployment of EVs to advance equality between women and men and to improve the mobility conditions, employment opportunities and empowerment of women. Furthermore, through output 1.4, training materials and activities will

[53] CPI (2016). Market Report N° 9/2016. Transporte utilizado por trabajadores y estudiantes en Lima metropolitana. Accessed at http://cpi.pe/images/upload/paginaweb/archivo/26/transporte_taxi_201611.pdf

[54] Galiani, S. y Jaitman, L. (2016), "El transporte público desde una perspectiva de género. Percepción de inseguridad y victimización en Asunción y Lima", Banco Interamericano de Desarrollo, Nota Técnica No IDB-TN-11-24

[55] CEPAL (2019). Gender determinants in urban mobility policies in Latin America. FAL Bulletin 371.Accessed at https://repositorio.cepal.org/bitstream/handle/11362/44903/S1900405_en.pdf?sequence=1&isAllowed=y

be prepared and carried out targeting public and private decision makers in the transport sector regarding gender-responsive measures and considerations in public transportation, so as to raise awareness on the importance of integrating a gender approach and ensure there is significant support for the implementation of the e-mobility strategy's gender action plan.

- **The alignment of the quality of public transport services (including bus interior design) with women's priorities and needs, as well as those of other vulnerable groups (children, elderly, persons with disabilities, etc.) is addressed within component 2, output 2.1.**
  - The guidelines prepared for the project demonstrations are expected to influence future transport quality requirements by public authorities, and internal quality assurance procedures within transport operators. Although there is no information on the priorities of men and women regarding urban public transport quality in Peru, the discussion in the analysis above suggests that women have greater concerns with regards to safety and personal security; furthermore, women tend to have less access to different transport options, and tend to have multiple purposes for their trips, many times during off-peak hours, to accommodate their needs with respect to their economic activities, household duties and childcare. Issues related to transport quality of high relevance for women probably include off-peak service quality, convenient transfers, safety measures and protocols to prevent harassment, and good coverage of key destinations by the network beyond the traditional radial lines serving commuting trips. Through the guidelines for the EV demonstrations, the project intends to address these challenges to define future public transport quality priorities from a gender perspective, so that e-buses and 3-Wheelers incorporate gender-responsive services. This will also include, for example, the use of inclusive language within public transport services to ensure that gender stereotypes are not reinforced, reducing their negative impact on the achievement of gender equality.
- **Access to jobs in the transport sector is addressed within component 2, output 2.4, supporting the involvement of women in EV operations, and in component 3, output 3.4, including in the guidelines for transport operators considerations on the need of a corporate strategy to facilitate women's access to jobs in the future.**
  - Current barriers to jobs in the transport sector for women have been discussed during project design, although few conclusions can be advanced at this stage: these barriers are probably associated to tradition, conservative management and unfriendly working environments, as in many other countries. A pro-active strategy has been agreed, so that the introduction of electric vehicles (EVs) is taken as an opportunity to foster the participation of women in jobs in the public transport sector, by giving them priority to access the new jobs linked to EV (driving, maintenance and management) in the companies associated with the project pilots. The project should facilitate further streamlining of policies to facilitate the access of women to jobs in the transport sector based on these pilots and replication activities. Furthermore, and taking into consideration international evidence and good practice, the project will support participating transport operators in the revision of their current procurement, recruitment and training strategies, in order to facilitate the incorporation of women in the sector. The ToR for activities 2.2.3 and 2.3.3 will require from operators providing the demonstration services to prepare and implement a gender policy, which will be monitored during the project lifespan. Based on the project experience, it is expected that transport companies will be able to mainstream gender-responsive measures and increase the percentage of female employees, integrating these aspects within the business plans to be developed within the project and to be implemented after the project's conclusion. The public transport sector should benefit from providing a working environment more attractive for women, as their communicative and people-oriented skills can be of great value to improve the quality of the service.

Gender-oriented actions will be conducted by the following partners:

- The project team, particularly by the project manager with the support of the technical assistant. For both positions, previous experience in gender-mainstreaming in development projects will be considered as requirement for the selection process. The project manager will be responsible for annually monitoring the implementation of the gender action plan, as part of the project overall monitoring effort.
- The UNDP Peru Country Office in Peru, which will also be responsible for the annual monitoring and evaluation of the project gender marker.
- The Ministry of Environment as the project Implementing Partner, in close coordination with the Ministry of Women and Vulnerable Populations, the Ministry of Energy and Mines and the Ministry of Transport and Communications, as the governmental institutions responsible for the design of public policies.
- Local Transport Authorities in the pilot cities, as the main institutional partners involved in the implementation of the pilots where gender-responsive measures will be embedded.

Timelines and budgets are presented in the sections below.

2.3. Proposed activities

The following activities addressing gender issues have been included in the project:

- Project Component 1:
    - Output 1.2. The national e-mobility strategy will include a gender analysis and action plan to address gender gaps in the energy and transport sectors related to e-mobility deployment in Peru and include gender-responsive measures.
    - Output 1.4. Training of public and private decision makers on e-mobility policies and the regulatory and fiscal incentives will prioritize participation of women, and will include key information regarding gender gaps and gender-responsive measures in public transportation. Trainings will emphasize the capacity needs and priorities of women, based on the gender analysis conducted for the e-mobility strategy. Training materials on e-mobility will also be prepared, including gender considerations.
- Project component 2:
    - Output 2.1. The preparation of the guidelines for design of the e-mobility pilots will address considerations on the improvement of PT quality, and the potential contribution of EVs to it, with a specific consideration of the mobility conditions for women, such as bus interior design, prioritizing EV assignment to PT lines providing access to key social services (e.g. schools, health centers), and schedules that meet needs of women and men (during peak and non-peak hours). Quality improvement is addressed by the project from a participatory perspective, focusing on the PT traits more valued by women, as well as other vulnerable groups (children, elderly and persons with disabilities) and the population at large. This should result in higher satisfaction rates among women for the use of new EV services as compared to traditional non-EV public transport systems.
    - Output 2.2. and 2.3. Procurement of EVs for the two pilots will incorporate criteria to select vehicles with gender-responsive design and that address the needs of other vulnerable groups (children, elderly and persons with disabilities). Furthermore, the project will support the selection of women by public transport operators for EVs by ensuring that at least 50% of trainees on EV driving in Lima and Arequipa demonstrations are women. Furthermore, the pilot will incorporate measures against sexual harassment, theft, bullying and all forms of gender-based discrimination and harassment against women and against other vulnerable groups, such as the elderly, children, persons with disabilities and LGBT people. For example, multi-media campaigns targeted at riders will be implemented within EV buses addressing these issues, as well as security measures to prevent violence against female drivers and protocols for victims of violence and harassment in PT to report and receive adequate assistance.
    - Output 2.4. Technical training will be provided specifically targeting women and tailored to their specific needs, in order to facilitate their access to new jobs generated by the deployment of electric vehicles.

- Project component 3:
    - Output 3.4. Guidelines targeting transport operators will include considerations to incorporate gender-responsive measures within their EV procurement, recruitment and training strategies, with the aim of ultimately being fully integrated in companies' business models for future expansion. Worldwide experience shows that a gender-balanced staff results in PT services that are better valued by all users, including women.

This approach is expected to be sustained after project termination through the inclusion of the project's recommendations within the practice of the national government and urban public transport authorities.

2.4. Indicators, targets and proposed budget

The proposed project monitoring approach includes gender-responsive indicators within the Project Results Framework (PRF), which will provide the necessary information to monitor the achievement of gender-responsive results at the outcome level:

- Outcome Indicator 1.4 will provide the number of women and men who are public and private decision makers and receiving training on e-mobility policies.
- Outcome Indicator 4.3 will provide the number of women receiving professional training on end-of-life vehicle management.

Outcome indicators' baselines and targets are fully described in the PRF. Furthermore, activity-level indicators and targets have been identified for adequate monitoring on the implementation of gender-responsive activities that are included in the project and in this Gender Action Plan. The latter are detailed in the table below, where the budget allocated to gender-related activities is also noted:

Project component 1: Institutionalization of low-carbon electric mobility **Outcome: Public policy for promotion of e-mobility defined, with the support of all stakeholders**						
Gender-related activity	**Indicator**	**Target**	**Baseline**	**Budget**	**Timeline**	**Responsibility**
Development of a gender analysis and action plan for the national e-mobility strategy (Activity 1.2.5)	Availability of a gender analysis and action plan for incorporation into the national e-mobility strategy	A gender analysis and action plan fully incorporated into the national e-mobility strategy.	There are no gender analyses for e-mobility nor an e-mobility strategy in Peru.	20,000	Year 1	MINAM, project team
Conduct trainings for public and private decision makers on e-mobility policies and gender-responsive measures and considerations in public transport (Activity 1.4.2)	Number of public and private decision makers receiving training in e-mobility policies and gender responsive measures and considerations in public transport. % of female public and private decision makers receiving training in e-mobility policies and gender issues in public transport.	At least 50 public and private decision makers trained in e-mobility and gender issues in public transport. At least 50% of trainees on EV driving in Lima and Arequipa demonstrations are women.	0	40,000	Years 1, 2	Project team
Develop training materials on e-mobility policies that specifically address gender-responsive measures and considerations in public transport, targeted at public and private decision makers (Activity 1.4.2)	Number of training materials developed that address gender-responsive measures and considerations in public transport	At least 3 training materials developed that address gender-responsive measures and considerations in public transport	0	6,000	Year 1	Project team
Conduct technical trainings on EV operation and maintenance targeted at women, that address their specific capacity needs (Activity 1.4.3)	Number of technical trainings conducted on EV operation and maintenance targeted at women, that address their specific capacity needs.	At least 4 technical training workshops on EV operation and maintenance targeted at women and addressing their specific capacity needs	0	15,000	Years 1, 2	Project team, PT operators
Project component 2: Short term barrier removal through low-carbon e-mobility demonstrations **Outcome: Demonstrations on e-mobility in public transport empower stakeholders to scale-up e-mobility**						
Gender-related activity	**Indicator**	**Target**	**Baseline**	**Budget**	**Timeline**	**Responsibility**
Development and integration of guidelines on gender-responsive design in e-mobility public transportation pilots, based on participatory analyses (Activity 2.1.1)	Availability of guidelines on gender-responsive design for e-mobility public transportation pilots	Guidelines on gender-responsive design integrated in e-mobility public transportation pilots in Lima and Arequipa	0	5,000	Year 1	Local PT Authority, project team

Implement multi-media campaigns against gender-based harassment, discrimination and violence, including against other vulnerable groups (children, elderly, persons with disabilities, LGBT people) (Activity 2.1.3)	Number of riders who view multimedia campaigns that demonstrate zero-tolerance for gender-based harassment, discrimination and violence, including against other vulnerable groups.	At least 2 multi-media campaigns viewed by 200,000 riders in each city (Lima and Arequipa) against gender-based harassment, discrimination and violence, including against other vulnerable groups.	0	7,500	Year 1	The project provides the campaign materials, to be implemented by PT authorities in Arequipa & Lima
Conduct technical trainings on EV operation and maintenance targeted at professionals involved in the pilots, prioritizing female participation and that also address gender issues and gender-responsive measures in public transportation (Activity 2.4.1)	Number of persons who receive technical training on EV operation and maintenance and gender issues and gender-responsive measures in public transportation. Number of women receiving training on EV operation and maintenance.	At least 60 persons receive professional trainings on EV operation and maintenance, of which at least 40 are women.	0	22,500	Years 1, 2, 3	Project team, PT operators
Ensure hiring female drivers for EV pilots in Lima and Arequipa through the provision of training. (Activities 2.2.1 and 2.3.1)	% of female trainees for driving EV pilots in Lima and Arequipa	At least 50% of trainees on EV driving in Lima and Arequipa demonstrations are women.	0	*0*	Year 1	Project team
Project component 3: Preparing for scale-up and replication of low-carbon electric mobility **Outcome: Conditions in place to include EVs in the vehicle market and to accelerate their deployment**						
Gender-related activity	**Indicator**	**Target**	**Baseline**	**Budget**	**Timeline**	**Responsibility**
Development guidelines for EV procurement, recruitment and training strategies for urban public transport operators and authorities, that include recommendations for their adoption of gender-responsive measures. (Activity 3.4.1)	Guidelines for EV procurement, recruitment and training strategies that include recommendations for the adoption of gender-responsive measures delivered to urban public transport operators and authorities.	At least two (02) urban public transport operators or authorities integrate recommendations for gender-responsive measures in their EV procurement, recruitment and training strategies.	No guidelines	3,000	Year 3	Project team, MTC
Project component 5: Component 5: Project monitoring and evaluation **Outcome 5:The project monitoring and evaluation plan is implemented**						
Gender-related activity	**Indicator**	**Target**	**Baseline**	**Budget**	**Timeline**	**Responsibility**
Conduct a satisfaction survey for e-bus passengers in Lima and Arequipa, for three categories: women, men and persons with disabilities (Activities 2.2.3 and 2.3.3)	% of passengers who have positive perceptions of EV public transport pilot	At least 60% of each category of passenger (women, men and persons with disabilities) have positive perceptions of the EV public transport pilot in terms of	0	5,000	Year 2	Project team, PT operators

		responsiveness to their needs, safety and reliability. *64% of women in Lima consider public transport to be unsafe or very unsafe (BID 2016)				
Total budget allocation					***USD 104,000***	

3. References

IADB, 2016. The relationship between gender and transport. Consultado en https://publications.iadb.org/publications/english/document/The-Relationship-Between-Gender-and-Transport.pdf

INEI, 2017. Perú: Brechas de género 2017. Consultado en https://www.inei.gob.pe/media/MenuRecursivo/publicaciones_digitales/Est/Lib1444/libro.pdf

INEI, 2018a.Perú: Perfil sociodemográfico 2017. Informe Nacional. Consultado en https://www.inei.gob.pe/media/MenuRecursivo/publicaciones_digitales/Est/Lib1539/libro.pdf

INEI, 2018b. Indicadores de género. Consultado en https://www.inei.gob.pe/estadisticas/indice-tematico/brechas-de-genero-7913/

INEI, 2019. Los feminicidios y la violencia contra la mujer en el Perú 2015-2018. Consultado en https://www.inei.gob.pe/media/MenuRecursivo/publicaciones_digitales/Est/Lib1659/Libro.pdf

Lima Cómo Vamos, 2019. ¿Cómo vamos en Lima y Callao? Noveno informe de indicadores sobre calidad de vida. Consultado en http://www.limacomovamos.org/wp-content/uploads/2019/11/Informe-2018_web.pdf

Vallejo, E & M.P. Rivarola, 2013. La violencia invisible: acoso sexual callejero en Lima Metropolitana y Callao. Cuadernos de Investigación IOP-PUCP, n°4. Consultado en http://repositorio.pucp.edu.pe/index/bitstream/handle/123456789/34946/Cuadernos%20de%20investigaci%c3%b3n%204.pdf?sequence=1&isAllowed=y

WEF, 2018. The Global Gender Gap Report. Consultado en http://www3.weforum.org/docs/WEF_GGGR_2018.pdf

Annex 10: Procurement Plan

Budget item		List of Goods and Services required	Budget (US$)	Year	Brief description of anticipated procurement process
No.	Description				
	Staff & Personnel (Including Consultants)				
71801	Project Manager	42-month contract full-time	168,000	1	Open request for proposals
71803	Technical assistant	36 year contract, full time	72,000	1	Open request for proposals
	Contract Services				
74101	Independent financial audits	One independent audit to be completed every year	8,000	1	Open request for proposals
72111	Communication and capacity building plan	Consultancy services to design the plan on communication and capacity building	20,000	2	Open request for proposals
71201	Terminal evaluation	Terminal evaluation to be conducted following UNDP guidelines	30,000	4	Open request for proposals
72101	Draft e-mobility strategy, including economic, social, space and gender impacts, and gender plan	Consultancy services to deliver a draft e-mobility strategy, including economic, social, space and gender impacts, as well as a gender analysis and action plan (A.1.2.1, 1.2.2, 1.2.3, 1.2.5, 1.2.6)	60,000	3	Open request for proposals
72102	Design, implementation, monitoring and evaluation of demonstrations	Consultancy services to design demonstrations based on best practice, and to monitor and assess demonstrations (A.2.1.1. A.2.1.2, A.2.2.3, A2.3.3, A.5.1.2 and A.5.1.3)	40,000	2	Open request for proposals
72103	Consultancy services on technical, legal and fiscal regulations of vehicles (including technical guidelines)	Consultancy services to review, assess and provide proposals to reform regulations and standards on vehicle approval, EV charging infrastructure and a tax scheme incentivizing EVs and discouraging conventional vehicles (A.1.3.1. A.1.3.2. A.1.3,3)	110,000	3	Open request for proposals
72104	EV market analysis, scenario building, business models and financial schemes	Consultancy services to build future EV market scenarios in Peru (A,3.1.1), to assess to the potential of EV in urban PT (A,3.1.2), and to provide business models (A.3.3,1), and financial schemes for corporate (A.3.3,2) and PT (A.3.3,3) fleets, and recommendations to PT operators and authorities (A.3.3,4)	130,000	1	Open request for proposals
75701	Organization of	Consultancy services to support meetings of	24,500	1	Open request

Budget item		List of Goods and Services required	Budget (US$)	Year	Brief description of anticipated procurement
	meetings and workshops	the coordination space (A.1.1,2)			for proposals
75702	Workshops and training in e-mobility	Services to organize workshops and to provide workshops and training on e-mobility (A.1.4,3)	63,000	1	Open request for proposals
72105	Organization of meetings and workshops (inception workshop)	Services to organize and provide support to the inception workshop (M&E)	3,500	1	Open request for proposals
72106	Communication activities (knowledge management, communication plan, website management and publication of materials), The preparation of training materials is a separate "suplies", Note that the design of the communication plan is now in 1.2.	Consultancy for the provision of communication platforms and activities to support the dissemination of e-mobility among institutional stakeholders and the public (A.1.4.1), the dissemination of project demonstration results (A.2.5.1 and A.2.5.2), and the consolidation of the e-mob network (A.3.2.2)	50,902	2	Open request for proposals
75703	Professional and technical training on electric vehicles	Services to design and deliver professional training on EV driving and maintenance (A.2.4.1)	20,250	1	Open request for proposals
75704	Professional training on ELV management of EVs and their batteries	Services to design and deliver professional training on ELV management of EVs and their batteries (A.4.5.1)	9,000	1	Open request for proposals
72108	Provision of services with 2 e-buses in each city	Request for proposals to provide PT services with 4 e-buses (2 in Arequipa and 2 in Lima) for at least 24 months and providing at least 56,000 km per bus and year in exchange of a compensation per km actually travelled, with a maximum of USD 150,000, (A.2.2.1. A.2.2.2. A.2.3,1 and A.2.3.2)	600,000	1	Open request for proposals
72109	Provision of services with 10 electric 3-wheeler taxis in each city	Request for proposals to provide PT services with 20 electric 3W (10 in Arequipa and 10 in Lima) for at least 24 months and providing at least 14,000 km per vehicle and year in exchange of a compensation per km actually travelled, with a maximum of USD 5,000 (A.2.2.1. A.2.2.2. A.2.3.1 and A.2.3.2)	100,000	1	Open request for proposals
72110	Design and	Consultancy to provide technical assistance to	90,000	1	Open request

Budget item		List of Goods and Services required	Budget (US$)	Year	Brief description of anticipated procurement process
	implementation of a EPR approach to ELV regulations (including batteries)	undertake a diagnostic of ELV management in Peru, assessment of ELV management needs in Peru following the deployment of EV, review international best practice on regulation on ELV management for EV, draft regulations for Peru, review business models on ELV management and second-life battery use, and develop commercially viable models for Peru and a roadmap for implementation and provide support to secure approval.			for proposals
72107	NEW: Design and implementaton of multimedia campaigns on harassment, discrimitation and violence	Design and implementation of multi-media campaigns against gender-based harassment, discrimination and violence, including against other vulnerable groups Consultancy for the design of a multimedia campaign through a participatory process including women and other vulnerable groups, as well as public transport and 3-wheeler professionals including (1) Analysis of public transport users' profiles and selection of communication channels in Lima and Arequipa, (2) monitoring and assessment of impact (A.2.1.3).	10,000	1	Open request for proposals
72112	Project Administration	42-month (Profonanpe)	105000	1	Profonanpe
	Operating & Other Costs				
74501	Office operations	Miscellanous expenses for project office operations	7,250	1	Open request for proposals
	Supplies, commodities & materials				
72501	Publications on institutionalization of e-mobility	Design and publication of materials to support the dissemination and adotion of the e-mobility strategy (A.1.2.6 and A.1.2.7).	7,000	1	Open request for proposals
72502	Publications on project pilots	Design and publication of materials to disseminate contents and resuls of project pilots (A.2.4.1 and A.2.4.2).	2,250	1	Open request for proposals
72503	Publications on end-of-life vehicle management	Design and publication of materials to disseminate project contents on ELV management, focusing on Evs (A.4.5.3).	1,000	2	Open request for proposals
72504	Office supplies	Miscellanous expenses on supplies for project office operations	7,250	1	Open request for proposals
	Equipment, Vehicles & Furniture				
72201	Furniture	Basic office furniture	6,960	1	Open request for proposals
72801	Computers	Four computers and one printer for project team unit	5,000	1	Open request for proposals

Budget item		List of Goods and Services required	Budget (US$)	Year	Brief description of anticipated procurement
	Travel				
71601	Travel to attend working groups and workshops of the global e-mob programme	Travel to attend working groups and workshops of the global e-mob programme	28,000	1	
71602	Travel for meetings in Lima	Travel for meetings related to the demonstration in Lima	3,000	1	
71603	Travel for meetings in Arequipa	Travel for meetings related to the demonstration in Arequipa	3,000	1	
Total procurement			**1,784,862**		

Annex 11: GEF focal area specific annex (Estimates of Direct and Consequential GHG Emission Reductions and Beneficiaries)

GHG reductions and energy savings estimation for Peru	
Project information • Project duration: 3.5 years. Starting in 2021 and ending in 2024 • Time frame for indirect effects: 15 years. Starting in 2021 and ending in 2035 (Effects produced by the introduction of electric buses electric 3 wheelers in the public transport fleet in Peruvian cities) • Causality factor: 80%	
Total project emissions reductions, t CO_2	**415,117**
Total direct emission mitigation from demonstration projects, t CO_2	**178,460**
Primary direct emission mitigation	4,836
3-wheelers (considering end of life of vehicle as 5 years)	*195*
Buses (considering end of life of vehicle as 15 years)	*4,640*
Secondary direct emission mitigation (policy measures)	173,624
3-wheelers (considering end of life of vehicle as 5 years)	*247*
Buses (considering end of life of vehicle as 15 years)	*173,378*
Total indirect emission mitigation, t CO_2	**236,657**
Total direct emission mitigation at the time of measurement, t CO_2	
At end of month 24 (mid-term)	348
At end of month 42 (project termination)	871
Total project energy savings, MJ	**5,381,596,677**
Total direct energy savings from demonstration projects, MJ	**2,313,522,172**
Primary direct energy savings	**62,331,465**
3-wheelers (considering end of life of vehicle as 5 years)	*2,153,373*
Buses (considering end of life of vehicle as 15 years)	*60,178,093*
Secondary direct energy savings (policy measures)	2,251,190,707
3-wheelers (considering end of life of vehicle as 5 years)	*2,721,007*
Buses (considering end of life of vehicle as 15 years)	*2,248,469,700*
Total indirect energy savings, MJ	**3,068,074,505**

Methodology for the estimation of GHG reductions and energy savings benefits

A uniform methodology was applied in all countries project for assessing the short, medium and long-term project's benefits in terms of GHG reductions and energy savings. The methodology compares two scenarios, the "benchmark scenario" and the "electromobility scenario". The

benchmark scenario considers the baseline established in the mitigation measure T8 (Promotion of hybrid and electric vehicles at national level), within the NDC-2020[56], with a target of 5% of the bus fleet being electric by 2030 (there are no NDC targets for mototaxis), without any intervention from the project. The electromobility scenario envisages a country with higher participation of electric vehicles in the national fleet of 9-to-12-m buses and mototaxis in regular urban services. As a consequence of the project implementation, on top of the completion of the T8 measure, with an urban e-bus fleet increased by 133 vehicles and an electric mototaxi fleet increased by 1,417 vehicles in 2030. The e-mobility scenario uses the same projections with regards to vehicle fleet growth but assumes a higher penetration of electric vehicles within the new vehicle market, as a consequence of the project interventions including the adoption of EV policies, the use of business models and the existence of financial mechanisms. The scenarios use a "top-down approach" targeting the national vehicle market. Calculations are performed separately for the two modes (urban public transport buses and 3-wheelers) targeted by the project.

In order to be consistent with other projects within the e-mob programme, the emission factors (kgCO2/Lge) and specific consumption (Lge/100 km) used in this Annex slightly differ from those used in the National Emissions Inventory and other studies completed by MINEM[57].

Projections of fleet growth, energy use and GHG emissions are based on country specific data, and region-specific parameters. Projection of the vehicle fleet growth is based on the elastic relationship between per capita income and vehicle acquisition. Therefore, country specific scenarios for population growth (based on the UNDESA medium scenario) and projections for gross domestic product (GDP PPP) from the World Economic Outlook of the International Monetary Fund (IMF) are used. Vehicle fleet projections are based on vehicle sales and assumptions are based on technical life-time of vehicles. A comprehensive set of parameters describing the technologic and economic parameters of various vehicle technologies are used. Country specific grid emission factors for the carbon footprint of electricity are used. For petroleum-based fuels, well-to-wheel emission factors are used. Historic development of the vehicle fleet is based on country specific vehicle stock and sales data. Emission reductions which accrued during and after the project timeframe are taken into account. GHG emission benefits are classified as direct and indirect GHG emission reductions. This categorization follows the methodology suggested by the GEF.

> ***Direct benefits*** *correspond to the GHG emission reductions and energy savings obtained from 1.) The investments that are planned and executed during the project lifetime, i.e. the emission and energy use savings stemming from the demonstration of electric vehicles and EV supply equipment such as chargers purchased as part of the project; and 2.) emission reductions and energy savings as a result of investment in replication and upscaling (secondary direct benefits).*

[56] MINAM (2020). *Informe técnico de actualización del NDC.* Included in MINAM (2019) *Catálogo de Medidas de Mitagación* as measure ECM-30. Available at https://cdn.www.gob.pe/uploads/document/file/426371/CATALOGO_MITIGACION_baja_con_observaciones_levantadas.pdf

[57] TARYET (2019). Estimación de los rendimientos de una flota representativa de buses en el ámbito de Lima Metropolitana a través de la aplicación de encuestas y proceso de determinación mediante pruebas de laboratorio del contenido de carbono en combustibles (diésel y gasolina),

Indirect benefits *correspond to the GHG reductions and energy savings obtained during and beyond the project as the result of outputs and outcomes of the project. This includes in particular the adoption of policies, business models and financial mechanisms, which incentivize the uptake of electric mobility. Total emission reductions attributable to the project are based on the cumulative sum of annual emission reductions compared to the baseline scenario over a time frame equivalent to the lifetime of the demonstration assets purchased as part of the project or for a period of ten years after the end of the project*

Quantification of secondary direct and indirect benefits is based on an e-mobility scenario considering the maximum attainable electric mobility market (both in terms of size and pace of technology introduction). Causality factors are used to estimate the contribution of the GEF funded project to the projected large-scale and nation-wide introduction of electric vehicles. Guidelines issued by the GEF for the selection of the causality factor level are as following:

Level 5 = "The project contribution is critical, and nothing would have happened in the benchmark scenario," causality factor = 100%

Level 4 = "The project contribution is dominant, but some of this reduction can be attributed to the benchmark scenario," causality factor = 80%

Level 3 = "The project contribution is substantial, but modest indirect emission reductions can be attributed to the benchmark scenario," causality factor = 60%

Level 2 = "The project contribution is modest, and substantial indirect emission reductions can be attributed to the benchmark," causality factor = 40%

Level 1 = "The project contribution is weak, and most indirect emission reductions can be attributed to the benchmark scenario," GEF causality = 20%

In the case of Peru, Level 4 has been applied, as the project contribution is dominant in the targeted subsectors of public transport services by regular buses and 3-wheelers, where there has been very limited action thus far. Secondary direct and indirect emission reduction are based on a 30:70 split of the top-down emission reductions attributable to the project via the application of the causality factor.

Required input data for calculating the GHG reductions and energy savings.

A comparison in terms of GHG and energy between the projections under the benchmark scenario and the electromobility scenario is performed. To do this, demographic, economic and technology data must be provided. When local data is not available, default values are proposed based on expert experience or similar projects. The vehicle fleet data, vehicle operating data and scenarios correspond to the transportation mode affected by the project at national or city level. The required data are presented in the table below:

	Variable	Unit
Socio – economic	GDP PPP (2000-2018)	Billion USD

data	Population	Million habitants
	Annual growth of GDP	% of 2023-2030, and % 2031-2050
Vehicle fleet data	Vehicles stock (2000-2015)	Thousand vehicles
	Vehicles sales (2000-2015)	Thousand vehicles
	Technology share of stock	% of gasoline, diesel, hybrid, PHEV, BEV
Vehicle operating information	Annual Mileage	km
	Load factor	Passenger in a vehicle
	Technical lifetime	years
	Share of electric driving for PHEV	%
	Fuel economy (FE) by technology	Lge / 100 km, kWh / 100 km
	Annual FE improvement by technology	%
	FE gap (Real vs Type Approval)	%

Table 7: Data required by the E-mob Model

GHG and energy evaluation model for electromobility projects (Emob calculator)

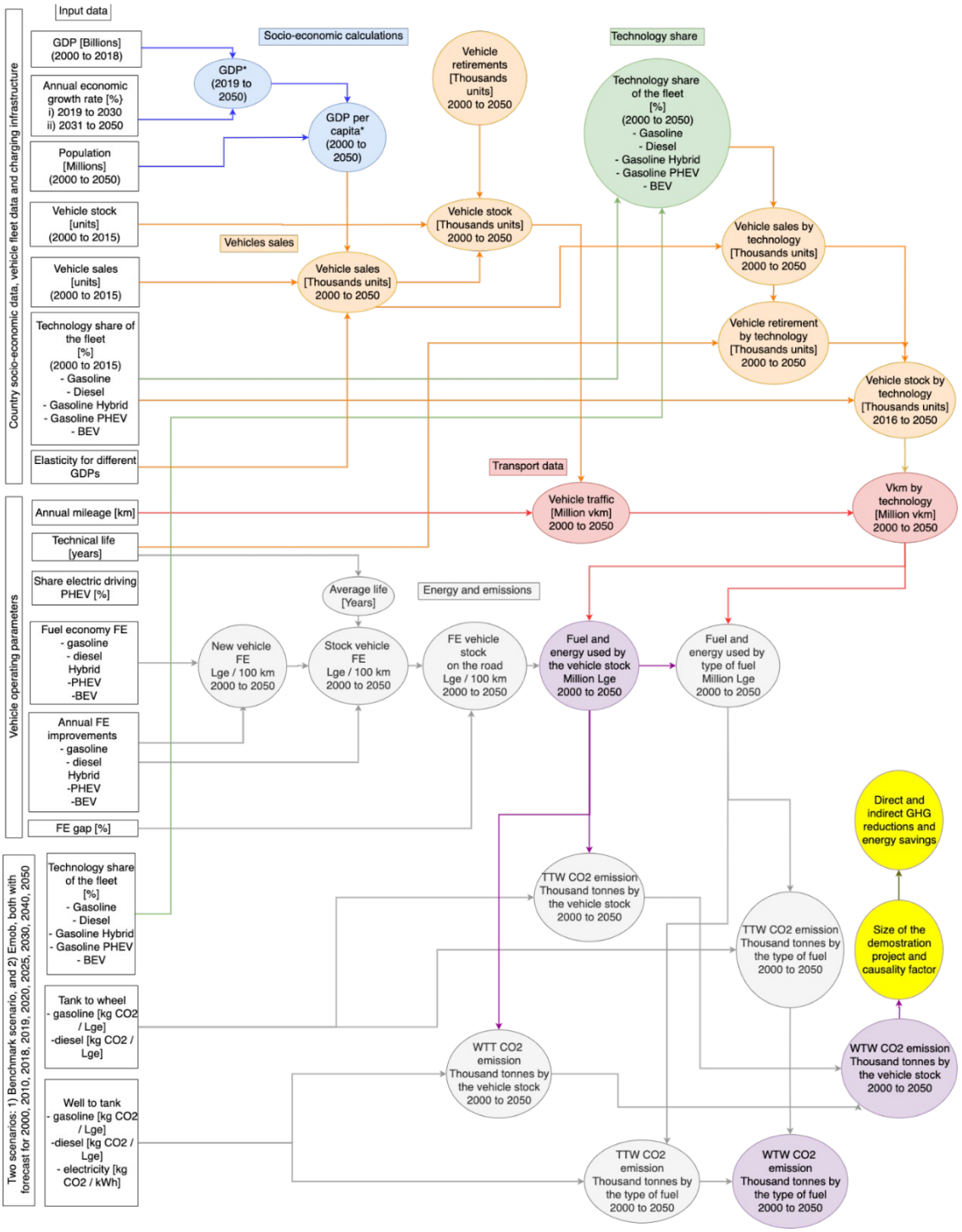
The country information provided as input data is used to model and forecast the behavior of the vehicle fleet until year 2050. The forecast must assume that size of the vehicle fleet and the vehicle technology would not remain constant in short, medium and long term. The following projections are established in the calculation process according to the specific scenario:

- Projection of vehicle sales and stock from 2000 to 2050
- Vehicles sales, retirements and stock by technology from 2000 to 2050
- Vehicle traffic activity by technology
- Projection of energy use by vehicle technology from 2000 to 2050
- Projection of GHG produced from 2000 to 2050

The last two projections are used to compute GHG reductions and energy savings produced by electromobility scenario respect to the benchmark scenario. The size of the demonstration project and the causality factor must be provided, for calculating the direct and indirect benefits, respectively. The formulas used to calculate the GHG reductions and the energy saving are

programed in a Microsoft Excel file developed by the Air Quality and Mobility Unit of the United Nations Environment Programme.

The obtained GHG reductions and energy savings results are compared with the UNFCCC National Communications, in which countries report the amount of GHG produced by different sectors, including transport. Then, a refinement of the calculation is performed if it is necessary. The following flow map presents the procedure to compute the direct and indirect GHG reductions and the energy saving obtained by the project.



Peru benchmark scenario

The model has used the GDP and population growth statistics from the government of Peru until 2019. Until 2022, the GDP growth follows the official estimates published at the beginning of 2020; is estimated at 6.1% annually; in 2023, the average growth of the previous four years is

considered; until 2030, annual growth is assumed to be 4.3%, annually and 4% in subsequent years.

Population trends follow the official forecasts, with a shrinking annual growth rate from 1.2% now to 0.6% in 2040. The impact of the global COVID pandemic on Peru's economy is not taken into account, due to lack of reliable data at the time of preparation of this document. The IMF has foreseen an 4.5% contraction of GDP in 2020, with no figures for subsequent years. A contraction in the economy would probably result in lower growth of the vehicle fleet in the country and higher public transport demand.

The model estimates the registration of new cars based on an elasticity to GDP per capita of 0.7 (until GDP per capita exceeds USD 20,000 in 2027), and an elasticity of 0.5 afterwards. However, the registration of new cars is not considered to be influenced by the project and therefore is not computed in this case.

Considering that the project is targeting full-size urban transport buses (above 9-m long) in regular lines and 3-wheelers providing regular for-hire public transport services, only registrations of urban buses and 3-wheelers have been included in the estimates. The estimate of the stock has considered a vehicle lifespan of 15 years for buses and 5 years for 3-wheelers.

The average fuel consumption is estimated at 52.0 liters diesel/100 km for buses (the average value obtained from public operators in Lima) and 6.0 liters gasoline/100 km for 3-wheelers (CO_2 GEF Manual). The average annual mileage is estimated at 54,000 km for buses and 14,000 km for 3-wheelers. A fuel efficiency improvement of 1% per year is assumed by the model.

An average consumption of 75 kWh/100 km has been considered for electric buses, and 8.7 kWh/100 km for electric 3-wheelers in the demonstration. As electricity in Peru is mostly generated from renewables, an average emission factor for electricity generation of 0.24 kg CO2/kWh has been assumed. This factor is expected to decrease to 0.21 kg CO2/kWh in 2030 and to 0.19 kg CO2/kWh by 2040, as a result of the implementation of the country's energy strategy.

Peru electromobility scenario

After the project starts with 4 buses and 20 3-wheelers (year 2), the number of electric vehicles introduced in the country in both categories is estimated to steadily grow: 16 more buses and 80 more 3-wheelers in year 3, 4% of bus sales (60 buses) and 2% (127) of 3-wheeler sales in year 4 and 5% of bus sales (76 buses) and 3% of 3-wheeler sales (162) in 2025; the share in sales increases steadily afterwards (for buses: 10% in 2032 and 15% in 2036; for 3-wheelers: 7% in 2032 and 10% in 2036).

Project beneficiaries

The direct project beneficiaries are estimated based on the services provided by the project's e-buses and electric 3W. Based on average values in Lima and Arequipa, it is assumed that each e-bus will provide 6 services per day, with an average bus occupancy of 40 passengers and a coefficient of exchange of 3.2. Each 3W is supposed to provide 28 services per day on average with an occupancy of 1.2 passengers.Each e bus or 3W use is assumed to make 500 trips per year.

This provides a total of 1,917 persons making use of e-bus services and 423 persons making use of electric 3W services, i.e. 2,340 beneficiaries in total in the first year of operation and 4,680 in two years. It is also assumed that 16 additional e-buses and 20 additional electric 3W will be operational by the end of the project, providing services to additional 9,360 persons per year. The total number of direct beneficiaries is therefore 14,040 persons during the project lifespan of which 57% (8,000) women. These are the main assumptions:

Average length bus line (one direction): 30 km
Services provided by each e-bus_ 6 services per day
Average e-bus occupancy[58]: 40 passengers
Coefficient of passenger exchange: 3.2
Number of trips per e-bus and day: 768 trips
Days of e-bus service per year: 312 days
Number of trips per e-bus and year: 239,616 trips
Number of e-buses in demonstration: 4 vehicles
Years of operation of pilot buses: 2 years
Number of additional e-buses from replication: 16 vehicles
Years of operation of replication buses: 1 year
Total trips during project lifespan: 5,750,784 trips
Average trips per beneficiary and year: 500
Beneficiaries of e-buses: 1,917 persons

Services provided by each 3-wheeler 28 services per day
Average 3-wheeler occupancy[59]: 1.2 passengers
Days of 3-wheeler service per year: 315 days
Number of trips per 3 wheeler and year: 10,584 trips
Number of 3-wheelers in demonstration: 20 vehicles
Years of operation of pilot 3-wheelers: 2 years
Number of additional 3W from replication: 80 vehicles
Years of operation of replication 3 wheelers: 1 year
Total trips during project lifespan: 1,270,080 trips
Average trips per beneficiary and year: 500
Beneficiaries of 3-wheelers: 423 persons

TOTAL
Number of trips provided by the project:.....7,020,864

[58] Considering COVID-related restrictions
[59] Considering COVID-related restrictions

Number of direct beneficiaries:.....14,042
Percentage of women in public transport:.....57%
Female project beneficiaries:.....8,004

The indirect project beneficiaries are estimated considering a 1-km wide corridor (500 m to each side) along the pilot e-bus route that will be implemented in Arequipa and Lima, and applying to the corridor the average population density in consolidated urban zones in Peru, resulting in 300,000 inhabitants. For 3W, the reference is one of the areas in Metropolitan Lima (Villa El Salvador, population close to 200,000 inhabitants) currently served by 3W. In both cases, it is considered that 5% of the potentially served population could experience some sort of indirect benefit from the new services (reduced noise and pollution, improved accessibility). In this case, they are equally distributed among men and women, resulting in a total of 53,000 beneficiaries. These are the assumptions applied:

Average length bus line (one direction): .....30 km
Area of influence of corridor (width):.....1 km
Average population density in urban zones:.....11,000 inh/km2
Population potentially served by e-buses in each city:.....330,000 inh
Population in areas served by 3-wheelers in each city:.....200,000 inh
% population with mobility patterns compatible with the services provided:.....5%
Project scale-up effect due to component 3.....
.....*Not considered (as additional EVs could be operating in the same areas)*
Total indirect beneficiaries.....53,000
Of which, women (50%):.....26,500

Annex 12: Additional agreements: such as cost sharing agreements, project cooperation agreements signed with NGOs (where the NGO is designated as the “executing entity”), letters of financial commitments etc..

There are no additional agreements in this project

Annex 13: GEF Core indicators

Core Indicator 6	**Greenhouse gas emission mitigated**					***(Tons)***
		Expected metric tons of CO_2e (6.1+6.2)			
		PIF stage	Endorsement	MTR	TE
	Expected CO2e (direct)	*480,949*	*178,460*		
	Expected CO2e (indirect)	*342,417*	*236,657*		
Indicator 6.1	Carbon sequestered or emissions avoided in the AFOLU sector				
		Expected metric tons of CO_2e			
		PIF stage	Endorsement	MTR	TE
	Expected CO2e (direct)				
	Expected CO2e (indirect)				
	Anticipated start year of accounting				
	Duration of accounting				
Indicator 6.2	Emissions avoided Outside AFOLU				
		Expected metric tons of CO_2e			
		Expected		Achieved	
		PIF stage	Endorsement	MTR	TE
	Expected CO2e (direct)	*480,949*	*178,460*	*348*	*871*
	Expected CO2e (indirect)	*342,417*	*236,657*		
	Anticipated start year of accounting	*2021*	*2021*		
	Duration of accounting	*2036*	*2036*		
Indicator 6.3	Energy saved				
		MJ			
		Expected		Achieved	
		PIF stage	Endorsement	MTR	TE
	Direct	*6,237,243,283*	*2,313,522,172*		
	Indirect	*4,440,675,868*	*3,068,074,505*		
Indicator 6.4	Increase in installed renewable energy capacity per technology				
	Technology	Capacity (MW)			
		Expected		Achieved	
		PIF stage	Endorsement	MTR	TE
	(select)				
	(select)				
Core Indicator 11	**Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment**				***(Number)***
		Number			
		Expected		Achieved	
		PIF stage	Endorsement	MTR	TE
	Female	*NA*	*8,000*		
	Male	*NA*	*6,000*		
	Total	*NA*	*14,000*		

Annex 14: GEF 7 Taxonomy

Level 1	Level 2	Level 3	Level 4
☒ **Influencing models**			
	☒ **Transform policy and regulatory environments**		
	☒ **Strengthen institutional capacity and decision-making**		
	☒ **Convene multi-stakeholder alliances**		
	☒ **Demonstrate innovative approaches**		
	☒ **Deploy innovative financial instruments**		
☒ **Stakeholders**			
	☐ **Indigenous Peoples**		
	☒ **Private Sector**		
		☐ Capital providers	
		☒ Financial intermediaries and market facilitators	
		☒ Large corporations	
		☒ SMEs	
		☒ Individuals/Entrepreneurs	
		☐ Non-Grant Pilot	
		☐ Project Reflow	
	☒ **Beneficiaries**		
	☐ **Local Communities**		
	☒ **Civil Society**		
		☐ Community Based Organization	
		☒ Non-Governmental Organization	
		☒ Academia	
		☒ Trade Unions and Workers Unions	
	☒ **Type of Engagement**		
		☒ Information Dissemination	
		☐ Partnership	
		☒ Consultation	
		☐ Participation	
	☒ **Communications**		
		☒ Awareness Raising	
		☒ Education	
		☒ Public Campaigns	
		☐ Behavior Change	
☒ **Capacity, Knowledge and Research**			
	☐ **Enabling Activities**		
	☒ **Capacity**		

Level 1	Level 2	Level 3	Level 4
	Development		
	☐ **Knowledge Generation and Exchange**		
	☐ **Targeted Research**		
	☐ **Learning**		
		☐ Theory of Change	
		☐ Adaptive Management	
		☐ Indicators to Measure Change	
	☐ **Innovation**		
	☒ **Knowledge and Learning**		
		☒ Knowledge Management	
		☐ Innovation	
		☒ Capacity Development	
		☐ Learning	
	☒ **Stakeholder Engagement Plan**		
☒ **Gender Equality**			
	☒ **Gender Mainstreaming**		
		☐ Beneficiaries	
		☐ Women groups	
		☒ Sex-disaggregated indicators	
		☐ Gender-sensitive indicators	
	☒ **Gender results areas**		
		☐ Access and control over natural resources	
		☐ Participation and leadership	
		☒ Access to benefits and services	
		☒ Capacity development	
		☐ Awareness raising	
		☐ Knowledge generation	
☒ **Focal Areas/Theme**			
	☐ **Integrated Programs**		
		☐ Commodity Supply Chains (Good Growth Partnership)	
			☐ Sustainable Commodities Production
			☐ Deforestation-free Sourcing
			☐ Financial Screening Tools
			☐ High Conservation Value Forests
			☐ High Carbon Stocks Forests
			☐ Soybean Supply Chain
			☐ Oil Palm Supply Chain
			☐ Beef Supply Chain
			☐ Smallholder Farmers

Level 1	Level 2	Level 3	Level 4
			☐ Adaptive Management
		☐ Food Security in Sub-Sahara Africa	
			☐ Resilience (climate and shocks)
			☐ Sustainable Production Systems
			☐ Agroecosystems
			☐ Land and Soil Health
			☐ Diversified Farming
			☐ Integrated Land and Water Management
			☐ Smallholder Farming
			☐ Small and Medium Enterprises
			☐ Crop Genetic Diversity
			☐ Food Value Chains
			☐ Gender Dimensions
			☐ Multi-stakeholder Platforms
		☐ Food Systems, Land Use and Restoration	
			☐ Sustainable Food Systems
			☐ Landscape Restoration
			☐ Sustainable Commodity Production
			☐ Comprehensive Land Use Planning
			☐ Integrated Landscapes
			☐ Food Value Chains
			☐ Deforestation-free Sourcing
			☐ Smallholder Farmers
		☐ Sustainable Cities	
			☐ Integrated urban planning
			☐ Urban sustainability framework
			☐ Transport and Mobility
			☐ Buildings
			☐ Municipal waste management
			☐ Green space
			☐ Urban Biodiversity
			☐ Urban Food Systems
			☐ Energy efficiency
			☐ Municipal Financing
			☐ Global Platform for Sustainable Cities
			☐ Urban Resilience
	☐ **Biodiversity**		
		☐ Protected Areas and Landscapes	
			☐ Terrestrial Protected Areas
			☐ Coastal and Marine Protected Areas
			☐ Productive Landscapes
			☐ Productive Seascapes
			☐ Community Based Natural Resource Management

Level 1	Level 2	Level 3	Level 4
		☐ Mainstreaming	
			☐ Extractive Industries (oil, gas, mining)
			☐ Forestry (Including HCVF and REDD+)
			☐ Tourism
			☐ Agriculture & agrobiodiversity
			☐ Fisheries
			☐ Infrastructure
			☐ Certification (National Standards)
			☐ Certification (International Standards)
		☐ Species	
			☐ Illegal Wildlife Trade
			☐ Threatened Species
			☐ Wildlife for Sustainable Development
			☐ Crop Wild Relatives
			☐ Plant Genetic Resources
			☐ Animal Genetic Resources
			☐ Livestock Wild Relatives
			☐ Invasive Alien Species (IAS)
		☐ Biomes	
			☐ Mangroves
			☐ Coral Reefs
			☐ Sea Grasses
			☐ Wetlands
			☐ Rivers
			☐ Lakes
			☐ Tropical Rain Forests
			☐ Tropical Dry Forests
			☐ Temperate Forests
			☐ Grasslands
			☐ Paramo
			☐ Desert
		☐ Financial and Accounting	
			☐ Payment for Ecosystem Services
			☐ Natural Capital Assessment and Accounting
			☐ Conservation Trust Funds
			☐ Conservation Finance
		☐ Supplementary Protocol to the CBD	
			☐ Biosafety
			☐ Access to Genetic Resources Benefit Sharing
	☐ **Forests**		
		☐ Forest and Landscape Restoration	
			☐ REDD/REDD+
		☐ Forest	

Level 1	Level 2	Level 3	Level 4
			☐ Amazon
			☐ Congo
			☐ Drylands
	☐ **Land Degradation**		
		☐ Sustainable Land Management	
			☐ Restoration and Rehabilitation of Degraded Lands
			☐ Ecosystem Approach
			☐ Integrated and Cross-sectoral approach
			☐ Community-Based NRM
			☐ Sustainable Livelihoods
			☐ Income Generating Activities
			☐ Sustainable Agriculture
			☐ Sustainable Pasture Management
			☐ Sustainable Forest/Woodland Management
			☐ Improved Soil and Water Management Techniques
			☐ Sustainable Fire Management
			☐ Drought Mitigation/Early Warning
		☐ Land Degradation Neutrality	
			☐ Land Productivity
			☐ Land Cover and Land cover change
			☐ Carbon stocks above or below ground
		☐ Food Security	
	☐ **International Waters**		
		☐ Ship	
		☐ Coastal	
		☐ Freshwater	
			☐ Aquifer
			☐ River Basin
			☐ Lake Basin
		☐ Learning	
		☐ Fisheries	
		☐ Persistent toxic substances	
		☐ SIDS : Small Island Dev States	
		☐ Targeted Research	
		☐ Pollution	
			☐ Persistent toxic substances
			☐ Plastics
			☐ Nutrient pollution from all sectors except wastewater
			☐ Nutrient pollution from Wastewater
		☐ Transboundary Diagnostic Analysis and Strategic Action	

Level 1	Level 2	Level 3	Level 4
		Plan preparation	
		☐ Strategic Action Plan Implementation	
		☐ Areas Beyond National Jurisdiction	
		☐ Large Marine Ecosystems	
		☐ Private Sector	
		☐ Aquaculture	
		☐ Marine Protected Area	
		☐ Biomes	
			☐ Mangrove
			☐ Coral Reefs
			☐ Seagrasses
			☐ Polar Ecosystems
			☐ Constructed Wetlands
	☐ **Chemicals and Waste**		
		☐ Mercury	
		☐ Artisanal and Scale Gold Mining	
		☐ Coal Fired Power Plants	
		☐ Coal Fired Industrial Boilers	
		☐ Cement	
		☐ Non-Ferrous Metals Production	
		☐ Ozone	
		☐ Persistent Organic Pollutants	
		☐ Unintentional Persistent Organic Pollutants	
		☐ Sound Management of chemicals and Waste	
		☐ Waste Management	
			☐ Hazardous Waste Management
			☐ Industrial Waste
			☐ e-Waste
		☐ Emissions	
		☐ Disposal	
		☐ New Persistent Organic Pollutants	
		☐ Polychlorinated Biphenyls	
		☐ Plastics	
		☐ Eco-Efficiency	
		☐ Pesticides	
		☐ DDT - Vector Management	
		☐ DDT - Other	
		☐ Industrial Emissions	
		☐ Open Burning	
		☐ Best Available Technology / Best Environmental Practices	
		☐ Green Chemistry	
	☒ **Climate Change**		
		☐ **Climate Change**	

Level 1	Level 2	Level 3	Level 4
		Adaptation	
			☐ Climate Finance
			☐ Least Developed Countries
			☐ Small Island Developing States
			☐ Disaster Risk Management
			☐ Sea-level rise
			☐ Climate Resilience
			☐ Climate information
			☐ Ecosystem-based Adaptation
			☐ Adaptation Tech Transfer
			☐ National Adaptation Programme of Action
			☐ National Adaptation Plan
			☐ Mainstreaming Adaptation
			☐ Private Sector
			☐ Innovation
			☐ Complementarity
			☐ Community-based Adaptation
			☐ Livelihoods
		☒ **Climate Change Mitigation**	
			☐ Agriculture, Forestry, and other Land Use
			☒ Energy Efficiency
			☒ Sustainable Urban Systems and Transport
			☐ Technology Transfer
			☐ Renewable Energy
			☐ Financing
			☐ Enabling Activities
		☐ **Technology Transfer**	
			☐ Poznan Strategic Programme on Technology Transfer
			☐ Climate Technology Centre & Network (CTCN)
			☐ Endogenous technology
			☐ Technology Needs Assessment
			☐ Adaptation Tech Transfer
		☐ **United Nations Framework on Climate Change**	☐ Nationally Determined Contribution
	☒ **Rio Markers**		
		☒ Paris Agreement	
		☒ Sustainable Development Goals	
		☐ Climate Change Mitigation 0	
		☐ Climate Change Mitigation 1	
		☒ Climate Change Mitigation 2	
		☐ Climate Change Adaptation 0	
		☐ Climate Change Adaptation 1	
		☐ Climate Change Adaptation 2	

Annex 15: Funding scheme to public transport operators participating in e-mobility projects

Background

Public transport (PT) is one obvious area for undertaking demonstrations within GEF-funded projects dealing with the deployment of electro-mobility. Electric buses are already available in the market, and there is sufficient experience of operations in some countries.

E-buses are substantially (some 40%) more expensive than conventional diesel buses with similar characteristics. Although they offer relevant operational and maintenance savings during their service lifespan, these savings are unlikely to be attractive enough for the bus operator. Most of them have limited financial capacity, and also limited access to loans at attractive rates.

In the case of a single public bus operator, it is feasible to identify it as a "responsible party" within the project; in this case, the bus operator, as a responsible party, can make use of the project budget to undertake the project activities (in this case the procurement and operation of the e-buses). If the bus operator is identified as a co-financing partner, its contribution to the purchase and operation of the e-buses are identified within the Project Document. In any model, the project is expected to finance only the incremental costs of the e-buses compared to the purchase and operation of the conventional technology in place in the country (Euro IV bus in the case of Peruvian cities).

When there are many bus operators, which can potentially participate in the project, it is necessary to establish a competitive mechanism to select the most appropriate partner. The project's priority in the selection of its partner is to keep the e-buses operating as much as possible, as this provides a higher number of beneficiaries and GHG emission savings. This can be done through a contract of services.

When engaging with a private sector as a responsible party:

- All engagement with the private sector by UNDP follows the principles outlined in the Guidelines on Cooperation Between the United Nations and the Business Sector.
- The selection of a private sector partner is based on a competitive procurement process or a challenge prize undertaken by an implementing partner. Procedures are found in the Contracts and Procurement Section of the Programme and Operations Policies and Procedures, and the Open Innovation Policy. Private sector partners are subject to due diligence policies and procedures, and the Guidelines on Cooperation with the Business Sector.

Purpose of the contract of services

In the contract of services, the Executing Agency (EA) is requesting the contractor to operate an e-bus on a regular urban line for a minimum period of time and a minimum of km travelled. The EA is regularly paying the contractor with an amount according to the verified km travelled.

The contractor

The contractor should own the required e-buses at the time operations are expected to start. Accordingly, the contractor could be a bus operator but also an entity owning the e-buses and leasing them to a bus

operator. Such entities could be bus manufacturers (this has been the case in China), electricity utilities (this has been the case in Chile) or purely financial institutions interested in developing their leasing businesses (this has been the case in the United Kingdom); it can also be a joint venture of some of these.

At the time of providing its offer, the offeror is not expected to already own an e-bus, but he is expected to have signed a Letter of Intent with the manufacturer concerning the bus model and the delivery schedule. If the offeror is not a bus operator, he will be asked to provide a Letter of Intent with at least one bus operator to run the e-buses.

In the case of three-wheeler (3W) mototaxis, the contractor could also be a 3W operator with the necessary local permits to provide the service or an entity owning the electric 3W and leasing them to operating companies, associations or individual drivers. The contractor will be requested to comply to a rigorous code of conduct (to be defined during the design of the pilot) regarding working conditions and quality of service, as these are not fully addressed in the existing local regulations.

The e-buses

The technical characteristics of the e-buses to be used in the project must be defined in the contract. They must be consistent with the characteristics established for bus operations by the relevant Public Transport Authority (vehicle dimensions, number of seats, maximum number of standing passengers, accessibility conditions, cameras and other security equipment, GPS tracking…). There are additional minimum characteristics to be requested concerning the electric technology, such as battery capacity, range, and vehicle manufacturing year.

Although the technical conditions of the e-buses will be defined within the technical assistance for the design of the demonstrations, the following characteristics can be anticipated:

- Bus length: 12 m.
- Maximum passenger capacity (seating and standing): 80 passengers.
- Maximum motor power: 2x150 kW
- Maximum climb gradient: 18%
- Battery capacity: 350 kWh
- GPS tracking system including information of km travelled
- Ticketing system allowing to monitor the number of passengers.

Incremental costs of e-buses

The actual costs of conventional and electric buses are subject to many conditions (market availability, number of vehicles to procure, changes in the specifications required by public transport authorities…). Based on the interviews conducted during project design with different stakeholders and recent published research[60], the budget is based on the figures presented below.

[60] Grauers, A., Borén, S., & Enerbäck, O. (2020). Total Cost of Ownership Model and Significant Cost Parameters for the Design of Electric Bus Systems. *Energies*, 13(12), 3262.
Göhlich, D., Fay, T. A., Jefferies, D., Lauth, E., Kunith, A., & Zhang, X. (2018). Design of urban electric bus systems. *Design Science*, 4..¡

	ICE (diesel) bus	Electric bus
12-m bus	USD 270,000	USD 330,000
Battery (300 kWh)		USD 75,000
Depot charger (DC, 50-100 kW)		USD 35,000
Annual mileage	54,000 km	54,000 km
Consumption	52 liters/100 km	75 kWh/100 km
Energy cost	0.939 USD/liter	0.118 USD/kWh
Annual maintenance cost	0.77 USD/km	0.37 USD/km
Bus and battery economic life	10 years	10 years
Scrapage value	0	0

The incremental cost of an electric bus compared to a diesel bus is therefore USD 150,000, including the cost of the battery and the infrastructure at the depot for slow night charging.

In accordance with these figures, e-buses would be an attractive investment, with an internal rate of return (IRR) of 23%. Considering a 9% discount rate (close to commercial lending rate in Peru), the PTO would recover the incremental cost of e-buses by the end of the fourth year of operation.

However, to overcome the existing barriers to the use of e-buses, it is necessary to provide a reasonable incentive to the PTOs participating in the project. The assumption at project design is a worst case scenario: the PTOs need to cover almost 100% of the incremental costs (USD 145,000) to sign the contract and provide the expected services. Under these circumstances, the IRR jumps to 44%.

Therefore, there is a logic in establishing a competitive bidding process in which the interested PTO can request a lower contribution for the project (eventually, liberating resources to introduce additional electric vehicles or other activities).

The service requested

Payments to the contractor are based on verified results: The service requested is to provide a minimum number of km of bus service during the demonstration (a period of 1 to 2 years), and the payments can be made every month. An advance for the first month can also be envisaged, subject to the signing of the contract and the delivery of a gender action plan and a monitoring plan by the contractor.

The km travelled can be monitored with a GPS system, to which the project management unit should have access. Additional information can also be gathered for monitoring purposes (e.g. energy consumption, off-line km, etc).

The payment per km travelled depends on the total maximum project contribution per bus and the expected km travelled during the demonstration. For example, if it is expected that the e-bus will travel 54,000 km per year, the expected project contribution per bus is USD145,000 and the monitoring period is 2 years, the project would pay USD 1.34/km. The competitive process can define a threshold and select the best value for money for total amount of km travelled and emissions mitigated according to the public transport operator's proposal.

The payment can be improved with a bonus-malus system. For example, it can be established that the contractor will receive each monthly payment only if during that month a certain threshold of km have been reached.

The following minimum specifications could be included in the request of services:

- Minimum period of demonstration: 24 months.
- Expected in-route km travelled per year during the demonstration period: 54,000 km
- Maximum project contribution per in-route km travelled: 1.34 USD/km
- Maximum project contribution in 2 years: USD 145,000

Sustainability

After completion of the project demonstration, the e-buses remain under the ownership of the contractor.

The contractor will have good reasons to continue providing public transport services with the e-buses, as their operating and maintenance costs are lower than those of a regular bus, and the project contribution will have served to compensate a substantial part of the additional purchase costs. In the case the contractor is a bus operator, the e-bus will be integrated in the company's fleet and will continue operations. In the case the contractor is a leasing company, it will be interested in keeping its asset active, offering the bus operator a competitive leasing deal to keep the e-bus providing regular public transport services.

3-wheeler taxis

The considerations made above for e-buses are also valid for 3-wheeler (3W) taxis with some specificities.

The technical characteristics of the 3W to be used in the project must be defined in the contract. As there are no clear specifications established at the local level (power, capacity, etc.), the technical conditions of the 3W will be defined within the technical assistance for the design of the demonstrations; however, the following basic characteristics can be anticipated:

- Maximum capacity: 3 passengers or 200 kg.
- Maximum motor power: 8 kW
- Maximum climb gradient: 15%
- Battery capacity: 6 kWh (needed to negotiate strong slopes)
- GPS tracking system including information of km travelled (this could be requested to the operators or provided by the project)
- Monitoring of the number of passengers based on monthly operators' reports.

Incremental costs of 3-wheelers

There are no official regulations on the technical requirements for 3-wheelers providing taxi services in Lima or Arequipa. Based on the interviews conducted during project design with different stakeholders, the budget is based on the figures presented below.

	ICE (gasoline) 3 wheeler	**Electric 3 wheeler**
3 wheeler, 3 passengers (200 kg)	USD 1,600	USD 1,500
Battery (6 kWh)		USD 2,400
Depot charger (DC, 50-100 kW)		USD 1,000
Annual mileage	14,000 km	14,000 km
Consumption	6 liters/100 km	1.2 kWh/100 km
Energy cost	0.939 USD/liter	0.118 USD/kWh
Annual maintenance cost	0.01 USD/km	0.005 USD/km
Bus and battery economic life	5 years	5 years
Scrappage value	0	0

Therefore, the incremental cost of an electric 3W compared to a gasoline 3W is USD 3,300, including the cost of the battery and the infrastructure at home or at the depot for slow night charging. On top of that there is the cost and operation of the GPS tracking system for two years, estimated at USD 1,700. Although this item must not be included in the financial analysis, it is relevant for adequate monitoring of the pilot. The total incremental cost is therefore USD 5,000 per vehicle.

In accordance with these figures, electric 3W would not be an attractive investment, as its internal rate of return (IRR) would be close to the commercial lending rate in Peru (9%); the 3W operator would recover the incremental cost of the electric 3W at the end of the fifth year of operation, but would not have any incentive in taking the burden and risk of a significantly bigger initial investment. It can be argued that the cost of the electric 3W is likely to decrease in the near future, that the battery capacity actually required could be significantly lower to 6 kWh and that the cost of the charger is not strictly necessary; however, all these elements are highly recommendable for a successful pilot.

Therefore, to overcome the existing barriers to the use of electric 3W, it is necessary to provide a reasonable incentive to the mototaxi operators participating in the project. The assumption at project design is a worst case scenario: the mototaxi operators need to cover 100% of the incremental costs (USD 3,300) to sign the contract and provide the expected services. Under these circumstances, the IRR jumps to 50%, and the operator starts receiving an additional benefit in the second year of operation.

Therefore, there is a logic in establishing a competitive bidding process in which the interested mototaxi operators can request a lower contribution for the project (eventually, liberating resources to introduce additional electric vehicles or other activities).

The 3W service requested

Payments to the contractor are based on verified results: The service requested is to provide a minimum number of km of 3W service during the demonstration (a period of 1 to 2 years), and the payments can be made every month. An advance for the first month can also be envisaged, subject to the signing of the

contract, the completion of a gender sensitization training course by the drivers and the acceptance of the monitoring plan by the contractor.

The km travelled can be monitored with a cell-phone application, to which the project management unit should have access. Additional information can also be gathered for monitoring purposes (e.g. energy consumption, off-line km, etc), as defined in the monitoring plan to be established during the demonstration design stage.

The payment per km travelled depends on the total maximum project contribution per 3W and the expected km travelled during the demonstration. For example, if it is expected that the 3W will travel 14,000 km per year, the expected project contribution per 3W is USD 5,000 (USD3,300 plus USD 1,700 for GPS tracking) and the monitoring period is 2 years, the project would pay USD 0.18/km. The competitive process can define a threshold and select the best value for money for total amount of km travelled and emissions mitigated according to the mototaxi operator's proposal.

The payment can be improved with a bonus-malus system. For example, it can be established that the contractor will receive each monthly payment only if during that month a certain threshold of km have been reached.

The following minimum specifications could be included in the request of services:

- Minimum period of demonstration: 24 months.
- Expected in-route km travelled per year during the demonstration period: 14,000 km
- Maximum project contribution per in-route km travelled: 0.18 USD/km
- Maximum project contribution in 2 years: USD 3,300 (plus USD 1,700 for GPS tracking)

The request for services launched by the project can be addressed to individual owners and fleet owners holding an official permit to provide 3-wheeler taxi services or to manufacturers or financial companies interested in leasing the vehicles to authorized individuals or associations.

Sustainability

After completion of the project demonstration, the 3W remain under the ownership of the contractor.

The contractor will have good reasons to continue providing the electric 3W operational, as their operating and maintenance costs are lower than those of a gasoline 3W. In the case the contractor is a 3W operator, the electric 3W will be integrated in the company's fleet and will continue operations. In the case the contractor is a leasing company, it will be interested in keeping its asset active, offering the 3W operator a competitive leasing deal.